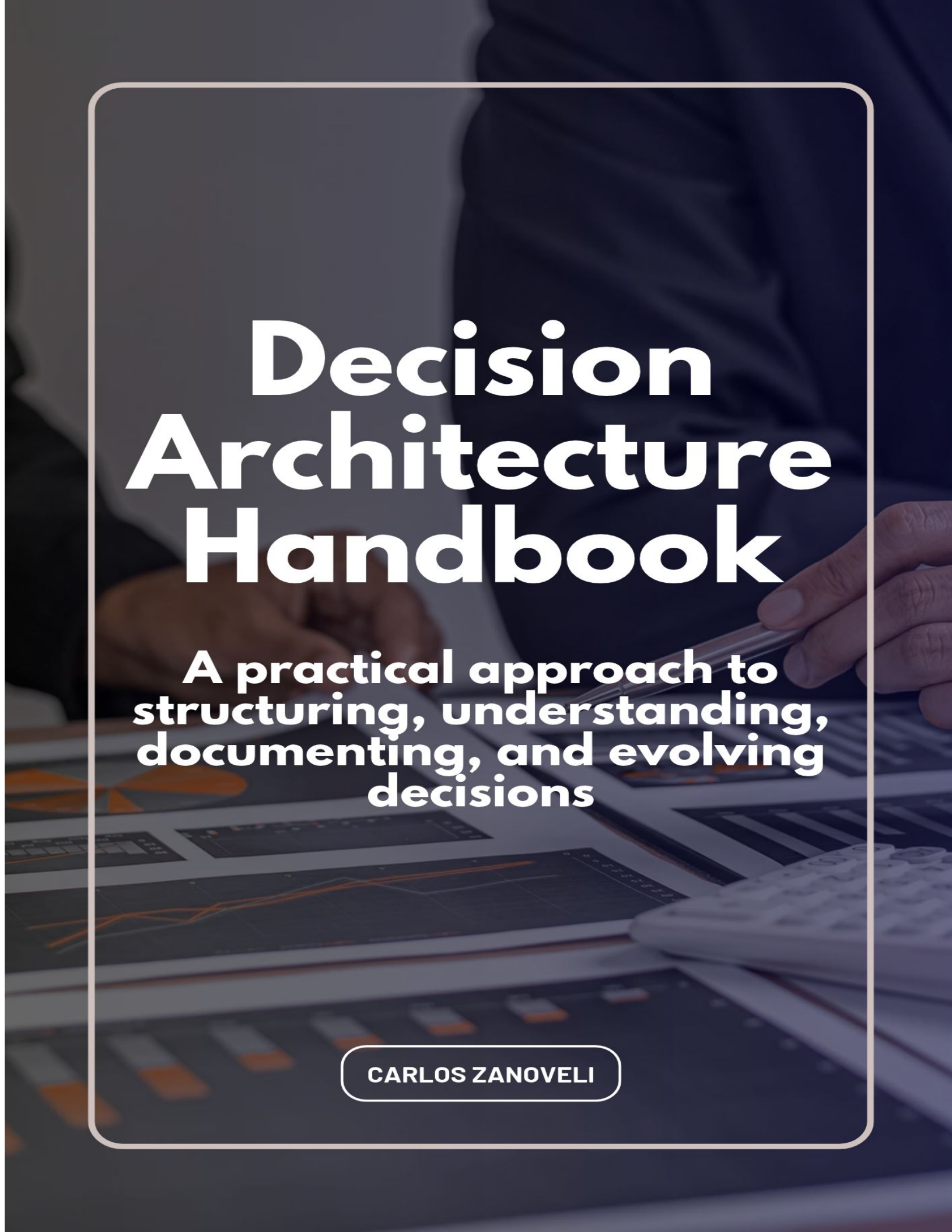


Decision Architecture Handbook

A practical approach to structuring, understanding, documenting, and evolving decisions

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Introduction

The initial idea behind this work was to develop a methodological reference for Solution Architecture that would play a role similar to that of the PMBOK in Project Management. I felt the need for an approach that would organize the discipline's knowledge and practices, providing solution architects with a reference to guide their work.

My education in Project Management and Solution Architecture, both at the MBA level, contributed to this perception. In the former, I encountered established methods and practices for organizing initiatives and managing resources, risks, and deliverables. In the latter, I found approaches for understanding business needs, evaluating technological alternatives, and structuring solutions. My entrepreneurial experience, particularly in the startup environment, added another perspective: the need to make decisions frequently with limited resources, incomplete information, and conditions subject to change.

As I moved between these fields, I began to realize that, despite their particularities, they dealt with similar fundamental issues. Managing a project, designing a solution, or developing a business requires understanding a situation, recognizing its conditions, establishing objectives, evaluating alternatives, dealing with uncertainties, and considering the consequences of choices. The methods employed are different, but many of the elements involved in decision-making are recurring.

This realization changed the direction of the work. Rather than developing a methodology specifically for Solution Architecture and, subsequently, others for Project Management and Entrepreneurship, I began investigating the concepts that precede these disciplines and help structure their practices. The objective was no longer to organize the work of a specific professional activity, but to understand the elements common to decisions made in different contexts. This shift gave rise to what I call, in this guide, Decision Architecture.

The proposal is not intended to replace existing frameworks or methodologies. References such as PMBOK, TOGAF, arc42, and Lean Startup address specific needs and have their own knowledge bases and practices. The focus of this work is on a layer that precedes their application: understanding which aspects need to be considered in a decision and how they relate to one another, regardless of the domain in which it takes place or the method used to carry it out.

This choice does not mean abandoning the possibility of developing dedicated works on the disciplines that motivated this investigation. Project Management, Solution Architecture, and Entrepreneurship continue to offer opportunities for further exploration. However, I considered it more important to begin with the broader perspective that makes it possible to understand their relationships, leaving discipline-specific studies as future possibilities.

Guide Structure

The guide is organized into two complementary perspectives. In the first chapters, I present the foundations of the proposal and explore each of them individually. In the final chapter, I reverse the approach: I start with concrete problems and demonstrate how the foundations can be used to analyze them.

The first chapter presents the six foundations of Decision Architecture: domain, initial context, objectives, risks and uncertainties, direct and indirect consequences, and evolution. They represent dimensions that make it possible to understand different aspects of a decision and the relationships between them. They do not constitute mandatory stages in a process, but rather perspectives that can be considered together, according to the characteristics of the situation being analyzed.

The following chapters are dedicated to exploring each foundation in greater depth. The second addresses the domain, identifying the field of knowledge or activity in which the decision is situated. The third deals with the initial context, considering existing conditions, available resources, constraints, assumptions, and dependencies that influence the alternatives. The fourth explores objectives, their definition criteria, and their importance in guiding choices.

The fifth chapter examines risks and uncertainties, including incomplete information, assumptions, and events that may interfere with outcomes. The sixth broadens the analysis to encompass the direct and indirect consequences of the alternatives, considering their impacts on people, resources, systems, and opportunities. The seventh addresses evolution, examining how decisions modify existing conditions and influence subsequent choices.

Throughout these chapters, the foundations are connected to practices and situations from different fields, particularly Project Management, Solution Architecture, Entrepreneurship, and User Experience. The examples seek to demonstrate how the same concepts can be applied in distinct contexts while respecting the particularities of each discipline.

The eighth chapter adopts a different perspective. Rather than starting with the foundations, it starts with problems and areas of activity to demonstrate how they can be identified and used in the analysis of concrete situations. Scenarios related to the disciplines already presented are explored, along with situations from professional and personal life.

This reversal brings the proposal closer to the way we typically deal with decisions. In practice, we rarely begin with an organized conceptual framework. We generally start with a problem, an opportunity, or a need for change and must identify which aspects deserve attention. The chapter demonstrates how the foundations can help guide this process without requiring the analysis to begin with any specific dimension.

This guide is also part of an open and evolving project. The concepts presented represent the current stage of development of the proposal and may be reviewed, further explored, or expanded as new applications, experiences, and lessons are incorporated. I do not consider this publication a definitive version, but rather a starting point for organizing and sharing the ideas developed so far.

The term Decision Architecture itself also remains open to revision. I chose this expression because I consider it appropriate for representing the proposal at this stage, particularly because of the idea of structuring the elements involved in a decision. I recognize, however, that the term may evoke associations with other concepts and fields of study, leading to interpretations different from the one I intend to develop. I remain open to finding a more precise designation as the work matures. The current name is a way of identifying the proposal, not a commitment to a definitive terminological definition.

Finally, I would like the development of these ideas to extend beyond the boundaries of an individual initiative. If the concepts spark the interest of other professionals and researchers, I would be very pleased to see them discussed, challenged, and refined collaboratively. Building a community around the proposal, bringing together experiences and perspectives from different fields, would be a valuable contribution to its evolution.

1. Foundations of Decision Architecture

This guide proposes six foundations for understanding decisions. They do not constitute a universal definition of Decision Architecture, nor should they be understood as stages of a process or as a list of mandatory elements. They are a conceptual proposition intended to make recurring characteristics of decisions and the relationships between them explicit.

A foundation, in this context, is a structural characteristic of a decision. It helps us understand a dimension that is present when a choice is considered, made, and produces effects. It does not, by itself, define how the decision should be conducted. Its function is to provide a reference for understanding the situation before determining how to act upon it.

This distinction is important because concrete decisions are often addressed through methods, techniques, and frameworks developed for specific contexts. These approaches organize ways of working and provide mechanisms for dealing with specific problems. The foundations proposed here are at a prior level: they seek to describe characteristics of the decision itself, regardless of the particular way used to conduct it.

The characteristics of a decision, however, do not appear in isolation. A decision takes place within some domain and starts from certain conditions. It is guided by objectives, but these objectives need to be considered in light of what we do not yet know. The choice produces consequences, and those consequences modify the conditions under which subsequent decisions will be made.

This structure can be observed in different types of decisions. An architectural decision, a business decision, a project decision, or a personal decision have distinct natures, but all of them occur in some situation, pursue some outcome, are made with incomplete knowledge, and produce effects that may alter what will be possible to do later.

The objective of this chapter is not to introduce new terminology to replace already established concepts. Many of the ideas presented here are well known and appear, under different names and levels of formalization, in disciplines such as architecture, management, engineering, entrepreneurship, and product development.

The proposition is different: to start from the most basic questions we need to answer in order to understand a decision and, from them, identify the needs that derive from them. Instead of starting with the available methods and trying to fit the decision into their structures, we begin by asking what needs to be understood about the decision itself.

What are the minimum conditions we need to know in order to understand a decision? What defines the field in which it takes place? From what situation is it made? What guides the choice? What do we still not know? What changes as a consequence of the decision? And how do these changes constrain subsequent decisions?

It is from these questions that we arrive at the foundations presented below. Each seeks to answer a more basic question about the structure of the decision and, from it, makes it possible to derive needs that can be addressed through different practices and approaches.

1.1. Every decision occurs within one or more domains

Every decision occurs within one or more domains of knowledge, activity, or problem. The domain establishes the field in which the decision is understood and provides concepts, knowledge, criteria, and practices used to interpret situations and evaluate alternatives.

A decision about a payments platform, for example, may require knowledge of architecture, security, finance, product, operations, and regulation. Each domain contributes its own references for understanding different aspects of the decision.

The participation of different domains does not simply mean bringing experts together in the same discussion. Each domain may see a different part of the situation and use different criteria to evaluate the alternatives. A technically suitable alternative may be financially unfeasible. A financially attractive solution may create operational risks. A decision suitable for the product may conflict with a regulatory constraint.

Therefore, the domain is not merely the subject we are talking about. It influences what we consider relevant, which concepts we use, and which questions need to be asked.

A software architecture decision mobilizes concepts such as components, interfaces, technologies, quality attributes, and dependencies. A project management decision may involve scope, schedule, budget, resources, risks, and dependencies. An entrepreneurial decision may involve customers, market, value proposition, business model, and resource allocation.

Specialized knowledge provides references for analysis, but it does not determine a single answer. Two decisions may occur in the same domain and arrive at different choices because they have different objectives, conditions, constraints, or alternatives.

This also explains why practices developed in one domain should not be automatically transferred to another. A practice may make sense because it responds to a specific need in that field and because its concepts, criteria, and artifacts were built for particular conditions.

The first need produced by this foundation is domain delimitation. We need to know which fields the decision belongs to, which knowledge is relevant, which participants need to be considered, and which boundaries define what is being analyzed.

Delimiting does not necessarily mean choosing a single domain. A decision may deliberately cross boundaries between areas. The objective is to make these boundaries visible so that we can understand which perspectives are part of the decision and which knowledge we need to mobilize.

This need can be addressed through practices such as scope definition, boundary identification, domain modeling, stakeholder identification, and the creation of a common language. Different frameworks organize these practices in different ways, but all are, to some extent, responding to the need to understand where the decision is situated.

1.2. Every decision starts from an initial context

Every decision starts from an existing situation. The initial context brings together the relevant conditions for understanding the decision at the moment it is considered.

Domain and context are not the same thing. The domain defines the field in which the decision makes sense and provides the references used to understand it. The context defines the particular conditions in which it occurs.

These conditions may include available resources, constraints, information, evidence, assumptions, commitments, dependencies, previous decisions, and other circumstances that influence the possibilities being considered.

The same domain may present completely different contexts. An architecture decision may occur in a new system or in a legacy environment. There may be an experienced team or a team that still needs to acquire knowledge. There may be an available budget or a significant financial constraint. There may be technological freedom or a contractual dependency that limits the alternatives.

These differences are not peripheral details. They can completely change the set of viable alternatives and how each alternative should be evaluated.

Context is also not static. New information may emerge, resources may be consumed, constraints may change, and previous decisions may produce unexpected effects. Therefore, the context considered at the beginning of a decision may not be exactly the same context encountered during its execution.

This characteristic also explains why previous experience cannot be treated as a ready-made solution. Past experience may provide useful references, patterns, and hypotheses, but its applicability depends on current conditions.

Experience provides references. Context determines their applicability.

The second need produced by this foundation is the explicit definition of the decision's conditions. We need to identify what exists, which resources are available, which constraints must be respected, which assumptions are being made, which dependencies exist, and which information is not yet available.

This need is addressed by practices involving diagnosis, constraint gathering, assumption identification, dependency analysis, understanding the existing environment, and identifying current conditions. The objective is not to describe all of reality, but to make visible the conditions that actually influence the decision.

This is particularly important when a method or framework will be applied. The practices of a framework were developed to address certain needs and assume certain conditions of application. Using them without verifying

whether these conditions exist can produce a situation in which we correctly follow the method but address a problem different from the one we actually have.

This is where one of the differences between knowing a method and knowing how to use it appears. The proper application of a practice depends on the ability to recognize the conditions for which it was designed and verify whether those conditions are present in the current situation.

1.3. Every decision is guided by one or more objectives

Every decision is related to one or more outcomes that are intended to be achieved, preserved, or avoided. These objectives guide the choice and provide a reference for determining what is intended to be obtained.

An objective may be clearly formulated or remain implicit. A person may decide to reduce costs without having previously defined by how much they intend to reduce them or within what timeframe. An organization may decide to modernize a system without having clarified whether the main objective is to reduce costs, increase capacity, decrease risks, or enable a new business strategy.

An objective does not exist independently of the domain and context. Every objective presupposes a reality to which it refers and conditions under which the intended outcome is relevant, even when these references are not made explicit.

Consider, for example, the objective of "reducing approval time." For this statement to have meaning, there must be some reality in which there is an approval process, a definition of what this time represents, and conditions under which its reduction is desired. "Reducing costs" presupposes costs of something. "Increasing availability" presupposes a system, service, or operation. The more these references are removed, the more generic and less informative the objective becomes.

This means that formulating an objective already contains assumptions about the domain and context of the decision. Analysis can make these assumptions explicit and verify whether they correspond to reality. Therefore, an investigation can begin with the objective, but the objective should not be treated as something semantically independent of the domain and context.

The relationship between these dimensions also does not imply a rigid sequence. It is possible to start with the objective, the domain, the context, or a situation perceived as problematic. What matters is recognizing that an adequate understanding of an objective involves the reality to which it refers and the conditions under which it is relevant.

When objectives are not identified or understood by the decision-maker, we have a blind decision. The decision still has a direction or purpose, but part of the logic guiding the choice remains implicit.

Making objectives explicit makes it possible to establish what should be considered in the choice. Without knowing what is intended to be achieved, it becomes difficult to determine which alternatives are relevant and which characteristics should be considered when comparing them.

A decision may also involve different objectives that cannot be met simultaneously to the same extent. Reducing costs may conflict with increasing quality. Accelerating a delivery may conflict with reducing risks. Increasing flexibility may conflict with reducing complexity.

Therefore, defining objectives does not simply mean producing a list. It is necessary to understand which objectives exist, how they relate to one another, and which of them are priorities when they cannot be met simultaneously.

Objectives may also change during the analysis. New information may show that what initially seemed important is no longer relevant, that a particular objective is not feasible under the existing conditions, or that another objective, previously not considered, is more important to the decision.

This creates an important relationship between objective and problem. A situation becomes a problem in relation to some intended outcome. If the objective changes, the interpretation of the situation may also change.

For example, "replacing the current system" may initially appear to be a technical problem. But if the objective is to reduce the time required to launch new products, replacing the entire system may not be necessary. The problem may be related to a specific capability, rather than to the technology as a whole.

In this sense, problem and objective should not be treated as completely independent elements. The definition of what needs to be solved depends, in part, on the outcome that is intended to be achieved.

Defining objectives also establishes the need for evaluation criteria. There must be references that make it possible to determine the extent to which an alternative meets what is intended to be achieved.

These criteria may take different forms, such as requirements, metrics, indicators, quality attributes, success conditions, or acceptable limits. Not every objective needs to be reduced to a metric. What matters is that there is a sufficiently clear way to evaluate whether what was chosen meets the objectives of the decision.

1.4. Every decision involves some degree of uncertainty

A decision involves choosing between possibilities without completely knowing their consequences or future conditions. The degree of uncertainty varies according to the quantity and quality of available information, accumulated experience, and the predictability of the situation.

Some decisions have a large amount of data and evidence available. Others must be made with scarce information and many unknowns. There are also situations in which the consequences of an alternative are largely known. In such cases, uncertainty may be small, but there is still a choice about how to act.

Uncertainty may also have different origins. We may not sufficiently understand the problem, not know how a solution will work, be unable to predict the behavior of users or customers, depend on external factors, or not know how certain variables will evolve.

This distinction is important because different types of uncertainty require different forms of investigation. When we do not understand the problem, we need to learn about the situation. When we do not know whether a solution works, we may need to experiment. When we know the alternatives but do not know which consequences will occur, we may need to analyze risks and scenarios.

Uncertainty does not need to be eliminated before acting. We can seek information, consult experts, test hypotheses, conduct experiments, build prototypes, or wait for new data.

Each alternative, however, has its own cost. Further investigation may reduce what is unknown, but it may also consume time, resources, or opportunities. An experiment may generate evidence, but it requires investment. Waiting for more information may improve a decision or simply delay a necessary action.

Deciding therefore involves evaluating not only which alternatives are available, but also how much it is worth learning before acting.

This evaluation also depends on the nature of the consequences. When a decision is easily reversible, it may be acceptable to proceed with a higher degree of uncertainty. When a choice creates commitments that are difficult to undo, the cost of a premature decision may be much greater.

The fourth need produced by this foundation is the treatment of the unknown. We need to identify what we still do not know, assess its importance, and decide how to deal with this uncertainty.

Hypotheses, evidence, experiments, prototypes, research, scenarios, risks, and opportunities are different ways of dealing with what we do not yet know.

Lean Startup, for example, structures practices for turning hypotheses into experiments and learning. Design Thinking uses investigation, prototyping, and testing activities to learn about needs and possible solutions. Risk management structures the identification and analysis of uncertain events and their possible consequences.

These approaches are not equivalent and should not be applied merely because a decision contains uncertainty. The issue is to understand what uncertainty exists, what knowledge is missing, and which practice can produce relevant evidence for the decision.

1.5. Every decision produces direct and indirect consequences

A decision produces more than an immediate outcome. It can commit resources, create dependencies, establish constraints, eliminate paths, or make certain changes more costly. Some of these consequences appear directly after the decision, while others emerge as indirect effects of what was changed.

Consequences can also propagate to other parts of the situation. A decision about technology can alter a team's work. A project decision can modify responsibilities, schedules, or resources. A product decision can affect customers, users, or partners. An organizational change can create new commitments for other areas.

When a decision produces effects on other people or groups, a need for communication also arises. It is necessary to make clear what was decided, why the decision was made, what consequences are expected, and what changes for those who will be affected or need to act as a result of it. In this sense, communication is not merely an activity that follows the decision, but part of dealing with its consequences.

A decision can also preserve options, generate new alternatives, or produce information useful for subsequent decisions. Its effects, therefore, are not limited to what happens immediately after the choice. Some consequences modify the conditions under which other decisions will be made.

This is how we can understand the space of possibilities. It represents what can be done from a given point in time, considering the existing resources, commitments, dependencies, and constraints. The consequences of a decision can alter this space, making some possibilities more accessible, others more difficult, and some infeasible.

Changing the space of possibilities does not necessarily mean reducing options. A choice can eliminate certain paths while simultaneously creating others. A new technological capability can open alternatives that did not exist. A commercial decision can create access to one market and close another.

Every choice also involves trade-offs. By favoring a particular outcome, a decision may require concessions in other objectives, criteria, or possibilities.

Improving performance may increase cost. Reducing time may increase risk. Increasing flexibility may raise complexity. Preserving compatibility may limit the ability to evolve. In many cases, there is no alternative that simultaneously maximizes all objectives.

This dynamic can be observed through the idea of optionality. Some choices preserve a greater capacity for future change. Others increase commitments and reduce room for maneuver. The consequence of a decision can therefore also be observed through the capacity it preserves or eliminates for subsequent decisions.

Reversibility is relevant at this point. A decision that is easy to undo produces different consequences from a decision that requires significant effort or cost to reverse. This does not mean that reversible decisions are always better, but that their consequence structure is different.

Time makes this dynamic even more evident. Delaying a choice may allow new information to emerge, but it may also cause an opportunity to disappear, consume resources, or allow other decisions to be made first.

Remaining in the current state does not mean remaining with the same alternatives. The context itself continues to evolve, and the absence of a decision may cause certain options to become more expensive, infeasible, or simply cease to exist.

It is in this sense that deliberately not deciding can also constitute a decision. There is a difference between consciously choosing not to act and simply failing to perceive that a decision needed to be made, but both situations can affect future possibilities.

The fifth need produced by this foundation is to understand, address, and communicate the consequences of the alternatives under consideration. This involves analyzing not only which alternative produces a given outcome, but also what direct and indirect effects it may generate, who may be affected, how these effects need to be communicated, and how the decision modifies the conditions for subsequent decisions.

Scenarios, alternative analysis, trade-offs, communication, coordination, reversibility, dependencies, opportunity cost, optionality, and path dependency are different ways of addressing this need.

The objective is not to find a universally better alternative. It is to understand how each alternative responds to the existing objectives and constraints, what consequences it produces, who may be affected, and how these consequences alter the space of possibilities for the future.

1.6. Every decision participates in a process of evolution

A decision modifies the subsequent situation because its consequences alter the space of possibilities. Resources may be consumed or created, commitments may be made, dependencies may arise, alternatives may be eliminated or opened, and new information may be produced.

This alteration of the space of possibilities changes the conditions that will be observed for subsequent decisions. The context may change, some objectives may become infeasible or gain importance, new uncertainties may emerge or disappear, and other alternatives may come to exist.

This is the sense in which this guide uses the term evolution. Evolution means a change of state over time as a consequence of decisions, actions taken, information produced, and changing conditions. It does not necessarily imply improvement.

A system may move closer to its objectives, but it may also accumulate constraints, dependencies, costs, or problems that make subsequent changes more difficult. An organization may learn from a decision while simultaneously taking on commitments that constrain its next choices.

Planning, execution, observation, and learning are part of this dynamic. A decision may be maintained, revised, or replaced as new data emerges and as the conditions for new choices change.

Models such as PDCA and OODA represent different ways of organizing cycles of this type. They are not Decision Architecture models, but they help illustrate a dynamic in which action, observation, learning, and change are related.

When this process is repeated, previous choices begin to influence subsequent ones. They generate dependencies, commitments, constraints, and learning that become part of the conditions for future choices.

This accumulation also appears in systems architecture. The current architecture can be seen as the result of many decisions made over time, some deliberate and others conditioned by the circumstances that existed.

An architectural decision, for example, may introduce a technology. That technology then influences the hiring of professionals, the choice of tools, operational costs, and integration decisions. An initial decision therefore participates in shaping the conditions for subsequent decisions.

This means that a decision should not be analyzed only based on the state existing at the moment it is made. We also need to consider the effects it produces on the space of possibilities, the conditions of future decisions, and the knowledge it generates.

The sixth need produced by this foundation is the observation, learning, and preservation of the knowledge produced by evolution.

We need to be able to observe what happened, compare the outcome with what we expected, and understand the reasons for any differences.

We also need to be able to retrieve relevant information about previous decisions when new decisions depend on them. This does not mean documenting everything. It means preserving what will be necessary to understand relevant decisions in the future.

Depending on the domain, this can be done through indicators, feedback, retrospectives, decision records, documentation, experiments, or other learning mechanisms.

Traceability emerges as a way of preserving this knowledge, but it is not the objective in itself. The objective is to maintain the ability to understand evolution and use the knowledge produced to guide future decisions.

1.7. How the foundations relate to one another

The six foundations proposed in this guide should not be interpreted as six independent subjects. They describe different dimensions of the same decision structure.

A decision occurs in a domain, but that domain always manifests itself within a specific context. The context defines conditions that influence what can be done. Within these conditions, there are objectives that guide the choice. Because the future is not completely known, there is uncertainty. The decision produces direct and indirect consequences, which can affect people, resources, systems, and the conditions for future decisions. These consequences alter the space of possibilities and, in doing so, modify the conditions under which new decisions will be made. This is the process through which evolution occurs.

Thus, we can visualize this relationship as follows:

Domain → context → objectives → uncertainty → consequences → evolution

The sequence does not represent a series of steps. It represents a relationship between dimensions that continuously influence one another.

New information about the domain can change our understanding of the context. A change in context can make an objective infeasible or reveal another that is more important. A different objective can change which alternatives are considered. New evidence can reduce uncertainty. A decision can produce consequences that eliminate a future alternative or create a new possibility. These changes modify the space of possibilities and, consequently, the conditions available for subsequent decisions. An unexpected outcome can also change the context again and require a new understanding of the situation.

A decision therefore does not take place within a static structure. It takes place within a structure that changes as we learn, choose, and act.

This relationship helps explain why different practices and frameworks can appear so different and yet address related needs.

We can establish a second relationship:

Characteristic → need → practice → framework

The domain produces the need for delimitation. This need can be addressed through practices involving boundary definition, modeling, scope, and a common language. DDD, for example, offers practices for understanding and delimiting domains, establishing models, and defining bounded contexts. Other architecture, management, and business analysis approaches also have practices intended to make explicit the boundaries of what is being analyzed.

Context produces the need to make existing conditions explicit. This may involve gathering constraints, assumptions, dependencies, the current situation, stakeholders, and available resources. arc42 explicitly addresses context and scope, constraints, and solution strategy. TOGAF also structures activities related to understanding the environment, architecture, and transition. In projects, planning and diagnostic practices fulfill similar functions.

Objectives produce the need for evaluation criteria. Requirements, metrics, indicators, quality attributes, and success criteria are some of the possible ways to address this need. Scrum uses Product Goal and Sprint Goal to guide the work. PMBOK addresses objectives, planning, delivery, and measurement. In architecture, functional requirements, quality attributes, and constraints help establish references for evaluating alternatives.

Uncertainty produces the need to deal with what we do not yet know. In this case, we can use risk analysis, hypothesis formulation, experimentation, prototyping, research, scenarios, or other investigative practices. Lean Startup uses experimentation and learning to test hypotheses. Design Thinking uses investigation, ideation, prototyping, and testing to learn about needs and possible solutions. Risk management practices address uncertain events and their possible consequences.

Consequences produce the need to analyze their effects and address those that need to be considered by the decision. This includes understanding direct and indirect impacts, identifying who may be affected, communicating relevant changes, coordinating resulting actions, and evaluating how the decision modifies the conditions for future choices. At this point, practices such as scenario analysis, alternative comparison, trade-off identification, reversibility analysis, dependency assessment, communication, and coordination emerge. In planning, different execution options can be compared according to their impacts, costs, risks, and commitments.

Evolution produces the need to observe results, learn, and preserve knowledge. Scrum incorporates inspection and adaptation. Lean Startup structures build-measure-learn cycles. Retrospectives make it possible to examine the experience of one cycle and adjust the next. ADRs preserve knowledge about decisions that continue to influence architectural evolution. Other methods and practices use different mechanisms to address the same fundamental need.

These examples do not mean that each framework belongs to only one foundation. That would be an excessively rigid interpretation. A framework may address several needs at the same time because a practice often operates across more than one dimension of a decision.

Scrum, for example, does not address only evolution. Its objectives guide the work, its inspection mechanisms produce information, and its adaptation allows decisions to be revised as new evidence emerges. Lean Startup does not address only uncertainty. Its cycles also connect objectives, evidence, outcomes, and evolution. arc42

does not address only documentation. Its structure relates context, objectives, constraints, quality, decisions, risks, and architectural knowledge.

The same applies to architecture practices. An architectural decision needs to consider the domain in which the system exists, current conditions, quality and business objectives, technical uncertainties, available alternatives, and the consequences that the architecture will carry over time.

Therefore, the framework does not need to be the starting point. It can be a response to a need that has already been identified.

This inversion is important. Instead of asking first, "which framework should we use?", we can begin by asking: "what decision are we trying to conduct?", "which characteristics of this decision need to be addressed?", "what needs arise from these characteristics?" and, only then, "which practices or frameworks can help us?".

The six foundations proposed in this guide therefore do not seek to replace DDD, TOGAF, arc42, Scrum, PMBOK, Lean Startup, Design Thinking, ADR, or other methods. They offer a way of seeing what these approaches are trying to address and recognizing that different disciplines can develop distinct responses to structurally related needs.

This perspective will be important in the following chapters. After understanding the general characteristics of a decision, we need to enter the concrete situation in which it takes place. Before defining which problem to solve or which solution to apply, we need to understand the domain, context, boundaries, and existing conditions.

It is from there that Decision Architecture ceases to be merely a conceptual structure and begins to guide the conduct of a real decision.

2. Understanding the Domain of a Decision

Imagine a biologist studying a species. To understand its characteristics and behavior, they do not observe only the individual. They also need to consider the ecosystem in which it lives, the organisms with which it interacts, the resources on which it depends, and the conditions that influence its existence.

The same principle applies to a decision. A decision may have a specific object, but that object is embedded in a broader reality. People, organizations, processes, systems, resources, rules, and other elements establish relationships with what is being decided and may influence its evolution.

The domain corresponds to the portion of this reality that is relevant to the decision. It does not represent everything that exists, but it is also not necessarily limited to the most obvious object of the decision.

This perspective helps distinguish domain from context. The domain provides a broader view of the reality related to the decision. Context represents the specific conditions under which it is considered. An organization, for example, may be part of the domain, while a budgetary condition, a regulatory change, or a market opportunity may characterize the context in which the decision occurs. The distinction lies mainly in the level of observation, not in the nature of the elements.

This difference between knowing a practice and knowing the reality in which it will be applied explains a recurring problem in professional practice. A professional may master a methodology, follow its rules, and correctly execute its practices. Even so, they may make inadequate decisions by applying them without considering the characteristics of the domain in which they are operating. This is the behavior of a professional who works "by the book": they transform general guidance into a prescription for a particular reality, as if the methodology also contained the knowledge necessary about the domain.

A project management methodology may guide the planning, monitoring, and control of a project, but it does not, by itself, know the organization's culture, processes, capabilities, or internal relationships. A solution architecture practice may guide the construction of a solution, but it does not automatically know the organization's business model, processes, rules, capabilities, or technological constraints. An entrepreneurship technique may guide the investigation of an opportunity, but it does not know the market, customers, competitors, or the specific conditions under which the business will have to exist.

Methods and practices provide general references for action. The domain provides the characteristics of the reality that need to be considered when applying those references.

Understanding the domain does not mean reproducing the entire reality. Just as the biologist does not need to describe the entire ecosystem to study a species, decision architecture needs to represent only what is relevant to the analysis. This work begins by mapping the ecosystem, proceeds to defining its boundaries, and advances to identifying the elements, their characteristics, and their behaviors.

2.1. Mapping the ecosystem

A decision does not exist in isolation within an organization. It is embedded in a broader reality formed by the organization itself, the industry in which it operates, the market in which it competes, and the economic, social, technological, and regulatory conditions that influence this environment.

This reality can be understood as an ecosystem: the set of elements and relationships that form the environment in which the decision exists. Not all of these elements will be part of the domain, but knowing them at a sufficient level prevents its boundaries from being defined in an artificially narrow way. The macroeconomic environment may also be part of this view. Inflation, interest rates, employment, income, credit availability, exchange-rate fluctuations, and other economic conditions can alter the behavior of an industry and, consequently, the conditions under which certain decisions are made. The same applies to technological, regulatory, or social changes.

The analogy with the biologist helps clarify this perspective. When studying a species, they consider the available resources, the other organisms, and the relationships they establish with one another. A change in the environment can modify the conditions for the species' existence, even if its biological characteristics remain the same. In decision architecture, the objective is similar: to recognize the broader reality in which the decision is embedded and identify the elements and relationships that may be relevant to what will be analyzed. It is not necessary to represent the entire ecosystem in detail.

Consider a retail company that decides to replace its sales system. The decision is related to the company's operations, but the relevant reality is broader. The company operates in a particular industry, competes for customers, depends on suppliers, uses payment methods, contracts logistics services, and is subject to the country's economic conditions. Consumer behavior, industry practices, labor availability, operating costs, credit conditions, and the level of competition can influence how the company sells its products and, consequently, the characteristics that a new system needs to support.

Within the organization there are also stores, salespeople, customers, products, inventory, payments, logistics, support, systems, and partners. These elements establish relationships with one another and form a reality much larger than the sales system in isolation.

Mapping the ecosystem broadens the field of observation before defining the domain boundaries. As new elements or relationships are identified, this understanding can be revised, also leading to a revision of the domain itself.

2.2. Establishing the boundaries of the domain

Recognizing the ecosystem does not mean analyzing all of its elements. The domain corresponds to the portion of this reality that needs to be considered to understand the decision. Establishing its boundaries means determining which parts of the ecosystem will be addressed directly in the analysis and which will remain outside it. This boundary is a conceptual construction: it defines the focus of the investigation, but does not eliminate the relationships that exist beyond the scope.

An element may remain outside the domain and still maintain relevant relationships with what is inside it. The boundary does not make the rest of the ecosystem irrelevant. In decision architecture, it needs to be broad enough to preserve relationships capable of changing the analysis, without incorporating the entire reality.

Consider the retail company again. If the decision involves replacing the sales system, the domain may include salespeople, customers, products, prices, orders, inventory, payments, and the sales system itself. It may also be necessary to consider who makes deliveries, who replenishes inventory, who provides customer support, or who checks suspicious transactions. These elements can directly influence sales operations, even if they are not initially perceived as part of the system.

Defining the boundaries may also reveal that we are dealing with more than one domain. Sales, inventory, logistics, payments, customer service, and fraud prevention may have their own responsibilities, structures, and ways of operating, even though they are related. Another decision then arises: whether to treat these domains jointly or decompose the analysis into related decisions.

This choice can alter the solution architecture. A capability such as fraud verification may be incorporated into a larger system or made available through its own service. Delivery may be part of an integrated solution or use specialized services from partners. Inventory control may reside in the same sales system or in another system related to it.

Therefore, the existing technology architecture does not, by itself, define the boundaries of the domain. A single system may contain capabilities belonging to different parts of reality, just as a single capability may be distributed across multiple systems or services.

Decomposition can also create new related decisions, each with its own boundaries. We therefore have a cycle in which the way a decision is structured influences how its parts will be analyzed.

Organizational culture can influence this choice. Authority structures, professional specializations, incentives, ways of working, and historical patterns of collaboration may favor a more integrated view or a greater separation between parts.

An excessively narrow boundary can hide relationships capable of changing the decision. An excessively broad boundary can make the representation difficult to use. The objective is to find a scope that preserves relevant relationships and allows for clear reasoning.

2.3. Identifying the elements of the domain

After establishing the boundaries, it is necessary to recognize which elements make up the domain. The objective is not to produce a complete inventory, but to identify the parts that have sufficient relevance to be treated separately in the decision. This identification depends on the level of abstraction. The same element

may be represented as a unit in one analysis and decomposed into smaller parts in another. The appropriate representation is the one that uses the level of detail necessary for the decision.

In the retail company, we can identify customers, salespeople, stores, products, orders, prices, inventory, payments, deliveries, support, fraud verification mechanisms, systems, and partners. The sales system can be treated as a single element when the decision is related to commercial operations as a whole. In a more specific architectural decision, it can be decomposed into applications, services, databases, and integrations.

The same applies to other elements. A store can be represented as a unit or decomposed into teams, equipment, processes, and spaces. A delivery service can be treated as a partner or analyzed in terms of its own capabilities and components.

Decomposition should not occur simply because it is possible to decompose something. It should occur when the current level of abstraction is insufficient for the decision.

When identifying the elements, their relationships also begin to become visible. A salesperson uses the sales system. The system queries product and price information. A sale generates an order. The order may trigger a delivery. Product availability depends on inventory. A transaction may go through payment and fraud verification mechanisms. These relationships may reveal that a particular element, initially considered peripheral, has significant involvement in the decision. In that case, the representation of the domain may need to be revised.

The result of this stage is a structured representation of the main elements of the domain and the level of abstraction appropriate for continuing the analysis.

2.4. Understanding characteristics and behaviors

Identifying the elements is still not enough. It is necessary to understand their main characteristics and how they act and interact.

Characteristics describe what the elements are. They may involve nature, purpose, function, capability, composition, structure, or other relevant properties. Behaviors describe how these elements act and interact. An element may respond to certain actions, perform a function, produce information, consume resources, or establish relationships with other elements.

In the retail company, the sales system can be characterized by its purpose, functions, and structure. The salesperson can be described by their activities and capabilities. The product has attributes that need to be presented and recorded. Inventory represents product availability. The delivery service has its own role in moving orders.

Behaviors appear in the interactions between these elements. The salesperson records a sale. The system queries the product and price. Availability is checked in inventory. Payment is forwarded for processing. A transaction may be submitted for fraud verification. Confirmation of the sale may initiate a delivery and update information used by other processes.

These interactions help explain the structural functioning of the domain. A system receives an input and produces a response. A participant performs an action that changes the state of another element. Information is produced in one part and used in another.

The analysis remains, at this point, at a structural level. Specific conditions, constraints, limitations, exceptions, and circumstances that may modify these behaviors belong to a later and more contextualized analysis. The objective is to build a basic understanding of the domain's structure: which elements exist, what their main characteristics are, and how they relate in general terms. This structure provides the foundation for investigating the specific conditions under which the decision will be made.

2.5. Application in project management

A project is an intervention carried out within an organization, a market, and a set of relationships that already exist before it begins. Project management practices therefore need to construct some representation of the reality in which the work will be performed.

This representation appears from the earliest activities. During initiation, it is necessary to understand the organization involved, the people affected, the participating areas, suppliers, existing systems, and their rela-

tionships. During planning, this knowledge guides the definition of the work, teams, deliverables, and necessary interfaces.

This need is present in different approaches, even though they use different forms and vocabularies. In the PMBOK, for example, identifying stakeholders, understanding the organizational environment, defining scope, and organizing the work contribute to representing what is involved in the project. In iterative approaches such as Scrum, the domain appears in understanding the product, users, stakeholders, and the reality in which increments will be used. In Kanban, it appears in understanding the workflow, types of work, and relationships between its stages.

The project structure also depends on how its elements are identified. A team can be treated as a unit or analyzed according to its roles and capabilities. A system may appear as an external dependency or be considered part of the work. A supplier may be merely a procurement or a relevant part of the operation that needs to be integrated into the project.

The adopted boundaries influence the very definition of the project. A sales system implementation may consider only the application and its technical team or also involve stores, salespeople, commercial processes, inventory, payments, logistics, and technology suppliers. What changes is not only the size of the project, but the reality within which it will be conducted.

Therefore, stakeholder identification, scope definition, work decomposition, process mapping, and interface identification also function as mechanisms for making the domain observable.

The level of representation required depends on the decision. An organization can be treated as a single element at a given moment and later be decomposed into areas, teams, processes, or systems.

The domain, therefore, is not an artifact specific to a methodology. It is the reality that project management practices need to represent for the work to have meaning.

2.6. Application in solution architecture

In solution architecture, the domain occupies a particularly evident position because a solution is built to operate on an existing reality. Systems, integrations, services, and components need to correspond to elements and relationships present in the environment in which they will be used.

Architecture therefore begins before the choice of technologies. The architect needs to recognize which parts of reality are related to the solution, which concepts exist in this space, and how they relate to one another.

Architecture practices use different mechanisms to construct this representation. Domain models, capability maps, process models, context diagrams, system maps, data models, and architecture diagrams represent different aspects of reality. The C4 Model, for example, works with different levels of representation of a system's structure. arc42 organizes architectural information to make decisions, structures, and relationships explicit. Approaches such as DDD place domain concepts and their boundaries at the center of architecture construction.

These representations also help establish boundaries. A business capability may be supported by a single system or distributed across several. A system may concentrate different capabilities. A functionality that appears to belong to an application may depend on processes, people, or external systems.

Consider again the retail company that intends to replace its sales system. The domain may involve sales, products, prices, inventory, payments, stores, salespeople, and other systems. This reality, however, does not automatically determine a technology architecture. The sales system may remain a single application, be divided into services, or depend on solutions provided by different partners.

The technology architecture is a representation constructed on top of this reality. Therefore, the boundaries of technological components should not be confused with the boundaries of the domain. A service is not necessarily a domain, just as a system does not necessarily correspond to a single business capability.

This distinction is fundamental to practices such as system decomposition, integration definition, data organization, and responsibility distribution. Before deciding how a solution will be structured, it is necessary to understand what exists in the reality that it intends to represent and transform.

2.7. Application in user experience

In user experience, the domain is often reduced to the interaction between a person and an interface. This representation is useful for some decisions, but insufficient for many situations involving a usage experience.

An interaction with a product occurs within a service and an organization. Behind a screen there are processes, systems, people, policies, information, and operations that influence what the user can do. Therefore, UX practices need to consider not only the point of contact, but the reality that produces and sustains that experience.

This perspective appears in different practices. User research seeks to understand people and their relationships with the product or service. Journeys represent sequences of interactions throughout an experience. Service maps broaden this view to include activities that occur behind the scenes. Usability testing analyzes interaction within a given scope. Design Thinking and Double Diamond organize investigation, definition, exploration, and development activities that work with different representations of the problem and its environment.

Consider again the sales system of a retail company. Research may show that salespeople have difficulty locating products or completing a sale. An analysis limited to the interface could identify navigation problems or an excessive number of steps. But the experience also depends on inventory, prices, business rules, payment methods, integrated systems, and activities performed by the salesperson. The behavior observed in the interface may result from the interaction among several of these elements.

The boundaries of the investigation therefore depend on the decision. In an analysis of a screen, the user, interface, and information presented may be sufficient. In an analysis of the shopping experience, it may be necessary to include the store, salespeople, inventory, payment, delivery, customer service, and digital channels.

The same applies to the level of abstraction. A user may be treated as a participant or analyzed according to tasks, needs, and forms of interaction. A service may be considered a single experience or decomposed into stages, channels, and backstage activities.

Personas, journey maps, service maps, task flows, prototypes, and usability tests are different ways of representing parts of the domain relevant to an experience. Each makes certain elements visible and leaves others outside its scope.

The quality of an experience therefore does not depend only on the interface. The interface is one element of a larger reality, and UX work needs to determine which part of that reality is relevant to the experience being studied.

2.8. Application in entrepreneurship

In entrepreneurship, the domain is the reality in which a business opportunity can exist. It involves customers, competitors, suppliers, channels, partners, technologies, regulations, available capabilities, and market characteristics.

For this reason, entrepreneurship practices often begin by constructing representations of this reality. The Business Model Canvas organizes elements related to customers, value proposition, channels, activities, resources, partners, revenues, and costs. The Value Proposition Canvas focuses on the relationship between a proposition and particular customer segments. Customer Development structures activities for investigating and learning about customers and the market. Lean Startup uses hypotheses, experiments, and learning cycles to obtain evidence about particular parts of the business.

Each approach works with a different scope, and none automatically represents the entire domain of a venture.

Consider a company that intends to create a solution for small retailers. The domain may involve the behavior of these retailers, the systems they use, their suppliers, available payment methods, acquisition channels, competitors, cost structure, and industry conditions.

The boundaries can alter the very interpretation of the opportunity. A solution for large retail chains may belong to a different reality from that found in small establishments, even if the product appears similar. Likewise, an opportunity may exist in a particular industry or region and not present the same conditions elsewhere.

The elements can also be analyzed at different levels. “Customer” can be treated as a segment or decomposed into different profiles. “Competition” can represent a category of alternatives or specific companies. “Market”

can be treated as a broad industry or delimited by region, channel, or customer type.

The hypotheses used in entrepreneurship are statements about the domain. When someone states that a particular group has a need, is willing to pay for a solution, or that a particular channel allows acquisition at scale, they are formulating a hypothesis about an external reality.

Experiments, interviews, value proposition tests, and initial launches make it possible to observe parts of this reality. The knowledge produced is associated with the investigated scope: evidence obtained from a particular segment, region, or channel does not automatically represent all possible customers or markets.

Entrepreneurship frameworks and practices thus provide structures for observing different parts of a business's domain. The adopted representation influences what can be perceived as an opportunity, market, customer, competitor, or business model.

2.9. Application in professional career and personal life

Professional and personal decisions also take place within domains that can be represented and investigated. A career decision, for example, is related to a reality composed of skills, experience, professions, organizations, the labor market, education, location, resources, and professional relationships.

This reality can be represented in different ways depending on the decision. A skills inventory can represent the person based on knowledge and capabilities. An analysis of the labor market can represent opportunities, organizations, professions, and demand. A career map can organize possible paths. A scenario analysis can represent different environments in which a path could occur.

The same principle applies to personal decisions. Moving to another city may involve work, housing, transportation, relationships, services, and resources. Starting a business may involve the market, customers, suppliers, capital, capabilities, and professional relationships. Choosing an educational program may involve institutions, courses, costs, recognition, location, and professional alternatives.

The boundaries vary according to the decision. To choose between two professional specializations, it may be sufficient to represent skills, institutions, costs, and career opportunities. To decide whether to move to another city, it may be necessary to include the labor market, housing, commuting, family relationships, and available services.

The elements can also be decomposed as needed. "Career" can be treated as a trajectory or analyzed in terms of professions, organizations, skills, and opportunities. The "labor market" can be viewed as a general category or divided by industry, region, profession, and experience level.

This perspective does not turn life into an abstract model. It recognizes that every choice takes place within a reality composed of elements that already exist and relate to one another.

Career planning practices, skills analysis, market research, scenario construction, and decision matrices are ways of making parts of this reality more explicit. Each produces a representation appropriate to a particular type of investigation.

The concept of domain, therefore, does not belong exclusively to a technical or business discipline. Project management, solution architecture, user experience, entrepreneurship, career, and personal life use different ways of representing the reality in which their decisions take place. What changes across these areas are the elements considered relevant, the boundaries adopted, the levels of abstraction, and the practices used to construct the representation.

The foundation remains: before deciding about something, it is necessary to recognize the reality to which that decision belongs.

3. Discovering the Initial Context

Imagine a journalist who receives an assignment about a company that intends to expand its operations into a new region. The decision seems promising: there are signs of demand, the market shows growth potential, and the organization believes it is in a position to seize the opportunity. Before producing the story, however, the journalist needs to gain a better understanding of the situation. Where are the signs of demand coming from? Who are the potential customers? What characteristics make this region attractive? What conditions could facilitate the expansion, and which could make it more difficult? The opportunity that prompted the assignment is the starting point of the investigation, not a conclusion about the feasibility of the initiative.

Discovering the initial context of a decision requires a similar approach. A demand may arise from a problem to be solved, an opportunity to be explored, a change in the environment, or an idea that sparks interest. In all these cases, there is a situation that deserves to be understood before deciding how to act. The initial expression may be legitimate and relevant, but it may still represent only part of reality. Investigating means exploring that initial perception in greater depth, confronting it with other perspectives, and gathering information that makes it possible to better understand the conditions involved.

If, in the previous chapter, we put on the biologist's hat to observe an ecosystem, identify its elements, understand their relationships, and delimit the domain of a decision, it is now time to change hats and take on the role of an investigative reporter. The biologist seeks to understand the structure and behavior of the environment being studied; the reporter starts with an assignment and investigates the events, circumstances, and perspectives that help explain what is before them. One perspective does not replace the other. They serve different and complementary purposes: first, we seek to understand the domain to which the decision belongs; now, we want to discover what is happening within that domain and why a particular situation has come to demand our attention.

This distinction matters because decisions rarely begin with a complete understanding of what is at stake. In general, they begin with a demand, a perception of discomfort, an apparent opportunity, or a proposed change. Someone identifies something that deserves attention, interprets what they observe, and suggests a direction. In a company, the request may be to replace a system, automate a process, reorganize a team, or launch a product. In personal life, it may be to change jobs, begin a training program, or start a business. In all these cases, the demand expresses an initial perception of reality and provides a starting point for the investigation.

In solution architecture, this distinction becomes apparent when a business request is treated as though it already contains all the understanding needed to guide a solution. A manager may propose replacing a sales system because they believe the current technology limits productivity. The proposal may be correct, but it is still necessary to understand what is happening, how the difficulty manifests itself, and what outcomes the organization intends to achieve. The perceived slowness may be related to the application's response time, queries to external services, the availability of inventory information, a manual approval step, or the organization of the sales process itself. Investigation makes it possible to evaluate these possibilities without prematurely dismissing the proposed solution or assuming that it is the only appropriate answer.

Experience in project management reinforces the importance of this distinction. A project may begin with formally defined objectives, deadlines, and deliverables, but these elements do not guarantee that the need that motivated the work has been understood. Similarly, entrepreneurial experience shows that a business opportunity may seem promising while its core assumptions remain unvalidated. In both cases, the ability to execute well does not compensate for a misinterpretation of the situation. It is possible to deliver a project on time or launch a technically sound product without producing the outcome that justified the investment.

Discovering the initial context therefore means investigating before prescribing. This does not require knowing every detail of reality or indefinitely postponing a decision until all uncertainty disappears. It requires building an understanding proportionate to the importance of the choice, making explicit the available facts, the interpretations in circulation, the relevant conditions, and the gaps that still need to be considered. The goal is not to produce a definitive description of reality, but to establish a sufficiently reliable foundation for the decision to move forward with an awareness of what is known and what remains uncertain.

3.1. Investigating the Initial Situation

Every investigation begins with some manifestation of reality. It may be a deteriorating metric, a recurring complaint, a market change, a regulatory requirement, or the perception that a practice no longer produces the

expected results. This manifestation draws attention to a situation, but it does not automatically explain what is happening. The first task is to understand what prompted the decision and how that motivation was formed.

It is useful to distinguish the observed event from the narrative developed to explain it. Consider a retail company whose managers claim that the sales system is harming store productivity. This statement may bring together legitimate experiences of salespeople, customer complaints, and unsatisfactory operational results. Even so, it contains several propositions that need to be examined: productivity has declined, the system contributed to the decline, and replacing the system is an appropriate response. Each proposition depends on different evidence. The existence of complaints does not, by itself, demonstrate the extent of the operational impact; a decline in productivity does not necessarily identify its cause; and identifying a cause does not automatically determine which intervention will produce the best outcome.

This separation makes it possible to investigate the situation without disregarding the person who raised it. The initial demand is valuable because it reveals what someone perceives as important, the outcome they hope to achieve, and the interpretation guiding their proposal. The problem arises when that interpretation is no longer treated as a perspective to be examined and instead begins to predetermine the boundaries of the analysis. When this happens, the investigation tends to seek justifications for the suggested solution rather than produce enough knowledge to compare different possible responses.

The starting point should be framed in a way that keeps the question open. Instead of assuming that the company needs to replace its sales system, we can investigate why completing sales is taking longer than expected and which conditions contribute to that outcome. This framing does not rule out replacement. It merely avoids turning a specific alternative into the definition of the problem. It also allows the investigation to uncover differences across stores, times of day, transaction types, or user profiles that might disappear in an excessively generic description.

The initial situation must also be understood in relation to what was expected to happen. A result can be interpreted as unsatisfactory only when there is some basis for comparison, whether a target, previous behavior, an operational need, a customer expectation, or an externally established condition. An operation may seem slow in absolute terms and still meet the business's needs; another may exhibit apparently acceptable response times but fail at critical moments when transaction volume increases. The meaning of the result depends on the purpose of the activity, the expectations involved, and the conditions under which it occurs.

This reference point is not always documented. In organizations, expectations may have been established informally, passed between teams, or embedded in practices that no one questions anymore. In a startup, the reference point may be a hypothesis about the behavior of a customer segment. In a career decision, it may be a personal expectation regarding compensation, autonomy, or quality of life. Making these reference points explicit is part of the investigation because it makes it possible to understand not only what is happening, but also why the situation has come to be considered relevant.

Investigating the initial situation should not involve seeking a single explanation before understanding the variety of conditions involved. The same outcome may result from independent factors, combined causes, or relationships that vary over time. Slowness in a sales operation, for example, may result from an unresponsive application under certain conditions and a manual approval process under others. Treating these manifestations as a single problem may lead to an overly broad intervention or one directed at the wrong factor.

The quality of this stage depends on the ability to keep plausible explanations open while gathering evidence to evaluate them. This does not mean treating all hypotheses as equally likely. It means avoiding giving the first available explanation an authority that the evidence does not yet support. From this standpoint, the next step is to organize the investigation around questions that allow the situation to be explored from different angles.

3.2. Formulating the Questions

Journalism has developed techniques for organizing investigations without assuming that the first account of an event is sufficient. One of the best-known is the 5W1H framework, which organizes questions around who, what, when, where, why, and how. Although it is used in different professional contexts, its contribution here is not to provide a mechanical checklist to be completed, but to help reveal dimensions of a situation that may remain hidden when the investigation is limited to the initial description of the problem.

The question of who involves more than identifying the requester or the person formally responsible for the decision. It is necessary to understand who noticed the situation, who is affected by it, who performs the activities

involved, who possesses relevant information, and who has the authority to decide on or implement changes. These people may have different perspectives because they occupy different positions within the operation. A director may observe consolidated metrics; a store manager may notice variations between shifts; a salesperson may know specific difficulties encountered during customer service; a customer may experience consequences that do not appear in internal reports. None of these perspectives necessarily represents the complete situation.

The question of what seeks to define the event or condition that prompted the investigation. It is necessary to clarify what is happening, which outcome is considered unsatisfactory, what opportunity has been identified, and what change is intended. The formulation should be specific enough to allow verification, but not so narrow that it prematurely incorporates an explanation. "Sales are taking too long" is an initial description that still requires clarification: does it refer to the time needed to register an order, complete payment, locate a product, or finish the entire customer interaction? Each interpretation leads to a different investigation.

The temporal dimension, expressed by the question of when, helps reconstruct how the situation evolved. When was the behavior first noticed? Was there a recent change? Did the problem already exist on a smaller scale? Did it coincide with a process change, a technology deployment, a regulatory change, or an increase in demand? Chronology does not prove causality, but it makes it possible to identify events that warrant investigation. If delays began after the introduction of a new validation step, for example, that change becomes a relevant hypothesis, although it is still necessary to verify how it relates to the observed outcome.

The question of where seeks to locate the situation. A problem may occur in a specific store, in certain regions, in a digital channel, at a particular stage of the process, or in the interaction between systems. The location may also be logical, organizational, or technological. A difficulty perceived in the sales interface may originate in a service shared by several applications; a failure apparently confined to one team may result from a policy applied across the entire organization. Defining where the phenomenon occurs helps distinguish a localized condition from a broader characteristic of the domain.

The question of why requires particular care because it often invites people to present a cause before it has been investigated. At this stage, it is more productive to ask why the situation is considered a problem, which outcomes are being impaired, and what evidence supports that interpretation. The causal explanation should remain a question to be verified. When a manager claims that system slowness reduces sales, it is necessary to understand whether there is an observable relationship between service time and sales performance, whether other factors are involved, and whether the difficulty affects all transactions or only some of them.

Finally, the question of how seeks to understand the mechanism through which the situation manifests itself. How is the work performed? Which steps are followed? What information is produced and consumed? Where do waiting, rework, manual decisions, or external dependencies occur? How do people work around existing limitations? This dimension brings the investigation closer to the concrete functioning of reality and helps transform a general perception into something that can be observed, described, and analyzed.

The six questions complement one another, but they do not need to be answered in a rigid order. An interview may reveal a condition that requires observing the process; observation may contradict an initial explanation; a document may indicate that the problem began before the date recalled by participants. The investigation progresses through successive approximations, revising the questions as new information emerges.

In addition to these dimensions, three questions run throughout the entire investigation: How do we know this? What evidence could contradict this explanation? What do we still not know? They help prevent the 5W1H framework from becoming a form filled out with assumed answers. The technique's value lies in directing attention to different aspects of the situation, not in automatically validating the recorded answers. The questions constitute the investigation's agenda; interviews, observations, documents, and operational data are instruments for producing and verifying the answers.

3.3. Identifying Sources and Evidence

Once the questions have been formulated, it is necessary to identify where the information capable of clarifying them will come from. Not every person involved knows every dimension of the situation, just as not every document or metric was produced to answer the question being investigated. The choice of sources should consider the type of information sought, the source's proximity to the phenomenon, the limitations of its knowledge, and the possibility of verifying what it reports.

People are important sources because they know practices, exceptions, decisions, and experiences that frequently do not appear in formal records. However, individual accounts are influenced by the speaker's position, objectives, memory, and the categories they use to interpret events. A manager may describe the process as it was defined; an operator may explain how it is actually performed; a technical team may detail the behavior of the systems involved. Differences between these accounts should not automatically be treated as errors or bad faith. They may reveal that people observe different parts of the same reality, or that actual practice has diverged from the established procedure.

Documents and records provide another type of evidence. Contracts, policies, meeting minutes, reports, specifications, incident histories, service records, and operational data can help reconstruct decisions and events. Their existence, however, does not guarantee that they fully represent the situation. A procedure describes how work should occur, not necessarily how it occurs in practice. A report may consolidate information in a way that conceals important variations. A metric may have been defined to measure financial efficiency and be insufficient to explain the user experience. It is necessary to understand the purpose, period, coverage, and criteria used to produce each record.

Direct observation is particularly useful when there is a gap between the described process and the work actually performed. Following a sale in a store, for example, may reveal that the salesperson switches between applications, consults colleagues to confirm information, or repeats steps because certain data are unavailable when needed. These activities may seem like exceptions to someone familiar only with the formal procedure, yet they may constitute a recurring part of the operation. Observing the work also helps identify behaviors that participants themselves have stopped mentioning because they consider them normal.

Quantitative data complement these sources by making it possible to verify frequency, distribution, variation, and trends over time. Instead of relying solely on the perception that the system is slow, the team can examine response times, periods of peak demand, total service duration, failure rates, and differences between stores. However, the numbers must be interpreted in relation to what they actually measure. An application's response time is not equivalent to the time required to complete a sale, just as an average can conceal extreme situations that disproportionately affect certain users.

The strength of the investigation lies in combining independent sources, a practice known as triangulation. If salespeople report frequent waits, records show an increase in transaction completion times, and observation identifies a step where customer interactions come to a halt, there is a more consistent basis for understanding the phenomenon. Even so, this evidence does not automatically demonstrate that a single cause explains the problem. It corroborates aspects of the situation and guides the next stage of the investigation. If accounts and records diverge, the discrepancy is also informative: it may indicate differences in time periods, distinct definitions, measurement limitations, or experiences that aggregated data do not capture.

The quality of evidence therefore depends on the relationship between the claim one wishes to support and what the source actually allows one to conclude. An interview may demonstrate that a person experienced a particular difficulty, but it cannot establish the frequency of that event across the entire organization. A metric may reveal a trend, but it cannot explain its causes on its own. A document may confirm that a decision was formalized without demonstrating that its assumptions were correct. Recognizing these limitations does not diminish the value of the sources; it makes it possible to use them more precisely.

The investigation must also be proportionate. Not every decision requires extensive research, a statistical study, or interviews with every participant. The effort should take into account the impact of the decision, the cost of an incorrect interpretation, the availability of information, and the existing degree of uncertainty. A reversible, low-impact change may be undertaken with limited evidence and close monitoring. A decision involving substantial investment, regulatory risks, or changes that are difficult to reverse requires greater care in verifying assumptions. What matters is that the depth of the investigation be a conscious choice, rather than an accidental consequence of haste or information availability.

3.4. Separating Facts, Interpretations, and Hypotheses

As information is gathered, the knowledge produced needs to be organized. Without this separation, observed facts, opinions, explanations, and predictions tend to merge into a single narrative. The result may appear coherent while concealing reasoning gaps that compromise the decision. Distinguishing these categories does not mean eliminating interpretation, which is impossible in any analysis, but making explicit the role each statement plays.

For the purposes of the investigation, a fact is a statement supported by identifiable evidence within known limits. This definition requires care because a record may also contain errors, and an observation may be subject to particular conditions. Even so, it is possible to state, for example, that the available records show an increase in the average time required to complete sales during a given period, provided that the metric, source, and analyzed interval are clear. This statement differs from saying that the system became slower, because completion time may include activities that do not directly depend on the application.

An interpretation assigns meaning to facts. Faced with an increase in completion time, someone may conclude that the sales process has become less efficient or that the salespeople's experience has worsened. These interpretations may be reasonable, but they depend on additional criteria and assumptions. The increase may reflect a new verification requirement, a change in the transaction mix, or a change in how time is recorded. To evaluate the interpretation, it is necessary to examine these possibilities and establish which outcome is considered relevant.

A hypothesis is an explanation or proposition that still needs to be tested. The idea that an external integration causes some of the waiting, for example, may be formulated as a hypothesis if there are indications that make it plausible but insufficient evidence to confirm it. The hypothesis guides the next stage of the investigation: identifying the necessary records, observing the integration's behavior, comparing transactions, and verifying whether delays occur under the predicted conditions. A useful hypothesis is not merely an imaginable possibility; it must make it possible to identify what evidence would strengthen or weaken it.

This distinction prevents the authority of the person making a statement from being confused with the strength of the evidence supporting it. A hypothesis formulated by an experienced executive remains a hypothesis. A perception shared by many people deserves attention, but may still require verification of its scope and cause. Similarly, an interpretation presented by a technical team does not become a fact merely because it uses precise terminology. Specialized knowledge is essential for guiding the investigation, but it does not replace the need to relate conclusions to the available evidence.

It is also important to recognize that a lack of evidence is not equivalent to evidence of absence. If records show no failures, this may indicate that none occurred, but it may also reveal that the recording mechanism does not capture certain situations. If no customers complain about a step, one cannot immediately conclude that it causes no difficulty; perhaps customers abandon the process before submitting a complaint. The meaning of silence depends on the ability of the sources to observe and record the phenomenon being investigated.

In real-world decisions, the separation between facts, interpretations, and hypotheses does not produce a perfectly stable boundary. An interpretation can be tested and gain support; a hypothesis can be disproved; a fact initially accepted may need to be reconsidered when a measurement problem is discovered. The objective is to maintain traceability of the reasoning: to know what was observed, how it was interpreted, which hypotheses are being considered, and why a particular conclusion seems more plausible than the alternatives.

This practice also helps reduce the influence of biases. When an organization has already invested in a technology, those responsible may interpret evidence in ways that preserve the previous decision. When a department requests a new solution, it may prioritize information that reinforces the need for change. In a startup, the founders' conviction about a market need may lead them to select interviews that confirm the value proposition. Explicitly separating statements does not eliminate these biases, but it makes them easier to question and helps identify what would need to be true for the adopted interpretation to be correct.

A consistent investigation should seek both evidence supporting the initial explanation and information capable of contradicting it. This principle does not require indiscriminate skepticism, but a willingness to revise the narrative when the facts do not fit it. The result is a representation that depends less on the first impression and is more useful for understanding the concrete conditions surrounding the decision.

3.5. Understanding the Conditions of the Decision

Understanding the initial context means more than explaining what happened. It means identifying the conditions that influence the decision that needs to be made. These conditions may include objectives, constraints, available resources, deadlines, dependencies, obligations, interests, risks, and uncertainties. Some arise from relatively stable characteristics of the domain; others have emerged from recent events; still others result from the way the organization itself has structured the decision.

Objectives help clarify which outcome matters. A company may want to reduce service time, increase sales capacity, reduce errors, or improve customer satisfaction. These objectives may be related, but they are not equivalent and may conflict. An intervention that reduces the time required for each transaction may increase the risk of errors; a solution that improves the customer experience may require additional investment or operational changes. If the investigation focuses only on the most visible symptom, it risks overlooking the outcome that actually justifies the change.

Constraints define the boundaries within which alternatives must be evaluated. They may be financial, regulatory, contractual, technological, operational, or related to team capacity. It is important to distinguish actual constraints from preferences or historical decisions that have come to be treated as immutable. An approved budget may establish a real limit for the initiative, whereas a preference for a specific vendor may be negotiable. Integration with a legacy system may represent a necessary dependency, but transition alternatives may also exist. The investigation should recognize existing conditions without automatically turning them into definitive prohibitions.

Available resources influence the feasibility of possible responses. Time, budget, skills, operational capacity, data, and organizational support determine what can be done and at what pace. In projects, a technically appropriate alternative may be infeasible if it requires skills the organization does not possess and cannot acquire within the required timeframe. In a startup, a commercial hypothesis may be promising but require an acquisition channel whose cost makes the business model unviable. The analysis must consider not only what would be desirable under ideal conditions, but also what can be achieved within the available reality.

Dependencies reveal that a decision is rarely under the control of a single participant. A change to the sales system may depend on vendors, payment integrations, security policies, data availability, inventory processes, and the stores' ability to adopt new practices. These relationships may introduce deadlines, costs, and risks that do not appear in the initial description of the demand. Understanding dependencies also helps identify which information must be obtained from other departments and which conditions may limit the implementation of an alternative.

The interests and perspectives of those involved are also part of the context. One department may prioritize speed, another may value control, while a third is responsible for compliance or operational stability. These differences should not be reduced to personal conflicts. They often reflect distinct responsibilities, incentives, and success criteria. A decision that seems simple from one department's perspective may create costs or risks for another. Making these perspectives visible makes it possible to understand the tensions the decision will need to manage without assuming that all participants evaluate the situation using the same criteria.

Finally, uncertainty must be recognized as a condition inherent to the decision itself. Some information can be obtained through further investigation, such as the response time of an integration. Other uncertainties depend on future events, such as customer reactions to a process change or the evolution of demand. This distinction matters because not all uncertainty can be eliminated before acting. Some can be reduced through analysis; others require experiments, scenarios, monitoring, or reversible decisions that allow learning along the way.

The identified conditions do not need to be treated as an independent list of factors. They form relationships that alter the decision space. A regulatory deadline may limit the time available for investigation; a budget constraint may make an alternative infeasible; a lack of data may prevent comparisons of outcomes; a poorly defined objective may generate disagreements about what success means. Understanding the context means recognizing these relationships and their implications, preparing a foundation for representing the situation coherently.

3.6. Building a Representation of the Initial Context

The investigation produces different types of information: accounts, records, observations, metrics, interpretations, hypotheses, and conditions. For this information to support a decision, it must be organized into a representation that preserves what is relevant without indiscriminately reproducing everything collected. The initial context is not a transcript of interviews or a document repository. It is a well-supported synthesis of the situation that makes it possible to understand why the decision is being considered, which conditions influence it, and which questions remain open.

This representation should begin with the observed situation and the motivation for the decision. It is necessary to make explicit what drew attention, which outcome is considered unsatisfactory or desirable, and what change

is being considered. The description should distinguish the perceived need from the proposed solution. If the original request is to replace the sales system, the representation may record this proposal as an alternative under evaluation while separately describing the observed difficulties in completing transactions. This approach preserves the possibility of investigating whether replacement is actually necessary.

Next, the representation should link the main evidence to the statements it supports. It is not necessary to include all the data collected, but it is important to identify the relevant sources, the periods considered, and the limitations affecting interpretation. If the analysis is based on service-time records, it should be clear what the metric measures and which steps are included. If a conclusion depends on interviews with salespeople from a few stores, that scope must be known. The representation becomes valuable when it allows another person to understand not only the conclusion, but also the reasoning that led to it.

The conditions influencing the decision should be presented in relation to one another, rather than merely as isolated items. A deadline may restrict the set of alternatives; a technological dependency may determine the implementation sequence; a difference between stores may indicate that a uniform solution will not produce the same results everywhere. The objective is to make visible the factors that alter the interpretation of the situation and the feasibility of possible responses.

The representation must also preserve unverified hypotheses and knowledge gaps. This prevents an unanswered question from disappearing in the final write-up and a provisional explanation from being interpreted as an established conclusion. If it is still unclear whether the delay results from the system, the process, or a combination of both, that uncertainty should remain explicit. It may guide further interviews, data analysis, operational observation, or a controlled experiment.

The appropriate level of detail depends on the decision. A low-impact change may require only a brief synthesis of the situation, the main evidence, and the known risks. A large-scale organizational transformation or architectural decision may require a broader representation, including different perspectives, dependencies, and scenarios. Rigor lies not in the length of the document, but in the alignment between what was investigated, what is being asserted, and the potential consequences of being wrong.

This representation should not be treated as a definitive artifact. The context may change during the investigation, and new evidence may alter the initial understanding. Data analysis may reveal that the difficulty is concentrated in a specific group of transactions; an interview may indicate that an operational change preceded the problem; a test may show that the most likely technical hypothesis does not explain the observed behavior. In such cases, revising the representation is part of the work, not a sign that the investigation has failed.

It is also important to preserve the distinction between the initial situation and the desired situation. The former describes the relevant conditions that exist when the decision is being considered. The latter expresses what is intended to be achieved. Confusing them may cause objectives to be presented as facts or current limitations to be treated as permanent characteristics. A clear representation keeps these aspects connected but distinct: it describes the starting point, makes the expected outcomes explicit, and allows the distance between the two to be evaluated later.

The initial context thus serves as a reference for reasoning. It provides a common basis for discussing the problem, comparing interpretations, evaluating alternatives, and recording why a particular decision was made. It does not eliminate disagreements or automatically resolve the identified questions. Its contribution is to make the discussion more precise by reducing dependence on implicit assumptions and allowing people to discuss the same situation, even if they continue to disagree about its causes or the best response.

3.7. Application in Project Management

In project management, understanding the initial context influences the very justification for starting a project. A project exists not merely because a deliverable has been requested or because resources are available to execute it. It is an intervention carried out to produce a change in a particular reality. Before organizing activities, estimating costs, and defining schedules, it is necessary to understand which situation motivated the initiative, which outcomes must be achieved, and which conditions may influence its execution.

This concern appears in different ways in management practices. Developing a business case seeks to establish the justification for an investment and compare expected benefits with costs, risks, and alternatives. The project charter formalizes authorization and provides an initial reference for the project's objectives and boundaries. Identifying stakeholders helps establish who influences or is affected by the initiative, while analyzing the

organizational environment reveals conditions that may facilitate or hinder the work. These instruments serve different purposes, but they all depend on a reasonably reliable understanding of the situation to be transformed.

Investigating the context should not be confused with defining all deliverables in advance. In a system replacement initiative, for example, the initial scope may have been formulated based on the perception that the current application harms productivity. If this interpretation has not yet been verified, turning replacement into an irrevocable deliverable may compromise the project before detailed planning even begins. The investigation may reveal that part of the problem is related to manual procedures, insufficient training, complex business rules, or unstable integrations. The project decision may then change, whether through a scope adjustment, a combination of interventions, or the selection of a different alternative.

This does not mean that the project must remain undefined until all of reality is known. Initiation often occurs under uncertainty, and the organization may need to authorize resources to investigate the problem in greater depth. In such cases, it is important to distinguish the commitment to address a need from the commitment to implement a specific solution. A project may begin with the objective of reducing service time and include an initial diagnostic phase before defining the intervention. This approach makes explicit that part of the work involves producing the knowledge required for a subsequent decision.

Stakeholder identification also gains depth when conducted as an investigation rather than merely as the completion of a register. In a retail company, senior management may request the initiative, the technology department may be responsible for the solution, store managers may be accountable for operational results, and salespeople may directly experience the changes. Customers, payment providers, and support teams may also be affected. Listening only to sponsors and technical leads would produce an incomplete view of the situation, especially if the most significant difficulties emerge in the day-to-day work of the stores.

The chronology of events may be equally important. If problems began after a change in commercial policy, a partial deployment, or a shift in demand, these events must be considered before attributing the cause to the system. Incident records, operational metrics, interviews, and observation of the work can help reconstruct the sequence and verify whether the initial interpretation is consistent with what happened. The analysis does not need to prove a single cause immediately, but it should prevent the project proposal from becoming an explanation merely because it was presented first.

The context representation also contributes to risk management. Risks do not arise only during execution; some are present from the moment the initiative is formulated. A critical dependency may be outside the team's control, benefits may depend on behavioral changes, available data may be insufficient to measure outcomes, or the departments involved may disagree about what success means. Identifying these conditions from the outset makes it possible to establish proportionate responses, such as further investigation, technical validation, user testing, or scope reviews.

As the project progresses, the initial representation serves as a reference for evaluating changes and outcomes. If objectives are altered, it becomes possible to understand whether the change results from a legitimate reassessment of the context, new evidence, or a decision that has moved away from the original justification. If the expected benefits are not achieved, the organization can examine whether execution failed, whether the assumptions were incorrect, or whether external conditions changed. Without this reference, the project tends to be evaluated primarily on scope, schedule, and budget delivery, even when the initiative existed to produce a different outcome.

In project management, therefore, discovering the initial context is not a peripheral activity that precedes the work considered important. It is part of defining the work itself. The quality of planning depends on the quality of the understanding that supports it, and the ability to execute an initiative does not replace the need to verify whether it addresses the situation that actually needs to be transformed.

3.8. Application in Solution Architecture

In solution architecture, the initial context determines how the problem will be interpreted before decisions are made about components, integrations, platforms, or technologies. A solution is not appropriate merely because it meets formalized requirements or uses an architecture considered technically sound. It must address a real need, operate within existing conditions, and produce outcomes consistent with business objectives. To achieve this, the architect must investigate not only the system being changed, but also the events and conditions that led to the request.

Consider again the retail company that intends to replace its sales system. The request may have arisen because salespeople experience delays during customer interactions. An approach focused immediately on technology might begin by comparing platforms, evaluating architectural patterns, and estimating migration effort. These activities may be necessary later, but they do not yet answer the central question: which part of the operation is causing the delay, and what type of intervention could change the outcome? Without this investigation, the architecture risks reproducing the problem in a new technology or introducing complexity to solve a difficulty that lies outside the system.

The investigation must combine business, operational, and technological perspectives. Users can describe what happens during a transaction; process owners can explain rules and steps; operational records can reveal waiting times and failures; technical analysis can show how applications and their dependencies behave. Context diagrams, process models, system maps, incident records, and performance metrics can support this analysis, provided they are used to answer specific questions. A diagram shows known relationships, but does not by itself demonstrate how those relationships behave under real-world conditions.

Architecture practices such as those organized by arc42 help document a system's context and scope, its interfaces, constraints, and relevant decisions. Enterprise architecture approaches such as TOGAF provide ways to relate business objectives, the current state, the desired state, and the changes required. Domain-Driven Design contributes to investigating concepts, language, and domain boundaries. These frameworks can support an understanding of the situation, but they do not replace the investigation of the concrete circumstances. No architecture model, however comprehensive it may appear, automatically contains knowledge about the operational conditions under which the solution will be used.

The difference between expected behavior and observed behavior deserves particular attention. Documentation may indicate that an application queries inventory and routes payments to an external service. This describes an interaction structure, but does not clarify whether responses arrive within the required time, whether recurring failures occur, whether the information is reliable, or whether users must perform parallel activities to complete a sale. Contextual investigation complements the structural view of the domain with evidence about how it operates under specific circumstances.

It is also necessary to examine the conditions that limit architectural alternatives. A solution may need to coexist with legacy systems, comply with security and audit requirements, preserve vendor contracts, operate in stores with variable connectivity, or be deployed without interrupting commercial operations. Some conditions are mandatory constraints; others are organizational preferences that can be reconsidered. The architect must distinguish between them to avoid confusing the existing architecture with the only possible architecture. At the same time, actual constraints cannot be treated as secondary details because they influence costs, risks, timelines, and the feasibility of change.

The initial context also clarifies the relationship between cause and intervention. If the data show that the delay is concentrated in an external integration, a broad change to the sales application may not be the most efficient response. If the main difficulty lies in the repetition of manual steps, the solution may require changes to the process and operational responsibilities in addition to technological changes. If the system cannot support the expected volume, architectural evolution may be necessary. The investigation does not predetermine which alternative will be selected; it establishes a basis for evaluating which alternatives address the observed conditions.

There is also an important temporal dimension. The architecture must consider not only the current situation, but also changes already underway and those that may alter the relevance of the decision. A system replacement may be planned while the company is opening new stores, modifying its payment methods, or adopting a new operating model. These initiatives may create dependencies or make some alternatives more attractive than others. The initial context is not an isolated snapshot of the technological environment; it is a representation of the relevant conditions at the time the decision is being considered.

Once made, the architectural decision must preserve the relationship between context, reasoning, and consequences. Architecture Decision Records, known as ADRs, can document the situation that motivated the choice, the alternatives considered, the decisive factors, and the expected consequences. Their value depends on the quality of the investigation that precedes them. A record that presents only a decision and its technical justification may be insufficient if it does not explain which need the decision seeks to address and which assumptions support that need.

In solution architecture, discovering the initial context means avoiding a situation in which technology becomes

the first and only language used to describe a problem. Technical analysis is indispensable, but it must be connected to how the organization operates, people's experiences, and the outcomes that justify the change. Architecture becomes more consistent when its decisions respond to an investigated reality rather than merely to a demand prematurely translated into requirements.

3.9. Application in User Experience

In user experience, the initial context is essential because the difficulties perceived by people do not always correspond to the apparent causes of the problem. A user may say that an interface is confusing, that a service is slow, or that a feature should be added. These statements are relevant sources of information, but they need to be understood in relation to the tasks the person is trying to perform, the conditions under which the product is used, and the steps that precede or follow the interaction.

A UX investigation therefore begins by understanding the experience being affected. If salespeople report difficulty completing a sale, it is necessary to observe how the activity takes place, which information they need to consult, which decisions they must make, and which interruptions occur during the interaction. The difficulty may be related to the interface's organization, but it may also result from inconsistent data, unclear business rules, approval processes, or limitations in integrated systems. The interface is where the difficulty manifests itself, not necessarily where it originates.

Interviews and surveys help clarify participants' expectations, needs, and interpretations. Task observation reveals behaviors that may not appear in accounts, especially when informal practices have become habitual. Usability testing makes it possible to verify how people interact with a solution under defined conditions, while usage data analysis may reveal patterns of abandonment, repetition, or difficulty. Each technique answers different questions, and the choice should be based on what the investigation needs to clarify.

Design Thinking and the Double Diamond, for example, provide structures for exploring problems, developing an understanding of needs, formulating definitions, and investigating alternatives. These approaches reinforce the importance of not converging on a solution before understanding the situation. However, conducting interviews or workshops does not, by itself, guarantee that the context has been understood. Research quality depends on participant selection, question formulation, observation conditions, and evidence interpretation. A study that asks only whether users would like a particular feature may confirm a stated preference without revealing whether the feature would solve a real difficulty.

The distinction between reported experience and behavior is particularly relevant. A person may say they want fewer steps in a form, but observation may show that certain steps prevent significant errors. Another person may claim to use a feature frequently, even though records indicate sporadic use. These differences do not necessarily mean that the person is mistaken. They may reveal that the person describes their experience based on memory, an exceptional situation, or an expectation about how they should behave. The investigation must understand these differences before translating them into design requirements.

It is also necessary to examine the service beyond the interface. A user journey may include digital and in-person steps, interactions with employees, waits for approvals, communication across different channels, and behind-the-scenes activities. A problem perceived during a purchase may depend on inventory, payment, logistics, or customer service. Journey maps and service blueprints help represent these relationships, provided they are not treated as complete descriptions without verification. The journey documented by the organization may represent the intended flow, while the actual experience includes deviations, exceptions, and strategies users have developed to complete their tasks.

Understanding the initial context also helps determine which users and situations need to be investigated. An experience may vary according to familiarity with the product, accessibility needs, physical environment, connection quality, or task urgency. Testing an interface only under ideal conditions may conceal difficulties that emerge in a busy store, on a less powerful device, or during an operational interruption. The relevance of these conditions depends on the decision, but they must be considered before generalizing results obtained from a limited sample.

When the investigation reveals that the difficulty differs from what was initially imagined, the problem definition needs to be revised. Instead of redesigning a screen to reduce the number of steps, it may be necessary to improve information availability or change the process sequence. Instead of adding a feature, it may be more important to make a rule understandable or ensure that an operation can be completed reliably. Research should not

serve merely to validate a solution that has already been chosen, but to understand what people are trying to accomplish and which conditions facilitate or prevent that accomplishment.

The context must also be related to organizational objectives. Improving the experience does not necessarily mean optimizing a single interaction metric. Reducing form completion time may be undesirable if it increases errors; a feature appreciated by one group may create difficulties for another; a simplification may shift work to teams operating behind the scenes. The investigation should consider the relevant effects on the experience and the relationships between users, processes, and operations.

In user experience, discovering the initial context means understanding the situation in which the interaction occurs before concluding that the interface, user behavior, or absence of a feature explains the problem. This approach broadens the ability to formulate relevant problems and reduces the likelihood of optimizing one touchpoint while the difficulty persists across the experience as a whole.

3.10. Application in Entrepreneurship

In entrepreneurship, the initial context is often built around a perceived opportunity. A founder identifies a difficulty they consider important, notices a gap in the market, or imagines a better way to serve a particular audience. This perception may be the beginning of a business, but it does not, by itself, constitute evidence that a viable market exists. It is necessary to investigate who faces the difficulty, under what circumstances it occurs, which alternatives are already being used, and which conditions influence the willingness to adopt and pay for a solution.

A startup's initial narrative often brings several hypotheses together in a single statement. Saying that small retailers need an integrated platform, for example, assumes that this group faces a relevant problem, that current solutions do not adequately solve it, that a new platform would be preferable, and that customers would be willing to change their processes to use it. It also assumes that the business could reach these customers, deliver the value proposition, and operate with a sustainable cost structure. Each of these assumptions describes a different dimension of reality and needs to be examined appropriately.

Customer Development, Lean Startup, and Design Thinking practices offer instruments for investigating these questions, although they emphasize different aspects. Exploratory interviews help clarify problems and behaviors; observations may reveal how customers perform their activities; experiments allow specific hypotheses to be tested; prototypes and minimum viable products can produce evidence about usage, perceived value, and willingness to adopt. The build-measure-learn cycle is useful when what is built is linked to an explicit hypothesis and when the measurements make it possible to evaluate what was intended to be learned.

The investigation loses value when questions are formulated to obtain approval for the idea. Asking a potential customer whether they would like a system that saves time tends to produce favorable responses because the proposal is presented in desirable and abstract terms. Asking how they currently perform the activity, which difficulties they encounter, what happens when those difficulties arise, and which alternatives they have already tried tends to reveal more concrete information. Evidence about past behavior and actual decisions generally provides a more consistent basis than generic statements of future interest, although it too must be interpreted within its context.

The same care applies to market analysis. A broad segment may seem attractive because of the number of companies or the volume of spending, but these figures do not demonstrate that the audience is reachable, that its members share the same need, or that a specific solution can capture value. Small retailers, for example, may differ significantly according to their type of operation, region, sales volume, systems already in use, and capacity to invest in technology. The investigation must define the relevant segment and understand the conditions under which the opportunity may exist, rather than extrapolating conclusions obtained from a few participants to an entire market.

Existing alternatives deserve attention because the problem does not occur in a vacuum. Customers may use spreadsheets, manual procedures, competing products, outsourced services, or improvised combinations of tools. These alternatives may be inefficient in certain respects and still adequately meet customers' needs. Understanding why they are used, what costs they impose, and what makes them difficult to replace helps evaluate the value proposition realistically. A technically superior solution may fail if it requires a behavioral change that customers do not consider worthwhile.

The investigation must also examine the business model. Even if a genuine need exists and customers recognize the value of the proposition, the startup must reach its audience through viable channels, deliver the solution with quality, and sustain its costs. Customer acquisition cost, sales cycle, implementation requirements, support needs, retention, and willingness to pay are conditions that may alter the viability of the opportunity. A hypothesis validated in terms of customer interest does not automatically validate the other dimensions of the business.

Uncertainty is an inherent part of this process. The objective is not to eliminate every doubt before launching a product, but to identify which assumptions are most important and which need to be tested first. A hypothesis about the severity of the problem can be investigated through interviews and observation; a hypothesis about willingness to pay requires different evidence; a hypothesis about operational scalability may require delivery tests and cost analysis. The choice of experiment should reflect the question, not merely the ease of producing a demonstration or launching a feature.

A startup's initial context also changes as the business learns. Evidence obtained from early customers may reveal that the initially selected segment is heterogeneous, that the primary need differs from what was imagined, or that the solution requires a different distribution model. Revising these assumptions should not automatically be interpreted as failure or lack of vision. It is part of the process of building a business based on progressively more reliable knowledge of an external reality.

In entrepreneurship, discovering the initial context means treating the opportunity as something to investigate, rather than as a conclusion derived from the founder's conviction. Experience, intuition, and industry knowledge help formulate hypotheses and choose where to look for evidence. The viability of the business, however, depends on how these hypotheses withstand contact with customers, competitors, channels, costs, and concrete operating conditions.

3.11. Application in Professional Careers and Personal Life

Career and personal life decisions also frequently begin with a narrative about the situation. A person may conclude that they need to change jobs because they are dissatisfied, begin a training program because they believe they have lost their competitive edge, or start a business because they want more autonomy. These conclusions express legitimate needs and expectations, but they may bring different factors together under a single explanation. Investigating the initial context helps clarify what is causing the dissatisfaction, which outcomes are desired, and which conditions need to be considered before choosing a path.

A career decision, for example, may involve compensation, autonomy, professional relationships, type of work, growth prospects, stability, and quality of life. These factors do not carry the same weight for everyone and may conflict with one another. An opportunity offering higher compensation may require longer commutes; a position offering greater autonomy may provide less predictability; a career change may expand future possibilities while reducing income in the short term. Before comparing alternatives, it is necessary to understand which dimensions explain the current situation and which outcomes truly matter.

The investigation may begin by reconstructing the events that led to the decision. Did the dissatisfaction emerge after a leadership change, a change in responsibilities, a prolonged period of excessive workload, or the perception that growth opportunities had run out? Is the difficulty specific to the current job, or does it appear across different professional experiences? Is there a need that could be addressed through a role change, a negotiation, or training without requiring a complete transition? These questions do not presume that the person should stay where they are. They help distinguish the perceived problem from the possible responses.

Sources of information include personal experience, professional history, conversations with people familiar with the work, market data, job descriptions, and the experiences of professionals working in the areas under consideration. Each source has limitations. Personal perception is essential for understanding preferences and discomfort, but it may be influenced by recent events. Other people's accounts help reveal possibilities, but do not guarantee that the same experience will be repeated in another context. Market data may indicate general trends without representing the opportunities available to someone with a particular profile, location, and set of skills.

It is also necessary to distinguish facts from interpretations about one's own abilities. A person may state that they are not suited to a particular role because they experienced difficulties in one specific situation. This conclusion may reflect a skills gap, inadequate working conditions, a lack of support, or incompatibility between

the role and their preferences. Treating the interpretation as fact may unnecessarily restrict the available alternatives. Similarly, believing that a training program will guarantee access to a new career is a hypothesis that must be compared against actual job requirements, market demand, and the experiences of people who have made similar transitions.

Practical conditions delimit the decision space. Financial reserves, family responsibilities, location, available time, access to training, and the possibility of trying a new activity influence what can be done and at what pace. These conditions should not be confused with an absolute determination of the future, but neither can they be ignored in favor of an idealized alternative. A transition may be feasible in stages, through side projects, gradual training, or changes in responsibilities before a definitive departure. The way the context is investigated helps reveal not only which options exist, but also which paths are compatible with the person's actual circumstances.

The same reasoning applies to decisions such as moving to another city, starting a business, or choosing a training program. The desire to live somewhere else may be related to career prospects, cost of living, proximity to important people, or the pursuit of a different routine. Evaluating only housing prices or job availability would be insufficient if other factors are decisive for the expected outcome. The investigation must consider the dimensions of reality directly related to the choice, without trying to turn every aspect of life into a measurable variable.

In this context, there is a particular risk of confusing desire with prediction. Wanting a change does not demonstrate that it will produce the expected effects; fearing an alternative does not demonstrate that it will be harmful. Desire helps define objectives, while evidence and analysis help clarify the conditions for achieving them. A well-founded decision does not need to eliminate the subjective dimension of the choice, but it should recognize when an expectation depends on assumptions that have not yet been examined.

It is also unnecessary to conduct an exhaustive investigation for every personal decision. The effort should be proportionate to the relevance, impact, and reversibility of the choice. A small decision can be made on the basis of limited information, especially when it is easy to change course. A change affecting income, relationships, location, or long-term commitments deserves more careful analysis. In some situations, a reversible experiment, such as undertaking a side project or speaking with professionals in a particular field, may produce more useful information than prolonging reflection without engaging with reality.

In careers and personal life, discovering the initial context means understanding the situation before assigning it a definitive cause or committing to a solution. This allows objectives and preferences to be considered without confusing them with evidence, constraints to be acknowledged without treating them as permanent, and alternatives to be evaluated based on the conditions under which they can actually be pursued.

Across all these areas, investigation serves the same fundamental purpose. It does not choose on behalf of the individual, replace professional judgment, or provide guarantees about the future. Its contribution is to make the decision more aware of the surrounding conditions, the evidence supporting the adopted interpretation, and the uncertainties that remain. Before deciding how to change a reality, it is necessary to understand how it presents itself, why change has become necessary, and which assumptions still need to be examined.

4. Establishing Objectives

In the previous chapters, we laid the foundations for guiding a decision. First, we put on the biologist's hat to understand the domain in which the decision is situated, identify its elements, and examine their relationships. Then, we assumed the role of the investigative reporter to understand what happens in that domain and why a particular situation requires attention. Now, it is up to the navigator to establish the destination that will give direction to the decision and evaluate the possible routes to reach it.

Imagine a navigator who needs to plan a trip. Between the starting point and the destination, there may be countless routes, each with its own characteristics, constraints, and consequences. To choose among them, one must know the destination, understand the conditions along the way, and assess which paths are most suitable. Without a defined destination, there is no reference point for comparing routes. Without understanding the starting point and the conditions of the journey, on the other hand, even a clear destination can lead to inappropriate choices. In decision-making, objectives serve as the destination: they guide choices and make it possible to assess whether the paths under consideration lead to the intended results.

Not all decisions, however, have this reference point. Some proceed without explicit objectives, driven by immediate demands, individual preferences, or actions that seem necessary. Without clarity about the intended outcome, it becomes difficult to determine whether the alternatives contribute to something relevant or whether the resources employed are producing the expected progress. These are blind decisions, not because those responsible are incapable of analyzing the circumstances, but because they lack sufficiently clear direction to guide their choices and evaluate their results.

There are also decisions that start with an apparently well-defined objective but without sufficient investigation of the situation that should underpin it. A company decides to replace a system, a manager proposes automating a process, or an entrepreneur plans to launch a product before understanding the needs, conditions, and constraints involved. The objective may then reflect mistaken assumptions or an incomplete interpretation of reality. Knowing the destination does not guarantee a successful journey; similarly, a clear objective does not ensure that it is relevant to the situation that motivated the decision.

These two risks highlight the importance of relating an understanding of the context to the definition of objectives. Investigating the situation without clarifying the desired outcome can leave the decision without direction; defining an objective without sufficiently understanding the context can direct efforts toward an outcome that is of little relevance or inappropriate. Consistency depends on the relationship between the circumstances that motivated the decision, what is intended to be achieved, and the possibilities available for doing so.

This does not mean that objectives can only be considered after the entire investigation has been completed. A preliminary intention can guide questions and help define what needs to be understood. As new evidence emerges, that intention may need to be adjusted. Further investigation may reveal that the initially desired outcome does not address the actual need, while clarifying the objective may indicate which aspects of the situation deserve greater attention. The process therefore requires a willingness to revise both the understanding of the context and the direction established.

Establishing objectives means transforming an understanding of the situation into an explicit direction for the decision. This requires clarifying what outcome is intended, why it matters, and which priorities should guide the choices. It also requires distinguishing the desired outcome from the actions that may produce it, the solutions initially envisioned, and the metrics used to track progress. An initiative may be executed as planned and show good indicators without generating the value that justified its implementation. Therefore, monitoring execution does not replace evaluating the results achieved.

Objectives provide a reference for comparing alternatives that may lead to similar outcomes through different paths, with different costs, risks, timelines, and consequences. They do not predetermine which solution should be chosen, but they make it possible to evaluate the possibilities in terms of what truly matters. This reference keeps the decision on course throughout the process and helps identify when a path, although feasible, does not contribute sufficiently to the desired outcome.

4.1 From Context to the Desired Direction

Understanding a situation does not automatically determine what we should do about it. An investigation may reveal operational difficulties, unmet needs, growth opportunities, or changes in external conditions, but turning

these findings into a desired direction requires a new stage of reasoning. We need to interpret what has been discovered, assess its relevance, and decide which outcomes deserve to guide our efforts.

A difficulty identified in the initial context, for example, may admit different responses. If a company realizes that its sales representatives have difficulty accessing product information, it may conclude that it needs to replace its sales system. However, it may also discover that the problem stems from data organization, a lack of integration between applications, an inadequate update process, or insufficient training. Each interpretation points to different interventions. Before comparing these possibilities, it is necessary to clarify the outcome the organization intends to achieve, such as reducing service time, improving the quality of available information, or increasing the conversion rate of sales opportunities.

This distinction separates the need motivating a decision from the solution initially proposed to address it. A request usually arises alongside an interpretation of what needs to be done because people naturally seek to associate their difficulties with familiar actions. A manager who identifies delays may request additional staff; a team facing operational limitations may ask for a new tool; an entrepreneur who observes interest in a particular product may conclude that the offering needs to be expanded. These proposals may be appropriate, but they still need to be related to the outcomes that justify the effort.

Establishing objectives means shifting attention from the suggested action to the effect one wants to produce. Instead of assuming that the organization needs to replace a system, we can investigate whether the objective is to reduce the time required to complete a sale, decrease processing errors, or provide more reliable information for decision-making. The difference is not merely terminological. When the desired outcome is explicit, it becomes possible to consider alternatives other than the one that originated the request and evaluate each according to its contribution to what truly matters.

This transition also requires recognizing that not every situation should be treated as a problem to eliminate. An expansion opportunity, for example, may motivate the definition of objectives related to growth, entry into new markets, or revenue diversification. In this case, there is not necessarily an undesirable condition that needs to be corrected. Instead, there is a possible future that the organization considers relevant and wants to explore. In both cases, the objective establishes a direction, but its formulation depends on understanding the circumstances and making explicit the reasons why a particular outcome deserves to be pursued.

Context therefore provides the basis for defining objectives, but it does not replace the judgment required to choose them. Two organizations facing similar conditions may establish different objectives because they have different strategies, resources, responsibilities, and priorities. Similarly, two people facing the same professional opportunity may value different outcomes depending on their circumstances and aspirations. Decision Architecture does not assume that there is a single correct destination for every situation. Its purpose is to make explicit the relationship between what has been understood, what is intended to be achieved, and the reasons supporting that choice.

4.2 Defining What Is Intended to Be Achieved

An objective expresses a desired outcome that guides a decision. It describes a condition to be achieved or a change to be produced and provides a reference for evaluating the available possibilities. Its function is not to anticipate every detail of execution, but to clarify what must be achieved for the decision to deliver the expected value.

An objective must be formulated clearly enough to distinguish desirable outcomes from irrelevant ones, without unnecessarily restricting the paths that may lead to them. Consider a company that wants to improve its customer service process. If the objective is simply to implement a customer relationship management platform, the definition already incorporates a specific solution. The decision becomes driven by the execution of that initiative, even if other possibilities could produce better results. If, on the other hand, the objective is to reduce customer response times and increase the consistency of the information provided, different alternatives can be investigated and compared.

This distinction does not mean that objectives should be formulated abstractly or generically. Expressions such as improving efficiency, growing sustainably, or providing a better experience may indicate legitimate intentions, but they need to be clarified to guide a concrete decision. Improving the efficiency of which process? Growing along which dimension? Providing a better experience for which people and under what circumstances? The more important the decision, the greater the need to make the practical meaning of the desired outcome explicit.

At the same time, it is important not to confuse clarity of objectives with the advance specification of a solution. An objective can be precise about the intended outcome without determining how it will be achieved. An organization may establish that it wants to reduce inventory losses, improve product availability, or decrease the time required to process orders. These objectives guide the investigation of alternatives, but they do not require the answer to be a new application, a change of supplier, or a redesign of existing processes.

The distinction between objective and solution is particularly important because solutions carry assumptions about the causes of the situation and the effectiveness of particular interventions. When a solution is incorporated into the objective, these assumptions cease to be examined and begin to function as implicit conditions of the decision. This can limit the exploration of alternatives and increase the risk of investing resources in a response that does not adequately address the identified need.

We also need to distinguish objectives from activities and deliverables. An activity describes something that will be done; a deliverable represents a product, service, or immediate result produced by the work; an objective expresses what is intended to be achieved through those actions and deliverables. Developing an application, training a team, or reorganizing a process are initiatives that may contribute to a desired outcome, but their completion does not prove that the outcome has been achieved. An application may be delivered without improving operations, just as training may be completed without producing meaningful changes in performance.

This distinction makes it possible to evaluate a decision beyond the execution of a plan. The question is no longer merely whether we did what we had planned, but also whether what we did contributed to the outcome that justified the effort. In complex initiatives, this difference can separate a technically well-executed project from an effectively successful decision. The work may meet the planned scope, stay within budget, and adhere to the schedule, yet still fail to produce the benefit that motivated its implementation.

Defining objectives, therefore, means establishing a reference for judging choices. A consistent objective clarifies the outcome that matters, allows alternatives to be examined without prematurely committing to a solution, and creates the conditions for subsequently assessing whether the decision produced the expected effects.

4.3 Distinguishing Objectives, Metrics, and Targets

Once the intended outcome has been defined, the need arises to track progress and evaluate results. At this point, it is important to distinguish three related but different concepts: objective, metric, and target. Confusing them can undermine both the definition of direction and the assessment of a decision's success.

The objective expresses the desired outcome. A metric is a measure used to observe a particular dimension of reality. A target establishes a value or condition to be reached, usually within a defined time horizon. These elements complement one another, but they answer different questions. The objective clarifies what we want to achieve; the metric indicates what we will observe to assess relevant aspects of that outcome; and the target establishes a concrete reference for comparing observed performance with expected performance.

Imagine a company that wants to improve its customers' shopping experience. This objective can be tracked through different metrics, such as the shopping cart abandonment rate, the time required to complete a purchase, the frequency of complaints, and customers' reported satisfaction levels. Each measure offers a perspective on the experience, but none of them, in isolation, necessarily represents the objective in its entirety. If the company decides to reduce the shopping cart abandonment rate from 70% to 55% within six months, it will have established a target associated with a metric. It will still need to verify whether this reduction actually represents an improvement in the shopping experience and whether it has not come at the expense of other important outcomes.

The problem begins when a metric is treated as though it were the objective itself. This happens frequently because quantitative measures are easier to track, compare, and communicate than broad outcomes such as trust, experience quality, or business sustainability. When an organization turns an available indicator into the primary reference for its decisions, it risks directing efforts toward improving the measure without producing the change that truly matters.

A sales team evaluated exclusively on the number of sales may increase transaction volume through excessive discounts, compromising contribution margin and profitability. A support team guided solely by average handling time may close requests quickly without resolving customers' problems. A development team pressured to deliver more features may reduce the time devoted to quality, increasing defect rates and maintenance costs. In all these cases, the metric may show favorable progress while the broader outcome remains unchanged or deteriorates.

The risk lies not in using metrics, but in losing their connection to objectives. A measure is useful when it provides relevant evidence for understanding progress, identifying deviations, and evaluating the consequences of choices. When it begins to replace the objective, it ceases to be merely an observation tool and starts to determine behavior in ways that may be incompatible with the original purpose. The organization begins optimizing what it measures, even when improving the indicator does not represent a proportional improvement in the reality it intended to transform.

This difficulty is particularly important in decisions involving multiple dimensions of value. The shopping experience, for example, may depend on ease of navigation, product availability, information transparency, payment security, and delivery reliability. A single metric is unlikely to represent all these aspects. Therefore, broader objectives may require a set of complementary measures capable of revealing different dimensions of the outcome and identifying undesirable effects that would remain hidden if only one indicator were observed.

It is also necessary to recognize that not every relevant outcome can be measured directly. Some objectives involve changes in behavior, trust, organizational capability, or risk reduction, whose evaluation depends on qualitative and quantitative evidence. The absence of a simple measure does not make the objective less important, just as the availability of an indicator does not guarantee that it is appropriate. Metrics should be selected based on what is intended to be understood and evaluated, not merely on what is easiest to record.

Returning to the navigator metaphor, the destination is not the recorded speed, the distance traveled, or the number of stops made. This information helps track the journey, but it only acquires meaning in relation to the place one intends to reach. Similarly, in decision-making, measuring progress is necessary, but the value of that progress depends on the objective that gives it direction. Without this reference, we may improve indicators, increase productivity, and meet targets without producing the outcome that justified the decision.

4.4 Recognizing Priorities and Conflicts

In simple situations, it may seem possible to define a single objective that represents the desired outcome. In more complex organizational and personal decisions, however, different objectives often coexist and do not always point in the same direction. A company may want to grow, preserve profitability, reduce risks, and improve customer satisfaction. A project may need to accelerate delivery, keep costs under control, and meet rigorous quality requirements. An individual may seek professional growth, greater financial stability, and more time for personal life.

These objectives may be legitimate and important, but their coexistence does not guarantee that they can all be achieved simultaneously to the desired degree. Resources, time, and execution capacity are limited, and some choices produce consequences that favor one dimension while restricting another. Defining objectives must acknowledge these tensions and make explicit the priorities that will guide the decision when it is not possible to maximize every outcome.

Consider a company that intends to expand its operations into a new region. Expansion may increase the potential market and boost revenue, but it may also require investments, raise logistics costs, and introduce uncertainty about local demand. If the objective is defined solely as growth, the organization will have a broad direction but little guidance for evaluating the opportunity. If it also establishes that it wants to grow while preserving profitability and limiting its financial exposure, it will have made additional dimensions explicit that must be considered in the decision. The challenge then becomes understanding which expansion conditions meet these priorities and which trade-offs the organization is willing to accept.

Defining priorities does not mean ignoring secondary objectives or treating all other dimensions as irrelevant. It means recognizing which outcomes are essential, which can be negotiated, and which conditions must not be exceeded. In some decisions, certain conditions function as limits that alternatives must respect. In others, there are objectives that can be balanced in different ways depending on the circumstances and the preferences of those involved. Distinguishing between these situations helps prevent an apparently favorable choice in one dimension from producing unacceptable consequences in another.

The difficulty increases when different people or groups have distinct expectations about what the decision should achieve. A sales department may prioritize service speed, while the finance department emphasizes cost reduction and the operations team seeks to preserve process stability. Each perspective may reflect real needs, but the sum of these expectations does not automatically constitute a coherent set of objectives. It is

necessary to understand the interests involved, make disagreements explicit, and establish which outcomes will take precedence in the event of a conflict.

This task does not involve eliminating all disagreements through artificial consensus. In some situations, interests are incompatible, and the decision requires accepting consequences that cannot be avoided. Making objectives explicit allows these conflicts to be discussed more clearly instead of remaining hidden in contradictory requirements, implicit expectations, or inconsistent evaluation criteria. The decision then acknowledges not only what is desired, but also what one is willing to sacrifice or preserve to achieve that outcome.

Priorities may also change as the context evolves. An organization in a precarious financial position may prioritize cash preservation, while a company with available resources and a significant market opportunity may accept greater investment to accelerate growth. Similarly, a person may value stability at one point in life and place greater importance on exploring opportunities at another. Objectives must be understood in relation to the circumstances that give them meaning, rather than being treated as immutable declarations.

By making priorities and conflicts explicit, we create the conditions for comparing alternatives using more consistent criteria. Evaluation no longer depends solely on general impressions of which option seems better, but instead considers how each possibility contributes to the intended outcomes, which constraints it respects, and which trade-offs it requires. This clarity does not eliminate the judgment required for the decision, but it makes its foundations more transparent and its consequences easier to examine.

4.5 Establishing Success Criteria

Defining an objective clarifies the desired outcome, but we still need to determine how to recognize whether that outcome has been achieved. This question leads to success criteria, which establish the conditions and evidence used to evaluate the results of a decision. Without these criteria, the objective may remain a generic intention, subject to different interpretations when the time comes to evaluate what has been accomplished.

Success criteria must be related to the meaning of the objective, not merely to the activities performed during the work. If a company intends to improve product availability, implementing a new inventory system may represent an important deliverable, but it does not, by itself, demonstrate that products have become more available to customers. It will be necessary to observe the operational outcome, understand the changes produced, and verify whether they correspond to what motivated the initiative.

In some cases, the criterion can be expressed as a clearly verifiable condition. A decision may aim to reduce order processing time, with a target associated with the average time observed over a given period. In other cases, evaluation requires combining measures, observations, and qualitative judgments. Improving customer trust, increasing a team's innovation capacity, or strengthening the resilience of a technology architecture are outcomes that can hardly be represented by a single indicator. In these contexts, the criteria must reflect the nature of the objective and bring together sufficient evidence to support a reasonable assessment.

It is also important to establish when and under what conditions success will be evaluated. Some changes produce immediate effects, while others depend on adoption, learning, or gradual behavioral transformation. A new feature may be available at the time of delivery, but its impact on the user experience can only be understood after a period of use. Similarly, an initiative intended to reduce costs may require time before its operational and financial effects become observable. Evaluating prematurely may lead to the conclusion that a decision has failed before its results have had a chance to materialize; waiting indefinitely, on the other hand, may delay the identification of problems and the need for adjustments.

Success criteria must also account for potential side effects. A reduction in service time, for example, should be examined alongside resolution quality and customer satisfaction. A growth initiative may require monitoring profitability, resource consumption, and risk exposure. The purpose is to prevent favorable performance in one dimension from concealing significant deterioration in others. When multiple objectives are involved, the evaluation must reflect this plurality without assuming that all outcomes can be reduced to a single measure.

There is also a difference between recognizing that an objective has been achieved and concluding that the decision was good. Observed results depend not only on the quality of the choice, but also on the circumstances in which it was executed, external factors, and events beyond the control of those involved. A well-founded decision may produce an unfavorable outcome in the face of an unexpected change in the environment; a poorly founded decision may produce a positive outcome due to favorable circumstances. Therefore, evaluating a

decision requires considering both the results and the quality of the reasoning, evidence, and assumptions that supported the choice.

This distinction does not diminish the importance of results. On the contrary, it makes it possible to examine them more rigorously, separating what was expected from what actually happened and identifying the conditions that explain any differences. Success criteria help establish in advance what will be considered relevant, reducing the tendency to redefine expectations after the results are known.

By establishing success criteria, we turn objectives into useful references for evaluating alternatives and supporting subsequent learning. We do not need to predict every possible consequence, but we must clarify which outcomes justify the decision, what evidence will make it possible to track them, and which conditions would indicate that the chosen direction needs to be reconsidered.

4.6 Validating and Refining Objectives

Although objectives guide decisions, their initial formulation should not be treated as final. Understanding of a situation evolves as new information is discovered, the perspectives of those involved are compared, and relevant conditions become clearer. For this reason, objectives may also need to be revised. The challenge is to strike a balance between maintaining a sufficiently stable direction to guide choices and preserving the ability to adjust it in response to relevant evidence.

Imagine that a company begins investigating the need to replace its sales system because sales representatives report difficulties working with the available information. During the investigation, it is discovered that the main problem lies in updating inventory data, which is done manually and at different times by different teams. The initially formulated objective may have emphasized modernizing the platform, but the evidence points to a more specific need: increasing the reliability and timeliness of the information used during sales. Based on this understanding, the objective can be refined to better reflect the intended outcome, allowing the decision to consider different ways of achieving it.

This process does not mean that objectives should change whenever new information emerges or that every difficulty justifies abandoning the established direction. Revisions must be grounded in the relevance of the evidence, changes in conditions, or the identification of assumptions that do not hold up. Otherwise, the decision may lose coherence, with priorities that change continuously and success criteria that no longer provide a stable reference for evaluating choices.

Validating objectives also requires considering the people affected by the decision. An outcome that seems desirable to the person requesting an initiative may not represent the same value to those who will use the solution, carry out the process, or bear its consequences. A company may seek operational efficiency, but a change aimed exclusively at reducing costs may transfer work to customers or increase the complexity faced by employees. Making these perspectives explicit helps identify incomplete objectives and consequences that need to be considered before proceeding.

Another important aspect is verifying whether the objectives are compatible with known conditions. An outcome may be desirable but require unavailable resources, depend on factors beyond the organization's reach, or conflict with constraints that cannot be ignored. This does not mean that ambitious objectives should be discarded at the first sign of difficulty. It means that their formulation must distinguish the desired outcome from the conditions necessary to achieve it and the uncertainties that still need to be investigated. In some situations, the objective will need to be adjusted; in others, it will be more appropriate to investigate new possibilities or explicitly acknowledge the risks involved.

Validation also helps identify objectives that, although clearly formulated, do not justify the effort required to achieve them. An opportunity may seem attractive until costs, dependencies, and risks are considered. An initiative may produce a measurable improvement, but one of insufficient importance relative to the resources required. Establishing objectives does not mean assuming that every intention deserves to be pursued. It means making the intended outcomes explicit so that their relevance can be evaluated and a decision can be made about whether investing in their pursuit is worthwhile.

The result of this stage need not be an extensive statement or a rigid set of commitments. It needs to be a sufficiently clear representation of what is intended to be achieved, why that outcome matters, which priorities should guide the decision, and how success can be evaluated. This representation should remain open to justified revisions without losing its primary function: providing an explicit direction for comparing alternatives.

4.7 Building the Representation of Objectives

At the end of the objective-definition process, we need to consolidate the understanding developed into a representation that can guide the subsequent stages of the decision. This representation is not an execution plan, a solution specification, or a list of activities. Its purpose is to record the intended outcomes and the criteria that will make it possible to evaluate the available possibilities, while keeping the relationship between the initial situation and the desired direction explicit.

A consistent representation begins with the main objective, formulated to express the outcome to be achieved without unnecessarily anticipating a solution. When complementary objectives exist, they should be identified and related to the main objective, clarifying their contribution to the broader outcome. It is also important to make the priorities among them explicit and acknowledge any conflicts, so that the evaluation of alternatives does not depend on assumptions that remain hidden.

Relevant metrics and targets can be recorded alongside the success criteria, provided they remain clearly distinguished from the objectives. This separation makes it possible to understand which outcomes matter, which measures will provide evidence of their progress, and which values or conditions will indicate the expected performance. When evaluation depends on qualitative evidence, observations, or contextualized judgments, these needs should also be acknowledged, preventing the representation from being limited to what is easily quantifiable.

The representation must also consider the main conditions affecting the possibility of achieving the intended outcomes. Relevant constraints, dependencies, and uncertainties may alter the feasibility of certain alternatives or require specific trade-offs. Recording them helps preserve the connection with the context investigated in the previous chapter and prevents objectives from being treated as intentions independent of the circumstances in which they must be pursued.

In simpler decisions, this representation may take the form of a concise description accompanied by a few evaluation criteria. In complex decisions, it may require more detailed documentation involving different objectives, indicators, time horizons, stakeholder groups, and success conditions. The level of formalization should be proportional to the importance of the decision, its complexity, and the consequences of an incorrect interpretation. The purpose is not to produce documentation for its own sake, but to ensure that those involved share a sufficiently consistent understanding of what is intended to be achieved.

This consolidation also makes it easier to identify gaps in the reasoning. If we cannot explain why a particular objective matters, perhaps the underlying need has not yet been sufficiently understood. If we cannot establish any relevant evidence for evaluating the outcome, perhaps its formulation needs refinement or the situation requires further investigation. If different objectives appear incompatible, we need to make the conflict explicit before comparing alternatives. The representation thus functions both as a summary of the understanding achieved and as a tool for verifying its consistency.

With the objectives established, the decision now has a reference for examining the possible paths. This does not mean that all uncertainties have been eliminated or that the destination has been definitively fixed. It means there is an explicit understanding of the outcomes that justify the effort and the criteria that should guide the choices. This reference makes it possible to compare alternatives more coherently, recognize trade-offs, and assess whether a proposal addresses the identified needs.

4.8 Application in Project Management

In project management, defining objectives is essential for establishing the relationship between the work to be performed and the benefits that justify the investment. Projects often begin with a request, a set of expected deliverables, and an estimate of time and budget. These elements are important for organizing execution, but they do not guarantee that the need motivating the initiative has been understood or that the intended outcomes are clear.

Consider a project to implement a system that automates order processing. The deliverable may be described as making an application available that is integrated with existing systems and includes functionality for recording, validating, and tracking requests. However, the project's objective may be related to reducing processing time, decreasing operational errors, or increasing the capacity to meet demand. If this distinction is not made explicit, success may be associated with completing the implementation, even if operational results remain below expectations.

Clarity of objectives also contributes to scope management. When the intended outcome is well defined, it becomes possible to evaluate change requests according to their contribution to the objectives and their effects on established priorities. An additional feature may seem interesting but require resources and time without making a meaningful contribution to the expected outcome. Another change, not initially anticipated, may be necessary to ensure that the solution delivers the desired benefit. The objective provides a reference for evaluating these situations without turning the initial scope into a constraint disconnected from the project's needs.

Another important aspect is the difference between execution indicators and evidence of outcomes. Adherence to the schedule, budget control, and completion of deliverables help evaluate how the project is being managed. However, meeting these conditions does not automatically prove that the expected benefits have been produced. A project may be delivered on time and within budget and still fail to reduce processing time, decrease errors, or improve the user experience. Management must recognize these distinct dimensions and establish criteria that make it possible to monitor both execution and the outcomes that justify the investment.

In initiatives involving organizational transformation, this distinction becomes even more relevant. Benefits may depend on the adoption of new processes, behavioral changes, team training, and conditions outside the project's direct control. Delivering a tool does not guarantee that it will be used properly, just as completing training does not ensure that work practices will change. Objectives must reflect the desired outcomes and acknowledge the dependencies necessary to produce them.

Defining objectives also helps manage expectations among sponsors, managers, teams, and other stakeholders. When each group has a different interpretation of what success means, the project may progress with apparent alignment and reveal disagreements only at the time of delivery. Making the expected outcomes, priorities, and evaluation criteria explicit makes it possible to discuss these differences in advance and establish a more consistent basis for decision-making.

In this context, Decision Architecture helps establish the connection between the need that originated the project, the outcomes that justify its implementation, and the criteria that will guide choices throughout the work. Execution remains essential, but it is now guided by a clearer understanding of what must be achieved, allowing changes, constraints, and alternatives to be evaluated in relation to the value intended to be produced.

4.9 Application in Solution Architecture

In solution architecture, defining objectives establishes a reference for the technological and structural decisions that will be made. The diversity of available technologies, standards, platforms, and approaches can make choosing a solution a complex activity. Without clear objectives, there is a risk of shaping the architecture around familiarity with particular tools, a team's preferences, or the adoption of technical trends, without demonstrating how these choices contribute to the expected outcomes.

Consider a company that intends to replace its sales system because users report slowness and difficulty accessing information. The request may lead directly to the evaluation of new platforms, but investigating the context may reveal that response times are affected by inefficient queries, external integrations, data unavailability, or manual steps in the sales process. Each cause requires a different analysis, and choosing a new platform may not resolve the main difficulty.

Defining objectives makes it possible to direct this analysis toward the outcomes that matter. If the organization intends to reduce response times in sales operations, increase information availability, or improve data reliability, these outcomes can guide the evaluation of architectural alternatives. Completely replacing the system becomes one possibility among others, which can be compared with optimizing the existing application, revising integrations, modernizing specific components, or reorganizing information flows.

Objectives also help balance quality attributes that may conflict. A solution may need to provide greater availability, reduce operating costs, meet security requirements, and support frequent evolution. These needs cannot always be maximized simultaneously, and the architecture must reflect the priorities and constraints of the context. Without an explicit definition of what matters most, structural decisions may be made based on technical preferences that do not correspond to business needs.

The relationship between objectives and metrics deserves special attention in this domain. Indicators such as response time, failure rate, availability, infrastructure cost, and deployment frequency help evaluate different aspects of a solution. However, none of these indicators individually represents the total value produced by the

architecture. Reducing response time may be important, but its impact depends on processes and user needs. Increasing availability may be necessary for critical operations, but the additional cost must be evaluated in relation to the consequences of an interruption. A metric acquires meaning when it is related to the outcome the decision intends to produce.

This perspective also prevents architecture from being evaluated solely on compliance with standards or the adoption of technologies considered modern. A solution may use current components and recognized practices without adequately addressing the need that motivated the work. Similarly, a less sophisticated solution may be sufficient when it meets the objectives, respects the constraints, and has a cost commensurate with the expected value. Architectural quality is not limited to the choice of technologies; it also includes the suitability of decisions to the system's concrete needs and conditions.

Establishing objectives before comparing alternatives does not mean indefinitely postponing technical decisions. It means providing them with an explicit foundation. When the intended outcomes are clear, the architect can justify choices, identify trade-offs, and evaluate consequences more consistently. Architecture ceases to be a technological response to the initial request and becomes a set of decisions directed toward producing the outcomes that justify the solution.

4.10 Application in User Experience

In user experience, defining objectives requires connecting people's needs to the outcomes the organization intends to produce. An interface may be visually consistent, offer a wide range of features, and meet technical requirements, yet still make it difficult to perform the tasks that motivate its use. The quality of the experience depends on the relationship between what people need to do, the conditions under which they interact with the product, and the outcomes they can achieve.

Imagine a company that notices a high abandonment rate during the checkout process in its e-commerce business. An immediate response might be to redesign the screens, reduce the number of steps, or introduce new payment methods. These interventions may be useful, but defining objectives must start with a more precise understanding of the situation. Abandonment may be related to form complexity, delivery costs presented late in the process, lack of payment trust, product unavailability, or navigation difficulties on mobile devices. Each hypothesis suggests different possibilities and requires evidence to help identify which aspects deserve attention.

The objective should not simply be to redesign the checkout process, but to produce a meaningful improvement in the experience and the outcomes that justify it. This may involve making it easier to complete a purchase, increasing information clarity, reducing errors, or strengthening user trust. The formulation must consider both people's needs and the organization's objectives, without assuming that both will automatically be met by the same intervention.

The distinction between objective and metric is especially important in this context. Conversion rate, time required to complete a task, number of errors, and abandonment frequency provide useful information about user behavior. However, improving one of these measures does not necessarily improve the entire experience. Reducing the time needed to complete a task may be positive when it eliminates unnecessary steps, but harmful when it removes important information or makes it harder to understand the consequences of an action. Evaluation must consider the meaning of the metric in relation to the need being addressed.

It is also necessary to recognize that different user groups may have different needs. An interface simplified for experienced users may create difficulties for people who need additional guidance. A process optimized for most customers may exclude those who use assistive technologies or face connectivity limitations. Defining objectives requires clarifying for whom value is intended to be created, which needs are priorities, and which conditions of access and use must not be overlooked.

User experience also shows that objectives may conflict with apparently favorable business metrics. A strategy that increases impulse purchases may improve conversion in the short term but undermine customer trust. A process that makes cancellation difficult may reduce the observed churn rate without producing greater satisfaction or genuine loyalty. When a metric is treated as the objective, it becomes possible to improve the indicator through mechanisms that contradict the broader purpose of the experience.

Defining objectives helps establish which outcomes should guide the design and evaluation of interactions. Instead of starting with the screens or features to be developed, the team begins by investigating which tasks need to be made easier, which difficulties should be reduced, and what evidence will indicate a meaningful improvement.

This approach makes it possible to evaluate design alternatives against identified needs without confusing the delivery of an interface with the creation of a satisfactory experience.

4.11 Application in Entrepreneurship

In entrepreneurship, objectives are often defined under conditions of high uncertainty. A startup may identify a market opportunity, develop a hypothesis about customer needs, and propose a product it believes can address them. In this context, establishing objectives does not mean assuming that the business model has already been validated, but rather clarifying what needs to be achieved and what evidence will help assess whether the initiative is moving in the desired direction.

An entrepreneurial team may establish launching an application by a certain deadline as an objective. This objective organizes a deliverable, but it does not necessarily clarify the business outcome that justifies the launch. The application may be published without attracting users, solving a relevant need, or generating a willingness to pay for the service. If the purpose is to build a sustainable business, it will be necessary to distinguish product delivery from evidence that a real need exists, that the value proposition is recognized, and that the model can produce economically viable results.

This does not mean that launching is irrelevant. At certain stages, making an initial version of the product available may be necessary to observe user behavior and test hypotheses. The difference lies in the relationship between the activity and the objective. Launching serves as a means of generating evidence and learning about the market, while the objective may involve validating a need, demonstrating customers' willingness to pay, or identifying the conditions for sustainable operations. This distinction helps prevent completing a development plan from being confused with validating the opportunity.

A startup's metrics must be interpreted with the same care. The number of sign-ups, downloads, or visits may indicate initial interest, but it does not necessarily demonstrate recurring use, customer retention, or economic viability. A campaign may increase the number of registered users without attracting people who find value in the product. Similarly, rapid growth may conceal high acquisition costs, low margins, or dependence on conditions that cannot be sustained. The measures chosen must be related to the hypotheses and outcomes that truly matter at the business's current stage.

Defining objectives also helps establish priorities in the face of resource scarcity. A startup can rarely develop every desired feature, serve every market segment, and explore every identified opportunity at the same time. It is necessary to decide which outcomes deserve immediate attention and which hypotheses must be tested before increasing investment. A well-formulated objective makes it possible to concentrate efforts on what can reduce the most relevant uncertainties or produce the progress needed for the business to continue.

In this environment, objectives may change as hypotheses are tested against evidence. The team may discover that the identified problem is not important enough to customers, that the selected segment has needs different from those expected, or that the value proposition requires a different approach. Revising the objective under these conditions does not necessarily represent a planning failure. It may be a consequence of a learning process that revealed limitations in the initial understanding. The important thing is that the revision be grounded in evidence and that the reasons for changing direction remain explicit.

The entrepreneurial experience thus reinforces that defining objectives is not merely about setting growth targets or preparing financial projections. It is about clarifying which outcomes justify the initiative, which assumptions need to be validated, and what evidence will make it possible to decide whether to continue, adjust the proposal, or discontinue the investment. Under conditions of uncertainty, clear objectives do not eliminate risk, but they help direct learning efforts and prevent intense activity from being mistaken for real progress.

4.12 Application in Professional Careers and Personal Life

In professional careers and personal life, establishing objectives requires relating current circumstances to the outcomes we consider important. Individual decisions often begin with dissatisfaction, a perceived opportunity, or comparison with other people's choices. Someone considers changing jobs, pursuing further education, moving to another city, or starting a business. These possibilities may represent legitimate paths, but their suitability depends on what the person intends to achieve and the conditions under which the decision will be made.

Consider a professional who wants to change jobs because they feel their career has stagnated. Changing jobs

may be an appropriate response, but the initial dissatisfaction still needs to be understood. The difficulty may be related to compensation, lack of learning opportunities, absence of autonomy, the work environment, or incompatibility between the activities performed and personal interests. If the objective is simply to get another job, the decision may reproduce the same conditions that motivated the dissatisfaction. If the person clarifies that they seek greater autonomy, technical development, or a different balance between work and personal life, they will be able to evaluate opportunities according to criteria more closely aligned with their needs.

This distinction also helps separate personal objectives from expectations absorbed from the surrounding environment. A person may pursue a particular position, salary, or lifestyle because these outcomes are socially valued, without having examined whether they correspond to their priorities. This does not mean that external expectations should be ignored, but that they need to be understood in relation to what the person considers relevant. A consistent objective does not need to be independent of social, family, or professional influences; it needs to be sufficiently conscious for those influences to be evaluated rather than automatically determining the decision.

Metrics and targets can also create distortions when they begin to replace the outcomes that should guide personal life. Increasing income, accumulating certifications, expanding a professional network, or raising productivity can be useful intermediate objectives, but their value depends on what they make possible. A person may accumulate qualifications without developing skills relevant to their desired career, work longer hours without producing proportional results, or pursue higher compensation at the expense of conditions they consider essential to their well-being. The progress indicator may improve while satisfaction with the overall trajectory declines.

This does not mean that personal objectives need to be vague or that quantitative measures are inappropriate. It is possible to establish financial targets, deadlines for completing a course of study, or concrete criteria for evaluating professional opportunities. The key is to preserve the relationship between these targets and the outcomes that justify the effort. A target can serve as a practical reference without becoming the sole measure of success, especially when the decision involves dimensions such as autonomy, personal relationships, learning, everyday well-being, and the meaning attributed to work.

Personal decisions also involve objectives that change as circumstances evolve. A choice that made sense at a particular stage of life may no longer be appropriate in light of new responsibilities, opportunities, or preferences. Revising objectives does not have to represent inconsistency, provided the revision results from a clearer understanding of the situation and the outcomes that have become important. At the same time, continually changing direction without examining the reasons for each change can make it difficult to achieve any objective. The task of reflection is to distinguish a well-founded revision from a temporary reaction to difficulties along the way.

Defining objectives therefore offers an opportunity to make explicit what one wants to achieve and why it matters. Instead of following a path simply because it seems promising or meets other people's expectations, an individual can relate their choices to current circumstances, personal priorities, and the consequences they are willing to accept. The objective does not eliminate uncertainty or guarantee that the path will be easy, but it provides a reference for evaluating opportunities, recognizing trade-offs, and deciding when the direction needs to be adjusted.

5. Managing Risks and Uncertainties

Imagine a trader facing a trading opportunity in the financial market. Prices fluctuate, new information emerges continuously, and conditions that seemed favorable just a few minutes ago may change. There is no way to know for certain whether an asset's price will rise or fall, but the trader must decide whether it is worth taking a position. To do so, the trader examines the available information, assesses the fundamentals of the analysis, considers different scenarios, and determines how much they are willing to risk. They also establish conditions for closing the trade if the market behaves differently from what was expected. Their goal is not to infallibly predict the next move, but to make informed decisions and keep risk within tolerable limits.

This approach distinguishes trading from a bet based solely on the expectation that something favorable will happen. A trader may make a profit on a poorly founded trade through sheer luck and incur a loss on another that was carefully analyzed. An isolated outcome is not sufficient to assess the quality of the decision. What distinguishes a rational approach from a bet is the process used to evaluate information, recognize uncertainties, and determine acceptable exposure. Without this process, the trade ceases to be a structured decision and becomes predominantly dependent on chance.

Decision-making presents a similar challenge. We do not have complete knowledge of reality, nor can we predict with absolute precision how circumstances will unfold. Even when we gather reliable information and conduct careful analyses, aspects remain unknown or may change over time. This does not mean that all choices are equivalent or that making decisions is an exercise in guesswork. It means recognizing the limits of knowledge and learning to act without confusing what we know with what we merely assume.

In previous chapters, we put on the biologist's hat to understand the scope of a decision, the investigative reporter's hat to uncover the context, and the navigator's hat to establish objectives. Now, we assume the role of the trader to examine the uncertainty that remains even after these steps. Understanding a situation, investigating its conditions, and defining a desired outcome do not eliminate the possibility of error. We still need to assess the reliability of the assumptions underlying our reasoning, recognize the risks involved, and decide how to deal with what we cannot control.

In solution architecture, a technical choice may depend on assumptions about access volume, integration stability, infrastructure capacity, or the evolution of business needs. In project management, deadlines and budgets depend on estimates and conditions that may not materialize. In entrepreneurship, an initiative may be based on the expectation that sufficient demand exists to sustain a new product. In all these cases, planning requires us to work with assumptions, even though we cannot guarantee that they are true. The problem is not accepting uncertainty, but ignoring it or treating expectations as established facts.

Managing risks and uncertainties therefore requires examining the foundations of our reasoning, assessing relevant exposures, and choosing responses proportionate to existing conditions. Some uncertainties can be reduced through investigation and experimentation; certain risks can be avoided or mitigated; others must be accepted because no viable means of eliminating them exists. The same uncertainty may also reveal opportunities worth exploring. Just as a trader does not control the market, a decision-maker does not have complete control over the reality in which a decision will be implemented. What they can control, within certain limits, is the quality of the process through which they assess possibilities and decide how to expose themselves to them.

5.1 Recognizing the Limits of Knowledge

A trader does not need to know every factor influencing the market to execute a trade, but they do need to understand the limits of their analysis. Prices reflect a combination of information, expectations, decisions, and events whose interactions cannot always be predicted. Even an analysis supported by historical data, indicators, and market knowledge does not guarantee that the next move will follow the observed pattern. Recognizing this limitation is part of the job, not a deficiency that can be eliminated through greater confidence or more calculations.

Every decision begins with a partial understanding of reality. Some limitations stem from a lack of available information; others arise from the difficulty of interpreting existing data or from the dynamic nature of the environment itself. A decision made today may encounter tomorrow conditions that differ from those underlying its formulation. Therefore, the quality of an analysis should not be measured by the confidence with which its conclusions are presented, but by the consistency of its foundations and awareness of what remains unknown.

This distinction is particularly important when past experience creates an excessive sense of confidence. A trader may have correctly identified a market behavior pattern on several occasions and conclude that it will continue to repeat itself. Historical performance provides useful evidence, but it does not guarantee that the same conditions will persist. Changes in liquidity, the economic environment, or participant behavior may alter the observed dynamics. Experience helps formulate expectations; it does not turn those expectations into certainties.

Recognizing the limits of knowledge also does not mean assigning equal weight to every possibility. Some hypotheses are supported by stronger evidence than others, and certain estimates may be more reliable when relevant data and relatively stable conditions are available. The important point is not to confuse probability with a guarantee. An event may be more likely than its alternatives and still not occur in a particular situation. Likewise, an unlikely outcome may occur without necessarily meaning that the analysis that considered it unlikely was flawed.

In practice, recognizing these limits means avoiding both unjustified confidence and paralysis in the face of the unknown. We do not need to know everything to make a decision, but we do need to understand which gaps may compromise the choice and which can be tolerated. The trader translates this awareness into exposure limits and exit conditions; in other domains, it may translate into additional validations, safety margins, execution alternatives, or criteria for reviewing a decision. In all cases, the goal is to act in proportion to the available knowledge without demanding a level of certainty that reality cannot provide.

5.2 Examining Assumptions

Every analysis depends on foundations that we do not always examine with equal care. Some statements are supported by evidence, others are inferences drawn from what we observe, and many are assumptions we accept in order to move forward. The problem is not working with assumptions, which is unavoidable when knowledge is incomplete, but forgetting that they are assumptions and building on them without considering what would happen if they were false.

A useful analogy comes from mathematics, whose formal systems are built on axioms: propositions accepted as starting points and used to derive other statements according to logical rules. Conclusions depend on the adopted foundations and the inference rules used. Changing the axioms may lead to a different formal system, with distinct properties and conclusions. The analogy with decision-making is not exact, because the assumptions used to interpret the world are statements about a reality that must be investigated. Nevertheless, it highlights an important point: reasoning can be internally consistent and produce conclusions that follow coherently from its assumptions without those assumptions corresponding to reality.

Consider a trader who expects an asset to appreciate because they have identified a pattern that, in their analysis, indicates an upward trend. The decision may depend on several assumptions: the observed pattern remains relevant under current conditions, the available information is sufficient to support the interpretation, and no significant change will occur that invalidates the analysis before the trade is closed. Even if the reasoning is consistent, these conditions may not materialize. The market may react unexpectedly, and the asset may lose value. The problem is not necessarily a logical error, but the possibility that the assumptions used do not adequately represent actual conditions.

The same situation arises in business decisions. A startup may project its growth based on the assumption that demand for its product will continue increasing at the observed rate. The financial projections may be mathematically correct, and the plan may have been prepared rigorously. However, if demand slows because the market is nearing saturation or a competitor offers a more attractive alternative, the expected results will no longer be supported by actual conditions. The calculations do not need to contain errors for the decision to be inappropriate; it is enough for a fundamental assumption not to hold.

For this reason, it is important to distinguish facts, hypotheses, and assumptions. Facts are statements supported by available evidence, although they may need to be revised when more reliable information emerges. Hypotheses are explanations or possibilities that we formulate and can examine through observations, tests, or other forms of investigation. Assumptions are conditions that we provisionally accept to develop a line of reasoning or plan an action, even when their validity has not yet been confirmed. In practice, these categories may overlap, but distinguishing them helps us understand the level of confidence we should assign to each statement and which ones need to be verified before proceeding.

Not all assumptions require the same validation effort. Some have little influence on the decision, while others

underpin much of the reasoning. A trader may assume that the liquidity of a particular asset will be sufficient to close their position under the desired conditions. If this assumption fails in an illiquid market, exiting may be more difficult or may occur at an unfavorable price. In a company, a decision may depend on a supplier's continued availability or on the availability of a specific technology. Particular attention should be given to assumptions whose falsity would undermine the viability of the choice and for which reasonable means of obtaining additional evidence exist.

Examining assumptions also means recognizing that they can change over time. A condition considered reasonable at the beginning of a trade, project, or business initiative may cease to be valid following a market shift, regulatory change, or new evidence. Therefore, it is not enough to record assumptions during planning and treat them as permanent truths. We need to understand which conditions underpin the decision, monitor those that may change, and be willing to revise our reasoning when evidence conflicts with initial expectations.

5.3 Assessing and Managing Risks

In trading, identifying an opportunity is not enough to justify a trade. The trader must also assess how much they could lose if their analysis is wrong and whether that exposure is compatible with their resources and strategy. A favorable expectation does not eliminate the possibility of a loss, and a trade with high profit potential may be inappropriate when it requires taking disproportionate risk. The decision therefore depends not only on the expected outcome, but also on the conditions under which the risk will be assumed and the limits established to manage it.

Risk is associated with the possibility that an event or uncertain condition will affect the objectives of a decision. Identifying it requires considering what may happen, under what circumstances, and how significantly it could affect the intended outcome. The assessment may consider the probability of occurrence, the magnitude of the impact, the level of exposure, and the ability to detect or respond to the event. These factors help establish priorities, but they should not be treated as exact measures when the evidence does not support such precision. Assigning a numerical probability to an event does not automatically make the estimate reliable.

Consider the possibility of a meteor striking a company's headquarters and destroying it. The event is conceivable, and its consequences could be severe, but its probability for an ordinary corporate headquarters is extremely low. It would be possible to imagine specific protective measures against this scenario, but the costs would probably be disproportionate to the risk. The company does not need to ignore the possibility of a catastrophic event to recognize that devoting unlimited resources to preventing every imaginable threat would be incompatible with rational management. Risk assessment serves precisely to distinguish what deserves priority from what does not justify specific mitigation under the available conditions.

This does not mean that rare risks should always be disregarded. The decision depends on the combination of probability, severity, exposure, and recovery capacity, as well as applicable legal obligations and safety standards. In certain activities, even unlikely events require rigorous controls because their potential impact is unacceptable. In others, the most appropriate protection is not to try to prevent every specific event, but to reduce the organization's overall vulnerability. Maintaining backups in another region, for example, does not prevent the physical destruction of headquarters, but it can preserve essential information and facilitate the resumption of operations after different types of incident.

Responses to risks can take different forms. Avoiding a risk means abandoning or modifying the activity that produces it. Reducing a risk means lowering its probability, limiting exposure, or preparing mechanisms that lessen its severity. Sharing or transferring it may involve contracts, insurance, or agreements with other organizations, although these measures do not necessarily eliminate all possible liabilities and losses. Finally, accepting a risk means consciously deciding that the remaining exposure is tolerable in light of the expected benefits, available alternatives, and costs of other measures.

The trader applies this logic by determining position size, setting loss limits, and deciding under what conditions a trade will be closed. These mechanisms do not guarantee that the planned outcome will occur, nor do they eliminate all execution risks, such as abrupt price fluctuations or lack of liquidity. Their purpose is to establish discipline around exposure and reduce the possibility that a single decision will jeopardize more resources than considered acceptable. Risk management does not make a trade infallible; it makes the relationship between expected gains and the loss one is willing to bear more explicit.

Acceptance deserves particular attention because it can be confused with negligence. Ignoring a risk means

failing to consider it adequately; accepting it means recognizing it, evaluating treatment options, and deciding that none offers sufficient benefit to justify its cost or restrictions. This decision can be rational even when the event would have severe consequences, provided that the analysis is proportionate, tolerance limits are clear, and no safety obligations preclude the choice. In organizational contexts, it is also important for acceptance to be authorized by someone with the authority to assume the exposure, rather than remaining implicit in decisions made without assessment.

Risk management does not end when a response is chosen. The conditions underlying the assessment may change, new information may alter estimates, and measures that were initially sufficient may no longer protect the operation. Therefore, the most relevant risks need to be monitored, along with the assumptions on which they depend. In trading, this means observing whether the conditions supporting the analysis remain in place; in projects and organizations, it may mean monitoring indicators, critical dependencies, and environmental changes. Review does not need to be continuous in every aspect, but it should occur when relevant signs indicate that exposure has changed.

The purpose of this process is not to produce an endless list of threats or create the expectation that every eventuality can be predicted. It is to focus attention on relevant risks, choose proportionate measures, and recognize the limits of possible protection. In some situations, the most prudent decision will be to invest in prevention; in others, it will be to prepare for recovery or consciously accept residual exposure. What matters is that the choice results from deliberate assessment rather than from the assumption that something we do not expect to happen will necessarily not happen.

5.4 Recognizing Opportunities in Uncertain Scenarios

Uncertainty is often associated with threats because what we do not know can undermine plans and expected outcomes. However, the same lack of knowledge that prevents us from predicting certain events also means that opportunities may exist without having been identified. In financial markets, a change in economic conditions, prices, or participants' expectations may create opportunities, but it may also invalidate previous interpretations. The existence of a favorable possibility is not sufficient to justify a trade; the underlying rationale and the exposure required to capitalize on it must be assessed.

Recognizing opportunities in uncertain scenarios requires a different approach from simply looking for risks. While risk analysis seeks to understand what may undermine objectives, opportunity exploration aims to identify favorable conditions that could generate value. In both cases, it is necessary to work with hypotheses, evaluate evidence, and recognize the limits of estimates. The difference lies in the purpose of the investigation: rather than merely protecting the expected outcome, we seek to discover alternatives that could enhance it or reveal paths not yet considered.

In entrepreneurship, for example, a company may identify an emerging need before sufficient evidence exists to estimate the size of the market. The absence of certainty does not mean the opportunity should be dismissed, but neither does it justify a large investment based solely on enthusiasm. A more careful approach may involve small-scale experiments, interviews with potential customers, prototypes, or tests of willingness to pay. These initiatives make it possible to turn some uncertainty into knowledge, limit the initial investment, and create conditions for deciding whether it is worth proceeding.

This logic also applies to projects and technology solutions. A flexible architecture may allow future capabilities to be incorporated even when they have not yet been fully specified. A team may test a new approach in a limited context before adopting it across the entire organization. A company may set aside resources to explore a possibility without committing the entire budget in advance. In situations like these, the decision does not have to be between acting with complete confidence and remaining inactive. It can create conditions for learning, preserve alternatives, and expand commitment as the evidence becomes more favorable.

However, exploring opportunities also requires criteria. Not every possibility deserves investment, just as not every risk needs to be eliminated. It is necessary to assess the opportunity's potential significance, the resources required to investigate it, the reversibility of initial decisions, and the value of the information that can be obtained. When the cost of an experiment is limited and the learning can guide future choices, testing a hypothesis may be more appropriate than trying to predict every outcome in advance. When an investment is irreversible or exposure is high, it may be necessary to gather stronger evidence before proceeding.

Managing uncertainty involves balancing protection and exploration. An excessively defensive approach may

preserve existing resources but also prevent initiatives capable of generating value. An excessively optimistic approach may lead to disproportionate commitments to opportunities that have not yet been validated. The trader needs to consider both the possibility of loss and the expected return; in other domains, the same discipline helps compare the potential value of an initiative with the exposure required to realize it. In both cases, the decision should be based on explicit analysis without treating uncertain forecasts as guarantees.

5.5 Application in Project Management

In project management, uncertainties arise from the definition of scope through the estimation of timelines, costs, resources, and acceptance criteria. A schedule depends on assumptions about productivity, people's availability, requirements stability, and the execution of related activities. A budget depends on estimates of effort, prices, contracts, and supply conditions. Even when planning uses historical data and consistent methods, these conditions may change or prove different from what was expected.

Risk management makes these dependencies explicit and helps assess which ones deserve attention during execution. A critical activity may depend on a supplier for which no immediate alternative exists; a deliverable may assume that an integration will be available by a certain date; an effort estimate may rely on requirements that have not yet been fully detailed. By identifying these points, the team can seek additional information, create contingency buffers, prepare alternatives, or revise the plan before uncertainty compromises its ability to meet the project's objectives.

Not every risk can be eliminated through planning. Some conditions depend on external decisions, regulatory changes, or events beyond the team's control. In such cases, the response may involve appropriate reserves, contingency plans, monitoring relevant signals, or consciously accepting residual exposure. The choice depends on the nature of the risk and the project's ability to absorb deviations without compromising its objectives.

Assumptions also need to be monitored during execution. An approved plan does not turn its estimates into facts, and the occurrence of changes does not necessarily mean the original planning was inadequate. It means the initial decision was made using the information available at the time and that new evidence may require adjustments. Consistent project management does not merely compare actual results with the plan; it also examines whether the conditions that justified the plan remain valid.

5.6 Application in Solution Architecture

In solution architecture, many technical decisions depend on conditions that cannot yet be fully verified when the solution is designed. Choosing a technology may assume that it will continue to be supported, that it will be compatible with existing systems, or that it will meet anticipated performance requirements. A distributed architecture may depend on estimates of access volume, network latency, and the behavior of external services. An integration may be planned based on interface contracts that are still subject to change.

These assumptions need to be examined according to their relevance to the solution. When a choice depends on a little-known technology or a critical performance requirement, a proof of concept can provide evidence before a larger investment is made. When there is a dependency on an external service, it is advisable to assess its availability, operational limits, and alternatives in case it becomes unavailable. When future evolution is difficult to predict, decisions that preserve options for change can reduce exposure to premature commitments, provided that the additional flexibility does not introduce unnecessary complexity.

Risk management also requires distinguishing technical feasibility from operational suitability. A solution may work in a test environment and encounter difficulties when subjected to real-world volume, variability, and operating conditions. Therefore, load testing, monitoring, recovery strategies, and requirements validation are ways to obtain evidence and reduce certain uncertainties. Even so, it is impossible to guarantee that all future behaviors have been anticipated. The architecture needs to incorporate mechanisms to detect failures, limit their propagation, and restore operations when necessary.

In some cases, the most appropriate decision will be to accept a dependency or limitation because eliminating it would require disproportionate costs and complexity. This acceptance should be explicit and consistent with the organization's security, availability, and compliance requirements. The architect's role is not to choose a solution free of all risk, but to understand the relevant uncertainties, make dependencies visible, and guide choices whose limitations are known and manageable.

5.7 Application in User Experience

In user experience, uncertainties arise when interpreting needs, understanding people's behavior, and determining how they respond to an interface or service. Research and interviews can reveal expectations and difficulties, but participants' statements do not always correspond exactly to behavior observed in real-world situations. Similarly, a team may formulate hypotheses about whether a feature is understandable without knowing how different audiences will use it.

Treating these hypotheses as facts can lead to design decisions based on an incomplete view of the user. A team may believe that a particular step is unnecessary because it seems technically simple, while users struggle to understand its requirements. It may also assume that a new feature will increase engagement without evidence that it addresses a relevant need. Validation through usability testing, prototypes, and experiments makes it possible to examine these assumptions before committing resources to a broader implementation.

Not every uncertainty can be resolved before launch. Behavior may vary across segments, usage contexts, and conditions that were not reproduced during research. In such cases, the team can adopt a gradual rollout, monitor relevant indicators, and preserve the ability to adjust or reverse changes. This approach makes it possible to learn from real-world usage without assuming that results observed in a limited group will necessarily be replicated across the entire population.

Risk management in user experience also involves recognizing situations in which an error could undermine trust, exclude certain audiences, or make important tasks more difficult. The severity of these possibilities depends on the context and should not be assessed solely by expected frequency. A rare failure in a critical feature may require more attention than a frequent but easily reversible problem in a secondary feature. The analysis needs to consider the purpose of the interaction, users' needs, and the acceptable limits of the experience being offered.

5.8 Application in Entrepreneurship

Entrepreneurship involves decisions made under uncertainty because, in many cases, the business does not yet have enough historical data to support reliable projections. Market size, customers' willingness to pay, acquisition costs, retention, and the ability to scale operations may depend on hypotheses that can only be partially validated after the initiative reaches the market. A business plan can organize these hypotheses and produce coherent projections, but it does not turn estimates into guarantees of performance.

The analogy with mathematical axioms is particularly useful in this context. A financial projection may be structurally correct and still depend on an incorrect assumption about demand, competitor behavior, or customers' willingness to switch solutions. For this reason, it is important to distinguish calculated results from the conditions that must hold for those results to be plausible. The analysis should ask not only whether the numbers make sense, but also what evidence supports the assumptions used and how the initiative will be affected if they do not hold.

One way to deal with these uncertainties is to validate the most important hypotheses before making commitments that are difficult to reverse. Interviews, prototypes, pricing tests, and early versions of the product can provide evidence about whether the problem exists and whether the market is willing to adopt the solution. The goal is not to eliminate all doubt before starting, but to reduce the uncertainties that could invalidate the business model and limit exposure while learning remains insufficient.

Entrepreneurs also need to recognize that uncertainty can represent an opportunity. An underserved market may offer room for a new value proposition, although the absence of competitors is not, in itself, evidence of demand. An emerging technology may enable a different business model, but it may also introduce dependencies and costs that are not yet known. The decision requires balancing the potential to create value against the quality of the evidence, the cost of learning, and the ability to withstand unfavorable outcomes.

This logic favors progressive commitments: invest enough to test a hypothesis, evaluate the results, and expand the initiative when the evidence justifies additional resources. Rather than relying on a single forecast, the entrepreneur builds a learning process that allows assumptions to be revised and the business to adapt. The discipline lies not in avoiding every possibility of failure, but in understanding the main sources of uncertainty and not committing more resources than the situation warrants.

5.9 Application in Professional Career and Personal Life

Career and personal life decisions also depend on assumptions about conditions we do not fully control. Changing jobs may assume that a new organization will offer better development opportunities; pursuing further education may depend on the expectation that it will expand career prospects; moving to another city may involve assumptions about the cost of living, adaptation, and the quality of personal relationships. In all these cases, the decision combines available information, expectations, and aspects of the future that remain uncertain.

Examining these assumptions helps distinguish what has already been confirmed from what still needs to be investigated. Talking to people who know the organization, researching market conditions, assessing the available budget, or trying an activity before making a larger commitment can reduce certain uncertainties. These actions do not guarantee the desired outcome, but they make it possible to replace some assumptions with evidence and gain a better understanding of the conditions involved.

There are also risks that cannot be eliminated without giving up the initiative itself. Staying in a familiar situation may preserve stability, but it involves the risk of missing growth opportunities. Changing direction may open up new possibilities, but it may also require resources, adaptation, and tolerance for uncertain outcomes. There is no universally safe choice: even the decision not to act leaves a person exposed to conditions that will continue to evolve. The analysis needs to consider both the exposure associated with change and that associated with staying put.

In these situations, it may be useful to preserve alternatives and avoid irreversible commitments before the conditions are sufficiently understood. Building a financial reserve, developing transferable skills, testing an activity alongside existing commitments, or establishing criteria for reassessing a decision are ways to limit exposure without requiring absolute certainty. The suitability of these measures depends on personal circumstances, available resources, and the importance assigned to different objectives.

Managing uncertainty in personal life does not mean turning every choice into a formal risk analysis or attempting to calculate every possibility numerically. It means recognizing when a decision depends on fragile expectations, seeking information in proportion to its importance, and accepting that part of the outcome will remain beyond our control. Understanding this condition makes it possible to decide without demanding impossible guarantees and to change course when new evidence shows that the initial assumptions no longer support the chosen direction.

6. Predicting Direct and Indirect Consequences

Imagine a chess player facing a complex position on the board. They identify a promising move, but they cannot evaluate it solely by its immediate effect. They need to consider how the opponent might respond, which alternatives will remain available, and how each response might alter the position of the pieces. A move that initially appears advantageous may expose an important piece, create a vulnerability, or lead to a position that will be difficult to escape. Likewise, a move that produces no immediate gain may prepare a combination of moves capable of opening up new possibilities a few steps ahead. The chess player must therefore analyze not only what their move produces, but also what it makes possible or likely in subsequent moves.

This way of thinking helps us understand an important dimension of decision-making. A choice produces more than an immediate result. It can commit resources, create dependencies, establish constraints, eliminate paths, or make certain changes more costly. Its effects can also propagate to other parts of the situation, affecting people, processes, resources, and decisions that initially appeared independent. Evaluating an alternative requires considering not only the outcome it promises to produce, but also how it changes the conditions under which subsequent events will unfold.

If, in the previous chapter, we assumed the role of a trader to understand how to manage risks and uncertainties, we now put on the chess player's hat to examine the ramifications of our choices. The trader needs to assess the exposure they are willing to accept in the face of uncertain outcomes; the chess player needs to anticipate how a move changes the board and constrains subsequent responses. The two perspectives complement each other. Even after recognizing uncertainties and assessing risks, we still need to understand how alternatives can change the landscape, produce cascading effects, and influence future possibilities.

In solution architecture, this need arises when a technical choice is evaluated solely on the basis of the immediate benefit it offers. Replacing a component may improve an application's performance, but it may also introduce a new dependency, require skills the team does not yet possess, or make future upgrades more difficult. In project management, accelerating a delivery may require reallocating people, changing the sequence of activities, or compromising other commitments. In entrepreneurship, lowering a product's price may increase sales, but it may also change margins, customer expectations, and competitors' responses. In all these cases, the decision must be examined in relation to the set of conditions it changes, rather than only the most visible outcome.

Predicting consequences, however, does not mean knowing in advance everything that will happen. Just as a chess player cannot analyze every possible sequence indefinitely, decision-makers must work with an analysis horizon proportional to the importance of the choice and the complexity of the situation. The purpose is to identify relevant effects, understand how they may relate to one another, and assess how each alternative changes the conditions for future action. This analysis makes it possible to recognize trade-offs, preserve important options, and avoid having an apparently advantageous choice compromise objectives that will become evident only later.

6.1 Understanding Direct and Indirect Consequences

A direct consequence follows relatively immediately from the decision made. If a company contracts a cloud computing service, it begins using infrastructure provided by a third party and assumes the costs and conditions associated with the contract. If a project reduces its scope, certain deliverables are no longer part of the planned work. If a person accepts a new professional position, they assume the responsibilities and commitments associated with the role. These effects are usually easier to identify because they are closely linked to the choice and often form part of the reasons behind the decision.

Indirect consequences arise through intermediate relationships. Contracting the cloud service may require changes to the architecture, security, and team skill sets. Reducing a project's scope may change the usefulness of the delivered product or require renegotiations with other departments. Changing professional positions may alter family routines, commutes, and future development opportunities. These effects are not necessarily unpredictable, but they depend on connections that may not be apparent when the decision is examined in isolation.

The distinction between direct and indirect effects is not absolute. An effect considered indirect in one analysis may be treated as direct in another, depending on the boundaries established for examining the situation. The important point is to recognize that the effects of a choice do not end with the first change it produces. An analysis restricted to the immediate outcome tends to overlook relevant relationships and may overestimate the benefits of an alternative or underestimate its requirements.

The chess player faces a similar problem. A move may directly attack an opponent's piece, but its value also depends on the responses it provokes, the pieces that become vulnerable, and the positions that become accessible. The immediate effect is only part of the analysis. To understand the move, it is necessary to examine how it changes the relationships between the pieces and how those relationships constrain subsequent moves.

In practice, this means that each alternative must be evaluated at more than one level. First, we identify what it changes directly. Then, we examine which changes may follow from those alterations, which stakeholders may react, and which conditions will be modified. We do not need to construct an endless chain of possibilities. We need to deepen the analysis until the main decision-relevant ramifications are sufficiently clear, taking into account the time, information, and resources available.

6.2 Analyzing Scenarios and Sequences of Effects

In chess, analyzing a move involves considering different responses from the opponent. A sequence that appears favorable if the opponent makes a particular move may become disadvantageous in response to another. For this reason, the chess player does not examine only the continuation they hope to find; they seek to identify plausible alternatives and assess how each one changes the position. The more important the move, the greater the need to verify whether its apparent advantage depends on a specific response from the opponent.

In decision-making, scenario planning serves a similar purpose. A scenario describes a possible evolution of a situation based on certain conditions and choices. It is not a guaranteed prediction, but a way to explore how reality might unfold and what effects might arise in each case. A company considering entry into a new market may examine how the initiative would perform under strong customer adoption, demand below expectations, or a more intense competitive response. Each scenario makes it possible to observe different aspects of the decision and identify conditions that might require a different response.

The usefulness of scenarios depends on the quality of the assumptions used. If all analyses assume, for example, that customers will adopt the product quickly, they will not adequately represent the uncertainty surrounding the initiative. It is necessary to consider plausible alternatives, including those that challenge initial expectations. This does not require imagining every conceivable event, but rather exploring relevant variations in the conditions underpinning the decision.

It is also important to examine sequences of effects, rather than only isolated end states. A change may trigger a response that, in turn, alters the conditions at a later stage. A price reduction may attract new customers, provoke a reaction from competitors, and require further reductions to maintain market share. The final outcome will depend on the interaction between these moves, not just on the initial effect on sales. Analyzing sequences helps identify situations in which an immediate advantage may lose its strength or create subsequent difficulties.

However, the ability to analyze scenarios is limited. The greater the number of variables, stakeholders, and possible responses, the greater the number of sequences that can be considered. Attempting to examine all of them may consume resources without proportionately improving the decision. The challenge is to select scenarios that help distinguish between alternatives, reveal significant vulnerabilities, or could change the choice. Like a chess player who focuses on the most relevant lines of play, decision-makers need to direct their analysis toward ramifications that can genuinely make a difference.

6.3 Recognizing Trade-offs and the Space of Possibilities

A decision can improve one aspect of a situation while making another more difficult. This relationship is known as a trade-off: a concession made in favor of one objective, feature, or possibility at the expense of another. Improving a solution's performance may increase its cost. Shortening a project's timeline may require more resources or increase exposure to failures. Increasing an architecture's flexibility may introduce additional complexity. In many cases, no alternative simultaneously maximizes all relevant objectives.

The chess player deals with this type of choice continuously. A move may gain space on the board but leave a piece unprotected. Another may preserve the king's safety but reduce offensive initiative. The value of each move depends on the position and the strategic objectives being considered. It is not enough to identify a move's benefit; it is necessary to assess what is being given up in exchange and whether that concession is acceptable in light of future possibilities.

The same reasoning applies to the space of possibilities associated with a decision. This space consists of the

alternatives that remain available at a given moment, considering existing resources, commitments, dependencies, and constraints. A choice may reduce this space by eliminating paths or making certain changes more difficult. It may also expand it by creating capabilities, making resources available, or generating information that did not previously exist. Therefore, the consequences of a decision must be evaluated both in terms of what they produce and the possibilities they preserve, restrict, or create.

Consider a company that decides to adopt a proprietary technology platform to accelerate product development. The choice may provide ready-made tools, specialized support, and lower initial effort. In return, it may increase dependence on a vendor, restrict migration freedom, and raise future switching costs. An alternative based on open technologies might require more work initially but preserve greater flexibility. There is no universally better answer: the decision depends on the objectives, available resources, team skills, and importance assigned to future autonomy.

This analysis also helps explain optionality, that is, the ability to keep alternatives available for subsequent decisions. Some choices preserve greater freedom to adapt; others commit resources or establish dependencies that make future changes more difficult. Preserving options can be valuable when there is significant uncertainty about future needs, but this flexibility is not free. Maintaining multiple alternatives can increase costs, complexity, and coordination effort. The objective is not to maximize the number of options under all circumstances, but to understand which ones are worth preserving.

6.4 Considering Reversibility, Dependencies, and Time

Not all decisions have the same degree of reversibility. Some can be undone with little effort, while others require additional investments, renegotiations, or significant changes to reverse. A team can test a tool in a limited project and abandon it if the results are unsatisfactory. By contrast, adopting a platform that comes to support critical processes, store large volumes of data, and integrate multiple systems can make replacement far more costly.

Reversibility changes how a decision should be evaluated. When a choice can easily be revisited, it may be reasonable to proceed with incomplete information and learn from experience. When it requires commitments that are difficult to undo, it becomes more important to examine assumptions, compare alternatives, and understand the ramifications before proceeding. This does not mean that irreversible decisions should be avoided, but that they require analysis commensurate with the cost of a potential change.

Dependencies also influence future possibilities. A decision may tie an organization to a vendor, a technology, a contract, or a specific skill set. Over time, other decisions may be built on top of that choice, increasing the cost of replacement. This phenomenon is known as path dependency: previous conditions and choices influence which paths become accessible later. The current situation depends not only on what would theoretically be possible, but also on the path taken to reach it.

The chess player observes similar dynamics when a sequence of moves creates a position that favors some strategies and makes others more difficult. A poorly positioned piece may require several moves to reposition; an apparently minor exchange may permanently alter the balance of the game. Previous moves do not determine every future move, but they change the conditions under which those moves will be made.

Time, in turn, also changes the space of possibilities. Delaying a decision may allow new information to emerge and reduce uncertainty, but it may also consume resources, prolong a problem, or cause an opportunity to disappear. In other cases, acting too early may commit resources before conditions are sufficiently clear. The analysis must consider not only what happens if an alternative is chosen, but also what changes while the decision remains open.

Therefore, deliberately choosing not to decide can differ from simply failing to make a decision, but neither situation guarantees that the current state will be preserved. The context continues to evolve, other people may act, and certain alternatives may become more expensive or unfeasible. The decision to wait must also be examined in terms of what one expects to learn, how long it will take to obtain new information, and which opportunities may be lost during the wait.

6.5 Communicating Consequences and Coordinating Decisions

A decision that affects other people or groups cannot be treated merely as an individual choice. Its effects need to be understood by those who will have to implement it, adapt their work, or live with the changes it produces. Communication becomes an important part of managing consequences because it helps clarify what was decided, why a particular alternative was chosen, what changes are expected, and what responsibilities follow from it.

In project management, for example, a scope change may alter the work of different teams, require a new sequence of deliverables, and change previously established commitments. If these changes are communicated incompletely, each department may continue operating under different assumptions, generating priority conflicts and rework. The decision may be technically sound, but its implementation will be compromised if the affected people do not understand how it changes their responsibilities and dependencies.

Communication also makes it possible to reveal consequences that were not initially considered. An implementation team may identify operational constraints that were not apparent during the analysis. A user may point out difficulties resulting from an interface change. A business partner may clarify that a particular contractual condition affects the feasibility of an alternative. Listening to these perspectives broadens our understanding of the situation and may lead to revising the choice before commitments become difficult to reverse.

This does not mean that every decision must be negotiated with everyone involved or that communication eliminates disagreements. People and groups may have different objectives, interpret the same facts differently, and assign different values to trade-offs. The purpose is to make relevant effects understandable, explain the reasons behind the choice, and coordinate the actions needed to implement it. When legitimate conflicts exist, they must be acknowledged and addressed according to the responsibilities and criteria guiding the decision.

Communication must also keep pace with how the situation evolves. If conditions change, assumptions are no longer confirmed, or observed effects differ from expectations, the affected people need to understand what has changed and why the decision is being reconsidered. In this way, communication is not limited to announcing a decision that has already been finalized. It helps align expectations, coordinate responses, and maintain a shared understanding of the conditions guiding subsequent actions.

6.6 Application in Project Management

In project management, a decision rarely affects only the activity in which it was made. Changing the scope can modify the schedule, resource allocation, acceptance criteria, and coordination between teams. Shortening a delivery timeline may require parallelizing activities that would otherwise be sequential, increase the need for validation, or reassign professionals from other workstreams. An apparently localized change can therefore produce effects across the project as a whole.

Predicting these consequences requires examining dependencies between activities and understanding how a change propagates through the plan. Before approving a change, the project manager needs to assess not only the expected benefit, but also which deliverables will be affected, which commitments will need to be renegotiated, and which constraints may arise. The schedule is not merely a list of dates; it represents relationships between activities, resources, and the conditions necessary for work to progress.

Trade-offs become especially visible when schedule, cost, scope, and quality come into conflict. Accelerating a delivery may require more resources, reducing scope, or increasing exposure to problems. Preserving all deliverables within the original timeline may be unfeasible if additional requirements or technical difficulties arise. The purpose of the analysis is not to assume that all objectives can be maintained, but to make the necessary concessions explicit and assess which alternatives best preserve the highest-priority outcomes.

Reversibility also matters. A decision made early in a project, when few dependencies exist, may be relatively easy to adjust. The same change at an advanced stage may require rework, contractual renegotiation, and team reorganization. For this reason, decisions with significant potential to propagate deserve special attention before other activities are built on top of them.

Finally, the consequences must be communicated to stakeholders. Planning changes require teams, sponsors, suppliers, and customers to understand what has changed and how it affects their responsibilities. Consistent management combines alternative analysis, dependency assessment, and clear communication to prevent a local decision from creating misalignment elsewhere in the project.

6.7 Application in Solution Architecture

In solution architecture, technical decisions create relationships that can influence system evolution for many years. Choosing a technology, defining service boundaries, establishing a data model, or adopting a vendor affects more than the immediate implementation. These choices can determine how new features will be incorporated, how failures will be isolated, which skills will be required, and how much effort will be needed to replace components in the future.

An architecture based on independent services, for example, may allow parts of the system to evolve separately. However, this independence depends on well-defined contracts, observability, communication mechanisms, and the operational capacity to manage distributed components. The decision may increase flexibility in certain respects while introducing coordination costs and operational complexity. Evaluating only the benefit of separation, without examining its requirements, produces an incomplete view of the alternative.

The same reasoning applies to the choice between building and buying a solution. Purchasing an off-the-shelf product may reduce implementation time but create vendor dependency and limit the ability to customize features. Developing internally may provide greater control but require investment, maintenance, and skills that the organization will need to retain. The analysis must consider both current requirements and changes that may be necessary, without assuming that maximum flexibility is always desirable.

Architectural decisions can also create path dependencies. When multiple systems begin to depend on an interface, a data model, or a central service, replacing it may require coordinated changes throughout the solution. This coupling may be acceptable when it simplifies operations and meets business needs, but it must be recognized as part of the consequences of the choice. The issue is not to avoid all dependencies, which is impractical, but to understand which dependencies are deliberate, which constrain evolution, and how much it will cost to change them.

Anticipating these consequences does not require predicting every future requirement. It requires identifying decisions that may significantly limit the solution's evolution, evaluating alternatives, and preserving flexibility where it has concrete value. A proof of concept, a modular architecture, or gradual adoption of a technology can help generate learning before expanding the commitment. The most appropriate architectural decision is one whose benefits, costs, and future constraints are compatible with the organization's objectives and conditions.

6.8 Application in User Experience

In user experience, a change to the interface can alter more than the interaction directly affected. A new step in a flow may increase the amount of information collected, but it may also increase the effort required from the user. A simplification may reduce the number of actions needed, but remove guidance that is important for people unfamiliar with the system. An automated recommendation may speed up a task, but reduce transparency regarding how a particular result was produced.

Predicting these consequences requires examining how the change relates to user needs, other parts of the experience, and product objectives. A feature that improves an isolated metric may harm the experience elsewhere. Increasing the number of clicks on a recommendation, for example, does not necessarily demonstrate that it is useful; it may reflect curiosity, difficulty understanding it, or a lack of clear alternatives. Results must be interpreted in light of observed behavior and the conditions under which it occurs.

Indirect consequences may also emerge over time. An interface that encourages users to complete a task quickly may reduce errors in a particular context but make it harder to review important information. A change to the navigation system may improve the experience for frequent users while making the product more difficult to use for occasional users. Testing with different user profiles and contexts helps identify these effects before rolling out the change more broadly.

Reversibility is useful in this domain. Instead of immediately replacing an experience used by the entire customer base, the team can test the new approach with a limited group, monitor metrics, and gather qualitative evidence. This gradual implementation preserves the ability to adjust or reverse the change if the results reveal previously unanticipated problems. However, experimentation must respect privacy, accessibility, and applicable ethical constraints, especially when decisions affect vulnerable users or high-stakes tasks.

Communication is also part of the experience. Changes to features, rules, or flows need to be understandable to the people affected. When a change alters expected behavior, the absence of an explanation can cause confusion

and reduce trust in the product. Predicting consequences in user experience therefore means considering not only how the interface works, but also how people will understand, use, and incorporate the change into their activities.

6.9 Application in Entrepreneurship

In entrepreneurship, decisions about pricing, positioning, sales channels, hiring, and expansion can alter a business's trajectory. A price reduction may increase demand but also decrease the contribution margin and change the product's perceived value. Entering a new market may expand the potential customer base but require regulatory adaptations, additional support, and a different sales structure. Hiring someone may increase execution capacity while also raising fixed costs and reducing financial flexibility.

Evaluating these choices requires examining how immediate effects may trigger new needs or responses. A company that rapidly expands its customer base may need to invest in customer service, infrastructure, and processes that were unnecessary at a smaller scale. If growth is not accompanied by sufficient resources and operational capacity, expansion may put pressure on service quality and compromise retention. Increased demand is a favorable outcome, but its ramifications must be considered to determine whether the business can sustain it.

The responses of other stakeholders also matter. Competitors may react to a new pricing strategy, suppliers may change commercial terms, and customers may adjust their expectations based on the adopted positioning. Entrepreneurs do not make decisions in a static environment; their choices change the behavior of other participants, who in turn alter the available conditions. Scenario analysis helps explore these interactions without assuming that other stakeholders will remain passive.

Optionality can be particularly valuable for businesses that are still taking shape. Testing a market segment before expanding operations, launching a limited version of the product, or establishing partnerships that do not impose excessive commitments can preserve alternatives while the company learns. On the other hand, keeping too many possibilities open indefinitely can scatter resources and prevent the business from developing a consistent value proposition. The challenge is to preserve options that could change future decisions without compromising the ability to execute the chosen strategy.

Predicting consequences in entrepreneurship does not mean building a plan capable of anticipating every development in the market. It means understanding the main relationships between decisions, identifying commitments that may constrain the business, and assessing how different scenarios would change the strategy. This analysis helps distinguish opportunities that can be pursued progressively from those that require substantial commitments before sufficient evidence is available.

6.10 Application in Professional Careers and Personal Life

Personal decisions also produce effects that extend beyond the immediate objective. Accepting a job may increase income and expand development opportunities, but change daily routines, the time available for family, and where one lives. Starting a training program may open professional paths but require financial resources and reduce the time devoted to other activities. Moving to another city may offer new opportunities but change support networks, living costs, and personal relationships.

These consequences do not need to be treated as reasons to avoid change. They help clarify what each alternative requires and how it relates to a person's other objectives. A choice that favors professional growth may conflict with financial stability or availability for family commitments. The value of the alternative depends on the priorities involved and the concessions the person is willing to make, not just on the most obvious benefit.

Time and reversibility also influence these decisions. A change that can be tried temporarily offers different conditions from one that requires selling a property, giving up a source of income, or assuming long-term financial commitments. When possible, testing an alternative on a limited scale can reveal important information before a more difficult-to-reverse change. This may involve trying a side activity, talking to people who already live the desired reality, or concretely assessing the costs of a transition.

It is also necessary to consider how the decision affects other people. A career change may alter shared responsibilities, require reorganizing family routines, or modify commitments made to others. Communicating the reasons for the choice, clarifying what may change, and listening to the concerns of those involved does not

eliminate conflicts, but it allows the consequences to be better understood and the necessary adjustments to be coordinated.

Finally, the analysis must include the cost of remaining in the current situation. Delaying a change may preserve short-term stability, but it may also limit skills development or cause an opportunity to disappear. Inaction does not freeze the context or keep the same alternatives available indefinitely. Considering the ramifications of acting and waiting helps build a more complete view of the choices, without assuming that there is a path free of concessions or uncertainties.

7. Dealing with the Evolution of a Decision

Imagine a chronicler who follows the development of a city of which they are a part. They witness the construction of new roads, participate in discussions about changes to neighborhood organization, and observe how the arrival of companies transforms the daily lives of residents. When recording these events, they seek to preserve not only what happened, but also the circumstances in which each change occurred, the motivations that influenced it, and the effects that began to emerge. Because they are embedded in the reality they follow, the chronicler also experiences these transformations and must update their understanding as new facts emerge. Their records preserve the city's trajectory and help explain how each stage contributed to shaping the current situation.

Dealing with the evolution of a decision requires a similar perspective. A choice changes the situation in which subsequent choices will be made. Resources may be consumed or created, commitments may be undertaken, dependencies may emerge, alternatives may be eliminated or opened up, and new information may be produced. These changes alter the conditions of subsequent decisions, requiring the context, relevant objectives, and available alternatives to be reassessed. The current state of a decision is therefore the result of a trajectory, not merely the initial choice.

This perspective follows directly from what we examined in the previous chapter. If decisions produce consequences that modify the space of possibilities, then the situation on which we make decisions does not remain static. Consequences accumulate, conditions change, and new information begins to influence how the problem is interpreted. What seemed appropriate at a given moment may need adjustment, while an alternative initially discarded may become viable. A decision does not need to have reached its final state for its evolution to be relevant. In practice, many decisions go through successive iterations, each carried out based on the conditions and knowledge available at that moment.

If, in the previous chapter, we assumed the role of the chess player to anticipate the consequences of a move, we now put on the chronicler's hat to follow and record how the situation changes with each iteration. The chess player examines how a choice may alter subsequent moves; the chronicler participates in the process, observes the changes, and preserves the elements needed to understand why the current state differs from the previous one. The two perspectives complement each other. Anticipating consequences helps assess what may happen as a result of a choice, while tracking evolution makes it possible to understand what actually happened and use that knowledge to guide subsequent decisions.

In solution architecture, this relationship becomes clear when a technology initially adopted begins to influence subsequent decisions. A team may choose a particular database to meet a product's needs at its initial stage. As the application grows, that choice begins to influence hiring, tool selection, operational procedures, infrastructure costs, and integration strategies. The current architecture begins to reflect not only present requirements, but also the decisions accumulated during its construction and evolution. Understanding this state requires knowing the trajectory that produced it and the reasons certain choices were retained, adapted, or replaced.

The same dynamic appears in project management and entrepreneurship. A project may need to revise its plan because an earlier deliverable revealed an unidentified dependency or because test results changed the understanding of the requirements. A company may modify its business model after discovering that customers use the product differently than expected. In both cases, the ability to adapt depends on recognizing what changed, understanding why it changed, and using the knowledge produced to guide the next steps. Evolution does not simply consist of continuing to execute a plan, but of observing results, recording lessons learned, and assessing whether decisions remain appropriate under existing conditions.

7.1. Understanding the Evolution of a Decision

The term evolution is used in this guide to describe a change of state over time as a consequence of decisions, actions taken, information produced, and changing conditions. This definition does not presuppose improvement. A situation may evolve toward the intended objectives, but it may also accumulate costs, constraints, dependencies, and problems that make subsequent changes more difficult. An organization may acquire important knowledge by executing a strategy while simultaneously undertaking commitments that reduce its future flexibility. Evolution describes the transformation that occurred, not a positive assessment of its outcome.

This distinction matters because changes do not always represent progress, just as maintaining a decision does not necessarily mean stability. An initiative may formally remain in execution while its assumptions cease to be valid, its costs increase, or its expected benefits become less relevant. Similarly, a decision may be revised

without this representing an error in the original choice. If conditions have changed or new information has been produced, changing direction may be the most appropriate response to the current situation.

To understand this dynamic, we need to distinguish the initial decision from the state that emerges after its execution and the resulting consequences. The decision establishes a direction or authorizes a particular choice. The actions taken as a result change the situation, consume resources, produce results, and generate information. The resulting state then becomes the context for subsequent decisions. This sequence does not necessarily occur linearly: new information may emerge during execution, external changes may affect the results, and other decisions may alter the initially planned course.

For this reason, a decision should not be treated as an isolated event bounded by the moment an alternative is chosen. Depending on the domain, it may represent the beginning of a process of execution, monitoring, review, and adaptation. In other cases, the choice is a one-time event, but its effects continue to accumulate over a long period. In both cases, understanding evolution means observing the relationship between what was decided, what happened as a consequence, and the conditions that came to exist.

This perspective also makes it possible to distinguish three situations that may require different responses. The execution of a decision may need adjustments without changing its objectives or underlying rationale. The decision itself may need to be reconsidered because new evidence has weakened its assumptions or because its costs and benefits have changed. Finally, the context may change to the point where a new decision is required, with objectives or alternatives different from those originally considered. Recognizing these distinctions prevents every difficulty from being treated as an execution problem or every change in circumstances from triggering an unnecessary strategy review.

The current state, therefore, needs to be understood in relation to the trajectory that produced it and the conditions that exist now. History explains part of the situation, but it does not automatically determine the next choice. Previous decisions may contain assumptions that are no longer valid, commitments that no longer make sense, or constraints that can be removed. Knowing the past is necessary to understand the present, but it does not mean indefinitely reproducing the decisions that brought us here.

7.2. Observing Results and Learning from Experience

The evolution of a decision requires mechanisms to observe what happens after a choice is put into practice. Before execution, we work with expectations, hypotheses, and estimates. Afterward, we can compare those expectations with observed results, identify differences, and investigate their possible causes. This comparison produces information that may confirm part of the original reasoning, reveal limitations, or demonstrate that the situation was only partially understood.

Observation needs to be guided by the objectives and assumptions relevant to the decision. Measuring completed activities is not the same as verifying the results produced. A project may complete all planned deliverables without achieving the benefit that justified its investment. A product may record growth in the number of users without demonstrating that the problem it was created to solve is being addressed. A career change may provide higher compensation and still fail to meet expectations related to autonomy, learning, or quality of life. To learn from experience, it is necessary to observe what makes it possible to assess whether the decision is producing the intended effects.

Differences between expected and observed outcomes deserve attention because they may reveal problems of different kinds. A result below expectations may stem from inadequate execution, an incorrect assumption, a change in the environment, or an indirect effect that was not considered. Each possibility points to a different response. Improving execution does not necessarily resolve a problem caused by a flawed hypothesis, just as revising a strategy may be unnecessary when the difficulty stems from an isolated operational failure. The analysis must distinguish these causes before defining the intervention.

Planning, execution, observation, and learning cycles help organize this process. The PDCA cycle, for example, structures continuous improvement through planning, execution, verification of results, and action based on what has been learned. The OODA loop organizes adaptation through observation, orientation, decision, and action, emphasizing the importance of interpreting environmental changes before choosing how to respond. These are models with their own purposes and structures, not Decision Architecture models. Nevertheless, both illustrate how decisions and actions can form successive cycles of observation and adaptation.

These cycles also help explain why planning should not be confused with a definitive prediction of the future. Planning means establishing a direction and organizing actions based on the knowledge available at a given moment. Execution and observation produce information that may justify maintaining or revising that plan. The greater the uncertainty and the more dynamic the context, the more important it tends to be to incorporate learning into the decision-making process. This does not mean abandoning planning, but recognizing that its validity depends on the conditions that support it.

The frequency and depth of reviews should be proportional to the nature of the decision. A low-impact operational change can be monitored using simple indicators and frequent adjustments. A decision involving substantial investments, long-term commitments, or consequences that are difficult to reverse may require explicit monitoring criteria and formal reassessment points. In both cases, the purpose is to prevent execution from continuing merely through inertia when the evidence indicates that the course needs to be reconsidered.

7.3. Recording Iterations and Preserving Knowledge

The learning gained from the evolution of a decision can be lost when we fail to preserve enough information to understand what happened. A team may observe that a particular initiative did not produce the expected results, but without knowing the assumptions adopted, the alternatives considered, and the conditions that existed when the choice was made, it will struggle to explain the outcome. It may even repeat an approach that has already failed or discard an alternative that would be appropriate under different circumstances. The absence of records turns accumulated experience into fragmented knowledge that depends on the individual memories of those who participated in the events.

This is where the chronicler becomes important. Recording the evolution of a decision means preserving the elements that make it possible to understand each iteration and its relationship to previous ones. It is not enough to record that a choice was made or that a particular activity was completed. For significant decisions, it may be necessary to understand what problem motivated the choice, which objectives guided the analysis, which alternatives were considered, which assumptions supported the selected option, and which constraints influenced the process. Observed results, lessons learned, and the reasons for any revisions also help explain the trajectory.

The value of this record lies in the ability to recover the reasoning within its original context. A decision that appears inappropriate in light of current conditions may have been reasonable given the information available when it was made. The opposite also occurs: a choice that was initially defensible may continue to be maintained long after its assumptions have ceased to be valid. Without a record of the context and rationale, it becomes difficult to distinguish a decision that failed despite a sound process from one based on an insufficient understanding of the situation.

Because the chronicler participates in the events being recorded, the work also requires recognizing that the available knowledge changes throughout the process. An iteration may be carried out based on incomplete information and produce discoveries that change the interpretation of previous events. The record must preserve what was known at the time of the choice, without retrospectively replacing the original assumptions with explanations developed later. This distinction makes it possible to evaluate decisions according to the conditions under which they were made and understand how learning changed the direction taken.

Preserving iterations does not mean documenting every event. The volume of records alone does not guarantee useful knowledge. Extensive documentation can make it difficult to retrieve relevant information when there is no clear criterion for determining what deserves to be preserved. The goal is to record what may be needed to understand important decisions, explain changes, evaluate results, or guide future choices. The depth of the record should take into account the decision's significance, impact, reversibility, and the likelihood that other decisions will depend on it.

The mechanisms used may vary by domain. Indicators help monitor the evolution of results; feedback helps understand experiences and perceptions; retrospectives organize the analysis of what worked and what needs to change; decision records preserve the context and rationale of significant choices; technical documentation explains structures and dependencies; experiments produce evidence about specific hypotheses. These mechanisms serve different purposes and can complement one another. The choice depends on the type of knowledge that needs to be preserved and how it will be used in subsequent iterations.

Traceability emerges from the need to connect decisions, actions, results, and revisions over time. It makes it possible to recover not only the current state, but also the elements that explain how it was reached. In

complex systems, for example, knowing an application's technical configuration may not be sufficient to understand why a particular architecture was adopted. In many cases, it is necessary to recover the requirements, constraints, alternatives, and decisions that influenced its construction. Traceability is valuable when it makes these relationships understandable and accessible to those who need to make new decisions.

This knowledge also needs to survive people's departures, team reorganizations, and structural changes. When the rationale behind important decisions exists only in the memories of the people involved, the organization becomes vulnerable to knowledge loss and the repetition of investigations that have already been conducted. Preserving the history reduces this dependence and allows new people to understand the reasons behind an existing configuration before modifying it. The record does not replace the experience of those who participated in the process, but it makes part of that experience available for future decisions.

7.4. Understanding the Influence of Previous Choices

As decisions accumulate, they begin to influence the conditions of subsequent choices. A technology introduced into a system may require specific expertise, favor certain tools, and make some integrations simpler than others. A long-term contract may reduce immediate costs but limit the freedom to change suppliers. An organizational structure may facilitate the coordination of certain activities while creating obstacles for initiatives that depend on collaboration across departments. In each case, previous choices help shape the set of possibilities available in the present.

This cumulative influence is related to path dependency. The central idea is that the path taken affects which options become viable, their costs, and the conditions required to adopt them. An initial choice may create advantages that reinforce its continuation, encourage complementary investments, and make change progressively more expensive. Over time, an alternative may continue to be used not only because of its current characteristics, but also because the organization has built processes, capabilities, and commitments around it.

Path dependency does not mean that the past completely determines the future or that previous choices cannot be reconsidered. It means that any reconsideration must account for the structures and commitments accumulated since the original decision. Replacing a technology may require data migration, integration adaptation, professional training, and operational changes. Changing a commercial model may affect contracts, distribution channels, and customer expectations. The alternative of making a change needs to be evaluated by considering not only its future advantages, but also the costs and consequences of transitioning from the current state.

This perspective is related to the optionality discussed in the previous chapter. Some decisions preserve the ability to change direction, while others establish commitments that make certain changes more difficult. Over time, this difference can become more significant. An apparently simple choice may create dependencies that restrict future alternatives, while a decision that requires greater initial effort may establish a more flexible foundation for evolution. The effect depends on the context, objectives, and uncertainties that need to be considered.

For this reason, evaluating a decision requires examining both the outcome it produces and the conditions it leaves for subsequent choices. An alternative may achieve its immediate objectives and still create constraints that need to be considered. Another may produce a smaller initial benefit but preserve important options or generate knowledge that reduces future uncertainty. There is no universal rule that determines which configuration is better. What matters is recognizing that the effects of a choice extend to the decisions it constrains and that these effects are also part of its evaluation.

The history of a decision helps distinguish commitments that remain justified from those maintained merely because they have already been made. The investment made in the past is relevant information for understanding the situation, but it is not, by itself, a sufficient reason to continue investing. The current decision needs to consider the future costs and benefits of the available alternatives, including transition costs and the consequences of remaining on the existing path. Understanding how we reached the current state should expand our ability to make decisions about it, not turn the past into an obligation to continue.

7.5. Application in Project Management

In project management, the evolution of a decision can be observed through the relationship between planning, execution, monitoring, and review. The initial plan organizes deliverables, deadlines, resources, and dependencies based on the available information. During execution, however, data emerges about actual productivity, task

complexity, people's availability, and deliverable quality. This data may change the understanding of the project and require adjustments to the work sequence, resource allocation, or even the objectives themselves.

A team may, for example, decide to develop a set of features in a particular order because it believes that this sequence will reduce risks and facilitate product validation. During execution, a critical integration may reveal a technical limitation that affects several subsequent deliverables. The team must then reassess the plan, understand the consequences of the discovered dependency, and decide whether to bring the integration work forward, change the architecture, or revise the commitments made. The knowledge produced at this stage must also be preserved so that subsequent decisions take what has been learned into account.

In this context, indicators, change logs, retrospectives, and decision minutes help track the project's evolution. The goal is not to create documentary bureaucracy, but to maintain enough information to explain why the plan changed, which assumptions were revised, and what effects those changes produced. This record is especially important when team members change, scope changes occur, or decisions span different project phases.

Evaluating evolution also requires distinguishing deviations that call for correction from changes that justify a broader review. An isolated delay may be addressed through operational adjustments, while a persistent change in business conditions may require a review of the scope or expected benefits. Maintaining the original plan without considering accumulated evidence can turn execution discipline into resistance to adaptation. Conversely, continuous revision without clear criteria can undermine the stability needed to complete the work. The challenge is to use the information produced during execution to decide when to maintain the direction and when to change it.

7.6. Application in Solution Architecture

In solution architecture, evolution appears both in the transformation of systems and in the accumulation of decisions that shape their structure. A solution is rarely built all at once based on a complete understanding of every future requirement and constraint. It develops as new needs emerge, integrations are incorporated, technologies are updated, and limitations become apparent. The existing architecture therefore reflects a combination of deliberate decisions, responses to specific circumstances, and adaptations made over time.

Consider an application that starts as a relatively simple system, with few users and a small team. To reduce delivery time, the organization chooses a monolithic architecture and a technology widely known by the team. As the product grows, new requirements for availability, integration, and scale emerge. The initial choice may remain appropriate, but it may also impose limits that need to be evaluated. To understand the situation, it is not enough to compare the current architecture with technical standards considered desirable. It is necessary to understand the requirements that guided previous decisions, the problems they solved, and the dependencies that have formed since then.

Architectural decision records help preserve this knowledge. They can document the context, the alternatives considered, the determining factors, the expected consequences, and the conditions that would justify a review. When a decision is reconsidered, this history makes it possible to compare the current situation with the one that motivated the original choice. It also reduces the likelihood that a team will repeat discussions that have already been resolved or discard a solution without understanding the constraints that led to its adoption.

Architectural documentation needs to keep pace with the solution's evolution without attempting to reproduce every detail of its history. Diagrams, models, decision records, and information about dependencies serve complementary roles. The current model explains the existing structure; the record of iterations helps explain how and why it was built and modified. Both are necessary when the intention is to evaluate a change with an understanding of its effects and the commitments that will be affected.

This perspective is especially important when a solution undergoes significant transformations, such as migration to a new platform, replacement of a core component, or adoption of a new integration model. These initiatives do not start from a blank slate. They begin from a state built through previous decisions and must account for transition costs, available capabilities, existing contracts, and operational constraints. The ability to evolve an architecture depends as much on the technical quality of new choices as on understanding the path that brought the system to its current configuration.

7.7. Application in User Experience

In user experience, the evolution of a decision is related to how people interact with a product and the knowledge gained through these interactions. A team may adopt a particular interface solution based on research, behavioral hypotheses, and initial tests. After launch, actual usage may reveal difficulties that did not appear during conception, differences between user segments, or changes in expectations. This evidence may justify adjustments to the interface, the organization of workflows, or the very understanding of the problem the product seeks to solve.

Imagine a digital service that introduces a simplified registration process to reduce abandonment. The change may increase the number of people who complete registration, but it may also produce incomplete information that complicates subsequent stages of the service. Evaluating only the initial metric would lead to a partial interpretation of the result. It is necessary to observe how the change affects the entire user journey, what difficulties arise afterward, and whether the immediate benefit compromises other relevant objectives. The evolution of the decision depends on understanding these effects together.

Experiments, user research, behavioral analysis, and feedback help produce this knowledge. For it to guide subsequent iterations, it is also necessary to preserve the context in which each change was made, the hypothesis that motivated it, and the results observed. Without this record, a team may repeat similar experiments, interpret metric changes without considering previous changes, or lose track of why certain interface patterns were adopted.

The product's history also helps distinguish temporary preferences from persistent needs. A solution may have been adopted to address a technical limitation that no longer exists, or an interaction may remain in place because it was never reassessed after a change in the user profile. Understanding the trajectory does not mean retaining every previous decision. It means evaluating whether they still make sense in light of current evidence and recognizing which consequences need to be considered before changing them.

User experience evolves when teams turn observations into knowledge and use that knowledge to guide changes. The process requires more than collecting metrics: it requires relating results to the hypotheses, objectives, and decisions that produced the observed experience. In this way, each iteration can start from a more consistent understanding of what worked, what needs to be changed, and which conditions must be preserved.

7.8. Application in Entrepreneurship

In entrepreneurship, decisions are often made under conditions of uncertainty, with incomplete information and limited resources. The business model, value proposition, acquisition channels, and cost structure may need adjustments as the company learns from the market. A hypothesis about customer behavior may be confirmed, weakened, or replaced by another interpretation. Business evolution depends on the ability to turn this evidence into decisions without confusing persistence with insistence on assumptions that are no longer supported.

A startup may begin by offering a product to a specific segment and discover, during the first few months of operation, that another group shows greater interest or derives more value from the solution. This discovery may justify a change in positioning, but its effects go beyond commercial communication. It may change priority features, sales channels, pricing, required support, and the capabilities the team needs to develop. The decision to change the target audience modifies the state of the business and creates new conditions for subsequent decisions.

To understand this evolution, it is important to preserve the hypotheses tested, experiment results, market signals observed, and rationale behind changes in direction. This history makes it possible to distinguish an adaptation supported by evidence from a succession of changes motivated solely by passing impressions. It also helps identify which lessons remain valid even when a specific strategy is no longer used.

The record does not need to take the form of extensive reports. Depending on the stage of the business, it may consist of interview results, experiment metrics, records of hypotheses and decisions, or periodic analyses of indicator trends. The central criterion is to preserve information that makes it possible to understand what the company has learned and how that learning has changed its choices.

The trajectory of a venture also accumulates commitments. Contracts, investments, hiring decisions, brand positioning, and customer expectations may make some changes more costly than they would have been at the beginning. Recognizing these commitments does not mean avoiding them at all costs, but taking them into

account when evaluating new alternatives. A strategy that is appropriate at one stage may cease to be the best option at another, and the decision to maintain or replace it needs to consider both the knowledge acquired and the effects accumulated during its execution.

7.9. Application in Professional Careers and Personal Life

In professional careers and personal life, the evolution of decisions can also be understood as a sequence of choices that transforms the conditions of subsequent choices. A person may decide to pursue a course of study, accept a job opportunity, move to another city, or devote more time to a personal project. Each decision consumes resources, establishes commitments, develops skills, and changes the available alternatives. Over time, accumulated experiences alter both objective conditions and the way the person understands their priorities.

A career change, for example, may begin with the expectation of gaining greater autonomy or better development opportunities. After a few months, the actual experience may reveal differences between expectations and working conditions. The person may discover that they value aspects previously considered secondary or that certain transition costs were underestimated. This information may justify adjustments to the plan, a change of role, or a review of professional objectives. The initial decision does not need to be retrospectively classified as right or wrong merely because the results differed from expectations. It is necessary to understand what was known at that moment and what the experience made it possible to discover.

Preserving the history of these choices can help identify patterns and understand changes in priorities. Personal records, periodic reviews of objectives, and reflections on significant experiences make it possible to compare expectations with results and identify lessons that might otherwise go unnoticed in daily life. The purpose is not to document every everyday decision, but to retain information that helps interpret significant choices and prevents accumulated experience from being reduced to imprecise recollections.

This perspective also helps explain the commitments that form throughout life. A course of study may open up opportunities while requiring time that is no longer available for other activities. A move to another city may expand professional possibilities and change proximity to family and friends. A financial decision may provide security over a particular time horizon while reducing flexibility to take on new projects. The effects do not need to be entirely positive or negative: they may combine gains, losses, and new possibilities that only become clear over time.

Reassessing a personal decision does not necessarily mean abandoning the chosen path. It may mean adjusting how it is carried out, recognizing that objectives have changed, or consciously accepting the costs of continuing. The history of choices provides elements for this assessment, but it should not turn previous commitments into permanent obligations. The current context, present priorities, and future possibilities need to be considered together with what has been learned along the way.

8. More Practical Examples of Application

In the previous chapters, we examined six dimensions of Decision Architecture: understanding the domain, discovering the initial context, establishing objectives, managing risks and uncertainties, anticipating direct and indirect consequences, and dealing with the evolution of a decision. Now, we will bring these dimensions together in concrete situations, examining how they contribute to structuring decisions across different areas.

The examples follow the order of the previous chapters to facilitate understanding of the process. In each case, we will begin by defining the boundaries of the decision domain, investigate the situation that prompted the need to act, establish objectives, examine risks and uncertainties, compare the consequences of the alternatives, and track how the choice evolves. This organization makes it possible to recognize the contribution of each dimension without reducing decision-making to a mechanical sequence of activities.

In practice, these dimensions do not need to be addressed only once, nor always in the same order. Information discovered during risk analysis may change the objectives, just as a consequence identified when comparing alternatives may require further investigation. The sequence presented serves as a framework for reasoning and exposition, not as a rule that prevents revisions. The goal is to show how to build a sufficiently solid understanding to take action and how to use experience to guide subsequent decisions.

8.1. Project Management: Recovering a Delayed Project

A company is developing a new platform to serve its customers. The project was planned to last six months, but by the end of the fourth month, some important features remain incomplete, integration with an external system is proving difficult, and testing is revealing problems that had not been identified during planning. The launch date has already been communicated to strategic customers, and leadership must decide how to manage the project given the current situation.

Understanding the Domain

The first step is to understand the system in which the decision is embedded. The domain is not limited to the project schedule. It includes the contracted scope, available budget, team capacity, technical dependencies, quality criteria, customer expectations, and commitments made by the organization.

These elements are interrelated. Changing the scope may reduce the required effort but also affect the value of the deliverable. Expanding the team may increase execution capacity but requires time for onboarding and supervision. Delaying the launch may allow for additional testing but generate contractual costs or jeopardize commercial commitments. Likewise, preserving the original date without considering the other conditions may result in an incomplete or unstable release.

Understanding this domain makes it possible to recognize that the decision is not simply about recovering lost days. It is about reorganizing the project so that the most important outcomes are achieved within acceptable constraints of time, cost, quality, and operational capacity.

Discovering the Initial Context

With the domain defined, the team investigates what caused the delay. Comparing the plan with the work completed shows that external integration took longer than estimated. Requirements also changed frequently, and some deliverables stalled while awaiting validation from the business units. In addition, different managers prioritized features without adequately considering their dependencies.

The investigation therefore reveals that the delay does not have a single cause. Some of it stems from technical difficulties, some is related to requirements management, and some results from coordination between teams. Indiscriminately increasing development capacity would not solve all these problems.

The team also identifies that some features planned for launch are important for future stages but are not essential for customers to begin using the platform. This discovery creates an opportunity to reconsider the scope without necessarily compromising the project's purpose.

Establishing Objectives

Leadership needs to clarify which outcomes should guide the recovery effort. Launching on the original date remains desirable, but it cannot be treated as the sole objective. The platform must meet customers' essential needs, maintain an acceptable level of quality, stay within financial limits, and avoid placing an unsustainable burden on the team.

These objectives also help distinguish non-negotiable commitments from expectations that can be revised. A contractual requirement may demand formal negotiation to be changed, while a secondary feature might be delivered in a later version. The decision about what to preserve depends on the value of each deliverable and its contribution to the expected outcomes.

The team therefore establishes a priority: deploy a reliable version of the platform containing the essential features without exceeding the defined quality and budget limits. The deadline remains an important criterion, but it will be assessed alongside the other objectives.

Managing Risks and Uncertainties

The recovery alternatives depend on estimates that contain uncertainties. The team still does not know exactly how long it will take to stabilize the external integration, and the effort required to fix the problems found during testing may vary depending on the causes identified.

There are also risks associated with the recovery measures themselves. Hiring temporary professionals may not produce immediate gains because they need to become familiar with the system. Authorizing overtime may increase fatigue and the likelihood of errors. Reducing the scope without examining dependencies may compromise features that appear independent but are necessary for the platform to function.

The team examines these possibilities and recognizes that keeping the original plan unchanged also involves risks. An attempt to meet the deadline at any cost may result in failures after launch, undermining customer confidence and generating remediation costs.

To reduce the most significant uncertainties, the team conducts a focused technical analysis of the integration, revises estimates for critical activities, and establishes quality criteria that cannot be relaxed. Leadership also sets limits on the team's additional workload and identifies which features can be postponed without compromising the usefulness of the first release.

Anticipating Direct and Indirect Consequences

With the main uncertainties mapped, the team compares the alternatives. One option is to expand the team while retaining the full scope. Another is to reduce the features included in the first release. A third is to renegotiate the deadline to deliver the original scope more safely. It is also possible to combine a scope reduction with a limited schedule extension.

Expanding the team may help with independent activities but have little effect if the main bottleneck is concentrated in the external integration. Reducing the scope lowers the immediate workload, but dependencies between features must be considered. Delaying the launch allows more time for testing but may affect commercial commitments. Each alternative changes not only the schedule but also costs, expected quality, workload, and the conditions for subsequent deliveries.

The analysis indicates that some features can be postponed without compromising the platform's essential use. The team also identifies activities that can proceed in parallel while specialists work on the main integration. Leadership decides to retain the launch date as a reference point, remove secondary features from the first release, and assign the most experienced professionals to resolving the main technical bottleneck.

The plan includes frequent checkpoints to verify whether the recovery is working. This prevents the team from treating the revised forecast as a guarantee and preserves the option to revise the plan if observed progress differs from expectations.

Dealing with the Evolution of a Decision

In the following weeks, integration progresses, but testing reveals performance problems in one of the features retained in scope. The team must assess whether the fix can be completed in time or whether this deliverable

must also be postponed.

Because the quality and prioritization criteria were defined earlier, the discussion can focus on the consequences of each alternative. The team concludes that retaining the feature would require reducing the testing time for other components, creating a disproportionate risk for the launch. Leadership decides to postpone it, communicates the change to affected customers, and establishes a specific forecast for its subsequent release.

The platform launches with fewer features than originally planned, but its essential capabilities operate reliably. After delivery, the organization records the causes of the delay, the assumptions that proved incorrect, and the effects of the recovery measures. This information helps guide estimates, dependency management, and validation mechanisms in future projects.

The example shows that recovering a project does not simply mean accelerating delayed activities. It requires understanding what is happening, defining the outcomes that must be preserved, evaluating alternatives, and monitoring their effects. The decision continues to evolve during execution, and management quality also depends on the ability to recognize when the plan needs to be revised.

8.2. Solution Architecture: Modernizing a Legacy System

A company relies on a system developed more than ten years ago to manage orders, inventory, and billing. The system continues to support essential operations, but its limitations make maintenance and integration with new applications difficult. Few professionals understand its most critical components, and relatively simple changes require lengthy testing. Leadership is considering replacing the platform with a modern architecture capable of supporting business evolution.

Understanding the Domain

The decision is embedded in a domain that brings together operational systems, business processes, data, integrations, technical competencies, and continuity requirements. The legacy system does not operate in isolation. It participates in order processing, inventory updates, and the generation of financial information, connecting activities that must continue functioning throughout any transformation.

The architecture is also constrained by team capabilities, available resources, and the acceptable level of operational risk. A technically sophisticated solution may be unsuitable if it requires skills the organization cannot maintain. Likewise, a migration that interrupts order processing may have commercial consequences far greater than the expected benefits of modernization.

Understanding the domain means recognizing that the decision involves both the future architecture and the continuity of current operations. The goal is not merely to replace outdated technology but to create the conditions for the company to evolve without compromising the services on which it depends.

Discovering the Initial Context

The architecture team investigates the system's main problems. It analyzes incidents, maintenance costs, the time required to implement changes, dependencies between modules, and integration difficulties. It also speaks with the teams that use the platform to identify which limitations most directly affect their day-to-day work.

The investigation reveals that the main obstacles are concentrated in two modules: one responsible for pricing and another for integrations with sales channels. Other components, although old, are stable and require few changes. Full replacement would solve real problems but would also involve parts of the system that do not justify immediate investment.

The team also identifies that some business rules are undocumented. Part of the knowledge resides with experienced professionals and is embedded in the system's behavior. A new implementation would need to reproduce these rules correctly, including those that have not been formally specified.

The context shows that the difficulty does not stem solely from the age of the technology. The problem lies in the combination of technical limitations, accumulated dependencies, and concentrated knowledge. This understanding makes it possible to formulate a more targeted intervention.

Establishing Objectives

The company defines its objectives as reducing the time and cost of changes, facilitating integration with new sales channels, reducing dependence on knowledge concentrated in a few people, and preserving operational reliability. It also intends to improve the platform's ability to evolve without creating operational complexity that exceeds the team's capacity to sustain it.

These objectives help distinguish desirable characteristics from specific technology choices. Adopting independent services, migrating to the cloud, or replacing the programming language may be means of achieving certain outcomes, but they are not objectives in themselves. The technology choice must be justified by its contribution to the identified needs.

The company also establishes limits on investment, service interruptions, and the risk of data loss or inconsistency. These criteria will be used to compare alternatives and assess whether the transformation is producing the expected results.

Managing Risks and Uncertainties

The team recognizes that significant knowledge gaps remain. Some dependencies between modules may be undocumented, and the effort required to reconstruct business rules is uncertain. It is also difficult to predict in advance how a new architecture will behave under all operating conditions.

Full replacement concentrates risks within a single transformation period. Gradual migration reduces immediate exposure but may require temporarily maintaining both old and new components. Purchasing a commercial platform may reduce development effort but create vendor dependence and limit future customization.

To reduce uncertainties, the team identifies critical dependencies, documents known business rules, and conducts technical experiments with the most problematic modules. It also defines data validation mechanisms, performance criteria, and conditions for halting or reversing each migration stage.

The team recognizes that it will not be possible to eliminate all risks before starting. Some knowledge will emerge only when the new components are used under real-world conditions. The strategy must therefore enable progressive learning without unnecessarily exposing essential operations.

Anticipating Direct and Indirect Consequences

The company compares four alternatives: retaining the current system with targeted improvements, replacing the entire platform, adopting a gradual migration, or purchasing a commercial solution that covers part of its needs.

Keeping the current system requires less immediate investment but prolongs existing limitations. Full replacement offers greater freedom to reorganize the architecture but requires rebuilding business rules and concentrating validation efforts. The commercial solution reduces some development work but may require changes to company processes and create recurring costs. Gradual migration spreads the investment over time and allows technical choices to be validated, although it introduces a period during which both architectures coexist.

The analysis includes effects that extend beyond the replaced components. A distributed architecture may facilitate independent module evolution but requires observability practices, deployment automation, and failure handling. Cloud migration may change the cost structure and operational responsibilities. Replacing a component may also affect interface contracts, integrations, and procedures that depend on its current behavior.

The team decides to begin a gradual migration, starting with the pricing module. Before definitive replacement, it will run tests with representative data and compare the results of the new implementation with those produced by the current system. This decision makes it possible to validate business rules and assess the architecture's behavior without transforming the entire operation simultaneously.

Dealing with the Evolution of a Decision

The first stage produces positive results. The new module makes pricing rule changes easier and reduces the effort required to integrate with a sales channel. However, during operation, the team discovers that some processes depend on specific data formats and behaviors of the old system. Integration with billing requires an adaptation that had not been anticipated.

This discovery does not invalidate the gradual migration strategy. It reveals a dependency that needed to be understood before proceeding. The team documents the identified behavior, adjusts the tests, and revises the planned sequence for replacing the remaining modules. It also begins requiring an explicit dependency analysis before each new migration.

The company does not yet have an entirely new platform, but it has modernized a critical component and expanded its knowledge of the existing system. The next stages will be evaluated based on observed results, the cost of maintaining coexistence between components, and the team's ability to absorb further changes.

The architectural decision is no longer treated as a single, definitive choice. Instead, it guides a sequence of interventions that progressively transforms the solution while preserving the ability to learn and adjust the course as actual conditions become better understood.

8.3. User Experience: Reducing Abandonment in the Checkout Flow

An e-commerce company notices that many users add products to their carts but do not complete their purchases. Leadership requests a review of the checkout process, and the initial hypothesis is that the flow contains too many steps. The team considers consolidating the fields onto a single page and highlighting the payment button, but it needs to better understand user behavior before choosing an intervention.

Understanding the Domain

The decision belongs to the domain of the shopping experience, which involves user interaction with the interface, product and pricing information, shipping cost calculations, payment methods, transaction security, and the systems responsible for processing orders.

These elements are interrelated. Simplifying the interface may facilitate navigation but will not solve an unexpected-cost problem. Reducing the number of requested fields may decrease the effort required to complete them but compromise information needed for delivery. Making payment faster may improve conversion but must preserve security controls.

Understanding the domain makes it possible to recognize that cart abandonment is not merely a visual problem. It occurs within a journey that combines expectations, information, interactions, and operational constraints. The decision must improve the experience without compromising purchase reliability.

Discovering the Initial Context

The team analyzes browsing data to identify the stages at which the greatest number of users drop off. It compares behavior on mobile devices and computers, checks differences between product categories and payment methods, and examines error logs and complaints. It also speaks with users who started a purchase but did not complete it.

The data shows that abandonment increases on mobile devices, particularly while entering the address and after the total amount is updated to include shipping. Interviews also reveal that some customers avoid creating an account before they are certain they want to complete the order.

The initial hypothesis about too many steps explains only part of the problem. The investigation indicates that users encounter different sources of friction: difficulty entering information, surprise at the final cost, and registration requirements that seem premature. These findings help direct the decision toward concrete problems rather than assuming that a broad visual redesign will produce the desired results.

Establishing Objectives

The team defines its objectives as making purchases easier to complete on mobile devices, presenting the total cost predictably, reducing address-entry errors, and allowing users to purchase without being required to create an account.

At the same time, it must preserve transaction security, the quality of delivery information, and the ability to identify orders and provide customer support. Success will not be measured solely by an increase in conversion, because a change could increase completed purchases while simultaneously increasing fraud, delivery errors, or support requests.

The objectives must represent both the expected benefit to the user and the conditions required to maintain operations. The desired experience is one in which customers can complete purchases clearly and confidently, without unnecessary barriers.

Managing Risks and Uncertainties

The team recognizes that the available data helps locate the problems but does not establish with certainty how much each intervention will contribute to solving them. Observed behavior may vary according to device, region, product type, and user profile.

Combining all fields onto a single screen may reduce navigation but make data entry more difficult on mobile phones. Guest checkout may remove a barrier but requires decisions about how order tracking and customer communication will be handled. Address autocomplete may reduce errors but produce incorrect suggestions when the underlying database does not cover certain locations.

To reduce these risks, the team plans to test the changes in a controlled manner and monitor metrics beyond conversion. It also defines conditions for identifying negative effects and reversing a change if it harms the experience or operations.

Anticipating Direct and Indirect Consequences

The team compares different interventions: reducing the number of steps, presenting shipping costs earlier, enabling guest checkout, and simplifying address entry on mobile devices. Rather than redesigning the entire flow at once, it decides to combine changes directly related to the identified problems and evaluate their results in a controlled manner.

Presenting shipping costs earlier may reduce drop-offs near payment, but it depends on having sufficient information to estimate the cost. Guest checkout may facilitate the first transaction but must preserve appropriate ways to access the order. Autocomplete may speed up the interaction, but it must allow corrections and provide a manual alternative.

The team decides to display the estimated delivery cost before the final stage, allow checkout without mandatory registration, and simplify address entry. Implementation will be monitored using the completion rate, data-entry errors, payment failures, time required to complete a purchase, and customer support contacts.

This approach makes it possible to examine not only the immediate effects of the changes but also their potential impact on other parts of the journey. The desired outcome is not an interface with fewer steps at any cost, but an experience that facilitates purchasing without introducing new obstacles.

Dealing with the Evolution of a Decision

After implementation, the data indicates an improvement in purchase completion on mobile devices. Showing shipping costs earlier reduces drop-offs near payment, and guest checkout attracts users who did not want to create an account. However, address autocomplete presents difficulties in certain regions. Some users receive incorrect suggestions and must correct several fields.

The team analyzes the results and realizes that the feature helps when it finds an appropriate match but becomes an obstacle when it fails to recognize the address. Instead of removing the feature for all users or keeping its current behavior, the team decides to make it optional, allow manual entry, and improve the coverage of the data used to generate suggestions.

The team also begins segmenting metrics by region, preventing an improvement in the overall average from concealing difficulties experienced by specific groups. In the following weeks, errors decrease, although some regions still require further investigation.

The example demonstrates that user experience decisions depend on interpreting behavior, formulating hypotheses, and evaluating the effects of interventions. Implementation does not conclude the decision-making process. It generates information that makes it possible to better understand the journey and guide further adjustments.

8.4. Entrepreneurship: Choosing Between Growing and Consolidating the Business

A software company founded three years ago has built a recurring customer base and is beginning to receive opportunities to expand its operations. The founders are considering hiring a sales team, increasing marketing investment, and accelerating product development. At the same time, the operation still depends directly on them for several decisions, cash flow fluctuates, and some customer requests remain outstanding.

Understanding the Domain

The decision is embedded in the domain of business strategy and sustainability. It involves market demand, revenue, profitability, customer acquisition costs, operational capacity, product development, cash flow, and team competencies.

These elements influence one another. Increasing sales may raise revenue but also increase implementation and support costs. Developing new features may open up commercial opportunities but consume resources that would otherwise be needed to improve current operations. Hiring professionals may distribute responsibilities but create recurring expenses before the company has sufficient revenue to sustain them.

Understanding the domain makes it possible to recognize that the decision is not simply about choosing between growth and staying as things are. The company needs to determine how to combine the pursuit of opportunities with the development of the capabilities required to sustain them.

Discovering the Initial Context

The founders analyze recurring revenue trends, customer contribution margins, acquisition costs, the time required to recover sales investment, and the reasons some customers stop using the product. They also assess support capacity, platform stability, and dependence on centralized decision-making.

The investigation reveals that demand is growing, but the company faces customer service bottlenecks and still lacks some important features for new segments. The sales team can generate opportunities, but some do not convert because the product does not meet these customers' specific requirements.

The company realizes that an expansion opportunity exists, but its ability to capitalize on it is constrained by internal limitations. Growing immediately may increase business volume without resolving the factors preventing some sales. On the other hand, postponing all expansion may mean losing opportunities while competitors advance in the market.

Establishing Objectives

The founders define their objectives as sustainably increasing revenue, preserving service quality, reducing the founders' operational dependence, and maintaining a financial position capable of withstanding fluctuations in demand. They also intend to explore new segments without compromising their ability to serve existing customers well.

These objectives show that growth and consolidation do not have to be mutually exclusive paths. The question is to determine what pace of expansion is compatible with available resources and which capabilities must be developed to sustain growth.

The company also establishes limits on commercial investment and defines results that must be observed before expenses are increased. In this way, the desire to grow becomes guided by criteria that make it possible to assess whether the strategy is producing the expected results.

Managing Risks and Uncertainties

The company does not know precisely how quickly new opportunities will convert, how much it will cost to serve customers in other segments, or how much investment will be required to complete outstanding features. Sales forecasts are based on a market that may change, and the results of an initial campaign may not represent future behavior.

Immediate expansion carries the risk of increasing expenses before generating sufficient revenue. Consolidation reduces some of this exposure but may prolong the limitations that hinder growth. The company must also

consider the risk of continuing to depend on the founders for operational activities, which may restrict its execution capacity even if demand increases.

To reduce uncertainties, the founders analyze the segments in which they already have satisfied customers, determine which needs can be met with the current product, and establish a limited budget to test demand. They also identify operational improvements that could reduce implementation and support effort.

Anticipating Direct and Indirect Consequences

The company compares three alternatives. The first is to accelerate expansion immediately by hiring salespeople and increasing marketing investment. The second is to focus the next few months on consolidating operations before expanding customer acquisition. The third is to pursue selective expansion by investing in a specific segment while resolving internal bottlenecks.

Immediate expansion may capture market share quickly but increases financial and operational exposure. Consolidation may improve efficiency but does not guarantee that opportunities will remain available. Selective expansion spreads the commitment over time, allowing commercial assumptions to be tested before larger expenses are incurred.

The analysis also considers the effects on the organization. Hiring quickly may require the founders to devote more time to people management, reducing their availability for strategic decisions and product development. Postponing all expansion may prolong the founders' operational dependence because the company will continue to lack resources dedicated to distributing responsibilities.

The founders decide to invest in selective expansion. They choose a segment in which they already have satisfied customers and whose primary needs can be met without an extensive product overhaul. They establish a limited commercial budget, conversion and retention targets, and operational metrics that must remain within acceptable limits. In parallel, they prioritize improvements to the implementation process and documentation of activities that currently depend directly on the founders.

Dealing with the Evolution of a Decision

In the first few months, the initiative generates new contracts, but the sales cycle is longer than expected. The customers acquired show good retention, but they require more support during implementation. The additional revenue is positive, but the return on commercial investment takes longer to materialize.

The founders must decide whether to expand the team to capitalize on market interest or maintain the current pace while improving implementation. Because financial and operational limits were defined earlier, the discussion can be based on observed results rather than relying solely on enthusiasm about the new contracts.

The company keeps commercial investment under control and directs part of its efforts toward standardizing implementation, improving documentation, and transferring operational tasks to other team members. After another cycle, it finds that implementation time has decreased and service capacity has increased. With this evidence, it begins considering broader commercial expansion.

The decision was not simply about choosing between growth and consolidation. It was about determining the sequence and pace of initiatives while recognizing that the ability to capitalize on an opportunity depends on the business's internal conditions. By turning a broad bet into a bounded experiment, the company preserved the option to increase investment without prematurely assuming all the costs and risks of accelerated expansion.

8.5. Professional Career and Personal Life: Moving to Another City in Search of Opportunities

A 26-year-old professional lives in a mid-sized city, where they have worked for three years. They have a stable income, live near their family, and maintain a circle of friends built over the course of their life. Despite these favorable conditions, they feel that their professional opportunities are limited. Companies in the region offer few positions in their area of interest, salaries seem to grow slowly, and opportunities to experience new environments are limited. While talking to former colleagues, they realize that many have found opportunities in a state capital and begin considering a move.

Understanding the Domain

The decision involves more than choosing an address. Its domain includes career, income, cost of living, housing, commuting, personal relationships, professional development, and quality of life. These elements are interconnected, and a change in one may affect the others.

The capital offers a greater variety of companies and roles, but it also has potentially higher housing costs and a different daily routine. Distance may reduce spontaneous time with family and friends, while a new network of relationships may expand professional and personal opportunities. The decision is also constrained by financial reserves, the possibility of finding a job, and the willingness to face an adjustment period.

Understanding this domain makes it possible to recognize that the move cannot be evaluated solely by the number of available vacancies or the difference between salaries. It is necessary to consider the overall set of conditions that determine whether the new situation will be sustainable and compatible with what the professional wants from life.

Discovering the Initial Context

Before deciding, the professional investigates the causes of their dissatisfaction. They analyze their career path, identify which aspects of their current job cause the greatest discomfort, and speak with people working in similar roles at other companies.

The investigation reveals that the problem is not just the city. They want to work on more complex projects, develop skills that are not yet in significant demand in their current role, and broaden their growth prospects. The capital offers more opportunities, but there are also remote or hybrid work options, companies in the region that could offer more challenging roles, and training programs that could improve their competitiveness.

This discovery changes the initial question. Instead of deciding only whether to move to another city, the professional begins investigating how to expand career opportunities and improve quality of life without excessively compromising financial security. Moving remains a relevant alternative, but it is no longer treated as an objective in itself.

Establishing Objectives

The professional defines their objectives as securing a role with greater development potential, increasing income over the medium term, expanding professional and personal experiences, and building a more diverse network of relationships. At the same time, they intend to preserve sufficient financial reserves to handle unexpected events and avoid a prolonged period of instability.

Proximity to family and friends also has value, although it is not the main source of their current dissatisfaction. This aspect must be considered because professional improvement may come with personal costs that cannot easily be converted into money or compared using a single metric.

The professional recognizes that a higher salary may come with a higher cost of living and that a role with more responsibilities may require working hours that harm quality of life. The decision will be guided by the overall set of desired outcomes, not just the expectation of earning more.

Managing Risks and Uncertainties

The professional still does not know whether they will secure a suitable position in the capital, how much they will actually spend on housing and transportation, or how they will adapt to the new routine. Even information obtained through research and conversations with acquaintances does not guarantee that their experience will resemble that of others.

Moving without a job may require drawing on savings for an extended period and pressure the professional to accept an unsuitable position. An offer with higher compensation may not offset the increased cost of living. Distance from family may reduce the support available during difficult periods. On the other hand, remaining indefinitely in the current situation also involves risks: postponing skill development, missing opportunities, and accumulating dissatisfaction.

To reduce uncertainties, the professional researches salaries, rents, transportation, and everyday expenses in the capital. They speak with people who have made similar moves and analyze actual job openings to identify

the most sought-after skills. They also establish a minimum financial reserve, a monthly spending limit, and professional conditions that would justify the move.

The research reveals that their experience is valued in some positions, but the most attractive opportunities require knowledge they still need to develop. The professional decides to invest in these skills while searching for jobs, keeping their current position and avoiding permanent expenses associated with moving until they have more concrete evidence about their prospects.

Anticipating Direct and Indirect Consequences

The professional compares different paths. They can remain in their current city and invest in training and networking. They can seek remote or hybrid positions at companies in other regions. They can apply for positions in the capital and move only after receiving a suitable offer. They can also make a temporary move or relocate immediately, using their savings to finance their job search.

Staying preserves stability and the support network but may not provide the desired variety of opportunities. Seeking remote work expands the accessible job market without requiring an immediate move, although not all roles allow this arrangement. Moving after receiving an offer reduces financial uncertainty but may require a longer job search. Moving without a job provides the freedom to seek local opportunities but shifts a greater share of the financial risk to the professional.

The analysis also considers indirect consequences. A move may expand the professional network but reduce the time available to maintain personal relationships. Higher compensation may not increase savings capacity if expenses grow at the same rate. A more demanding role may accelerate professional development but also change daily routines and the time available for activities outside work.

The professional decides to start by applying for positions at companies in the capital and speaking with people working in their areas of interest. After a few months, they receive an offer that provides greater responsibility, higher compensation, and learning opportunities. Before accepting, they clarify the expectations for the role, confirm the hybrid work arrangements, and research housing options with reasonable access to the office. They calculate the initial costs and determine that they can make the move without using the financial reserve they consider necessary for unexpected events.

The decision to move is therefore supported by a concrete opportunity, predefined criteria, and a more realistic estimate of living conditions. The professional accepts the offer and organizes the transition.

Dealing with the Evolution of a Decision

In the first few months, the new job provides learning opportunities and contact with professionals from other fields. However, everyday experience reveals difficulties that the initial research had not fully anticipated. The chosen neighborhood makes commuting lengthy on in-office days, and housing costs reduce the ability to save. Distance from family and friends also requires more planning to maintain personal ties.

The move has produced some of the desired outcomes, but it has changed important aspects of the professional's life. Rather than evaluating the decision simply as a success or failure, they reassess the situation in light of experience. They identify which difficulties can be resolved without giving up the professional opportunity and consider negotiating greater flexibility on in-office days, moving to another neighborhood, or reorganizing visits to family.

The analysis shows that moving immediately to a different residence would require new expenses and commitments, whereas adjusting the work routine could alleviate part of the problem. The professional speaks with their manager and manages to group in-office days into consecutive periods, reducing commuting. They also research housing alternatives but decide to wait until the rental contract expires before making a new commitment.

Over time, they monitor changes in expenses, assess their adjustment to the city, and determine whether the role continues to offer the expected development. The initial decision is not treated as a commitment that must be defended at all costs. It remains subject to adjustments as professional and personal circumstances change.

The case demonstrates that complex personal decisions do not need to be treated as irreversible bets. It is possible to investigate the situation, define objectives, compare alternatives, establish exposure limits, and choose a path that allows learning from experience. The process does not eliminate costs or guarantee that all expectations

will be met, but it helps distinguish what can be known before making a choice from what can only be assessed after taking action.

The five examples show how the same dimensions of Decision Architecture can be applied to very different problems. The nature of the domain, objectives, risks, and consequences varies, but the need remains to understand the situation before prescribing a response, make the criteria guiding the choice explicit, and monitor the effects produced by the action. The sequence presented provides a framework for organizing this reasoning, while experience and new information determine when further investigation or revisions to previous decisions will be necessary.

Conclusion

Throughout this guide, I have sought to organize a perspective that makes it possible to understand decisions across different contexts, from project management, solution architecture, and business development to user experience, professional careers, and personal life. Although these fields have their own objectives and characteristics, they all involve situations in which we need to understand conditions, establish directions, evaluate possibilities, and deal with the effects of our choices. Identifying these common elements made it possible to bring together the foundations presented in this work.

I believe the main contribution of this approach lies in providing a framework for thinking before acting. Understanding a decision requires more than identifying alternatives and selecting one of them. It is necessary to recognize what is at stake, which conditions influence the available possibilities, and how a choice may affect not only the immediate outcome but also future opportunities and decisions. The foundations presented seek to make this analysis more conscious and consistent, without eliminating the need for judgment and adaptation to each situation.

This work represents a stage, not an ending. The concepts have been organized based on the experiences, studies, and reflections accumulated so far, but their maturity will depend on further applications and learning. Some foundations may be explored in greater depth, examples may be expanded, and aspects of the structure itself may be revised. Even the term Decision Architecture remains open to change if another expression better represents the proposal.

I hope this guide can serve as a starting point for new discussions and applications. If other professionals and researchers find these ideas useful, I would like them to feel invited to question, experiment with, and contribute to their development. The possibility of building a community around these concepts, bringing together different experiences and perspectives, would be an especially meaningful outcome.

Finally, I would like to thank you for the time dedicated to reading this guide and for your willingness to explore this proposal. Comments, criticism, and suggestions are welcome and can directly contribute to improving this work. I am available to receive feedback at cpzjunior@gmail.com. Rather than presenting a finished framework, my intention was to organize ideas that can continue to evolve, and any contribution in that direction will be welcome.